

week4

2학년 2반 2021041089_최태민

실습1 1입력 IN_X7_OUT_Y7 영상주소 : https://youtu.be/A_xMtlQAUy8

실습2 2입력 AND_gate 영상주소 : <https://youtu.be/JCM5srviBGA>

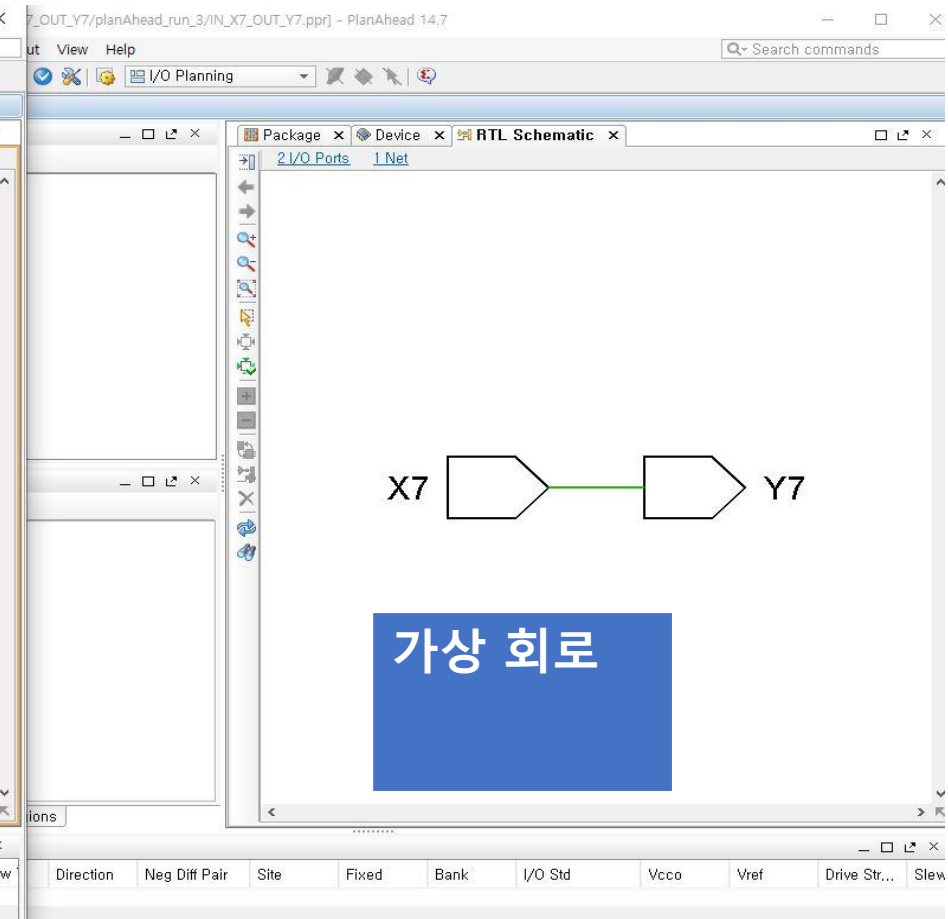
실습3 3입력 OR_gate 영상주소 : <https://youtu.be/ifIAiSAXOr0>

과제 1, 2, 3 입력 게이트 영상주소 : <https://youtu.be/fq0-KQaXvw0>

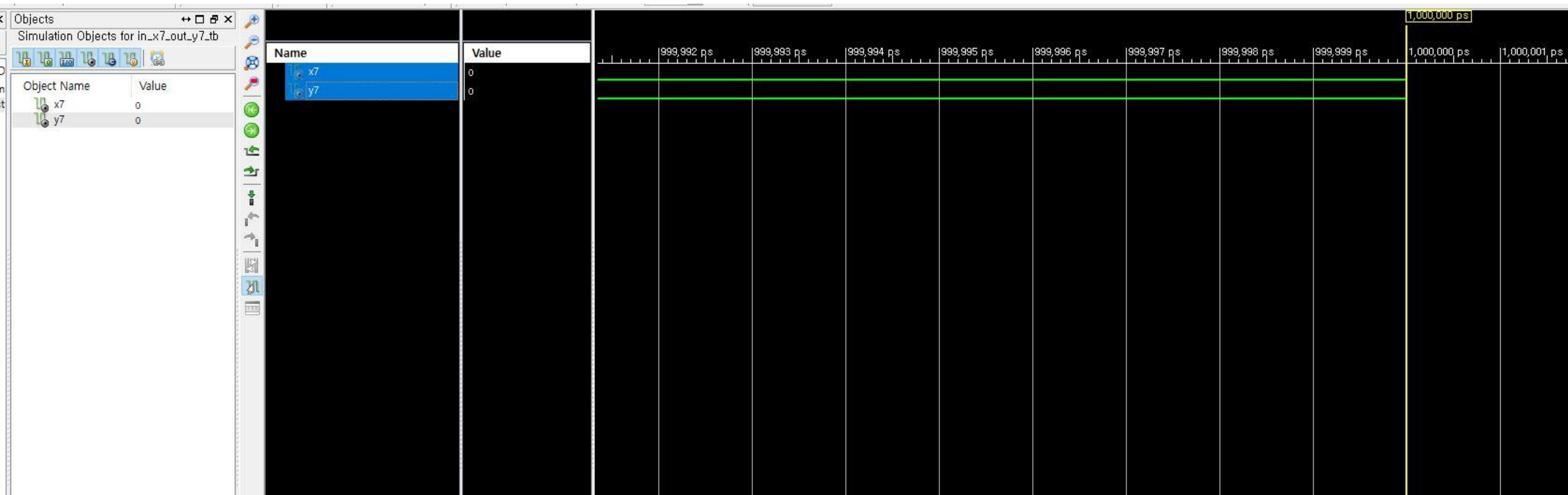
실습1 입력, 출력 (동작테이블)

입력 출력 동작 Table	
입력 스위치 X7	출력 LED Y7
0	0
1	1

실습1 입력, 출력 (가상회로, fpga 회로)



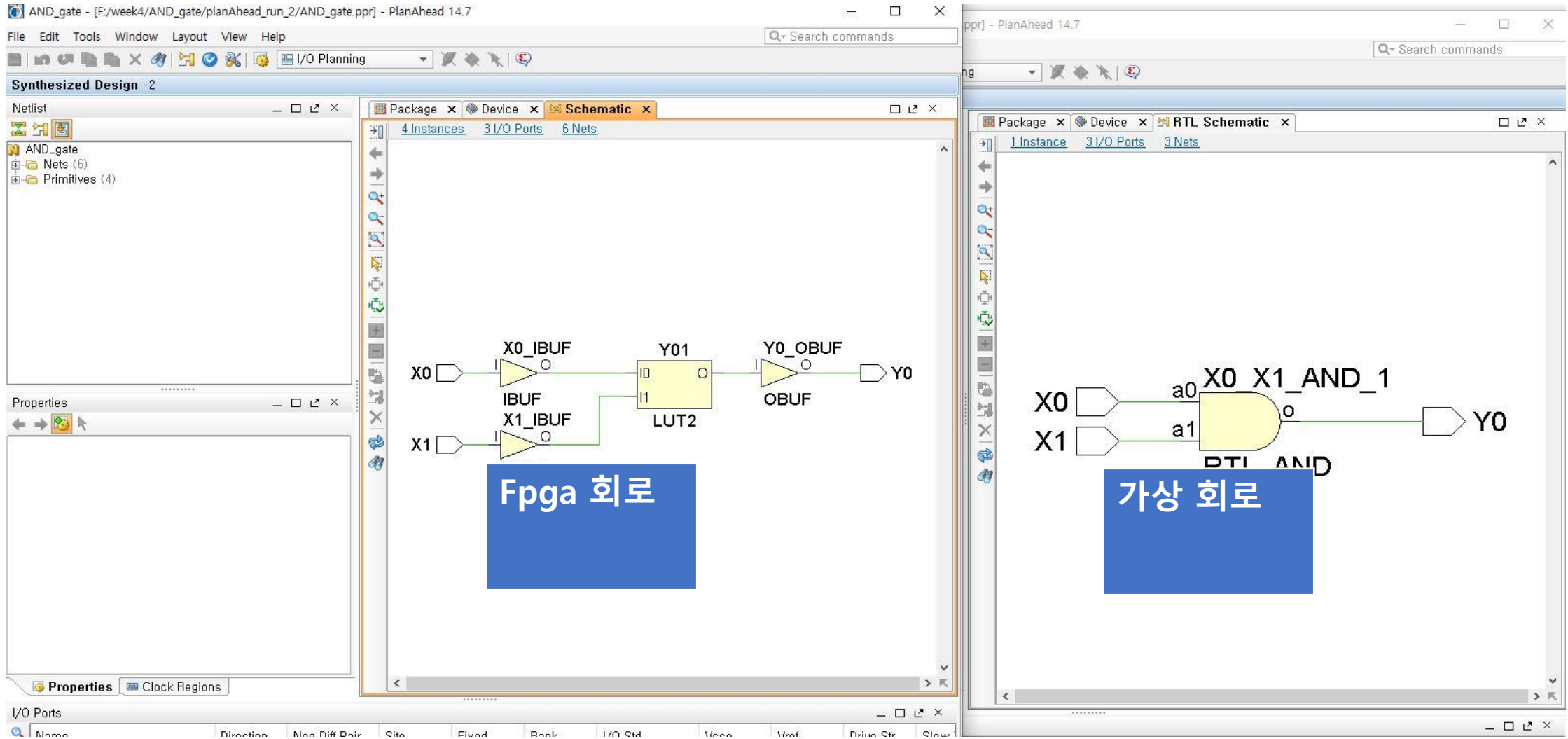
실습1 입력, 출력 (시뮬레이션 화면)



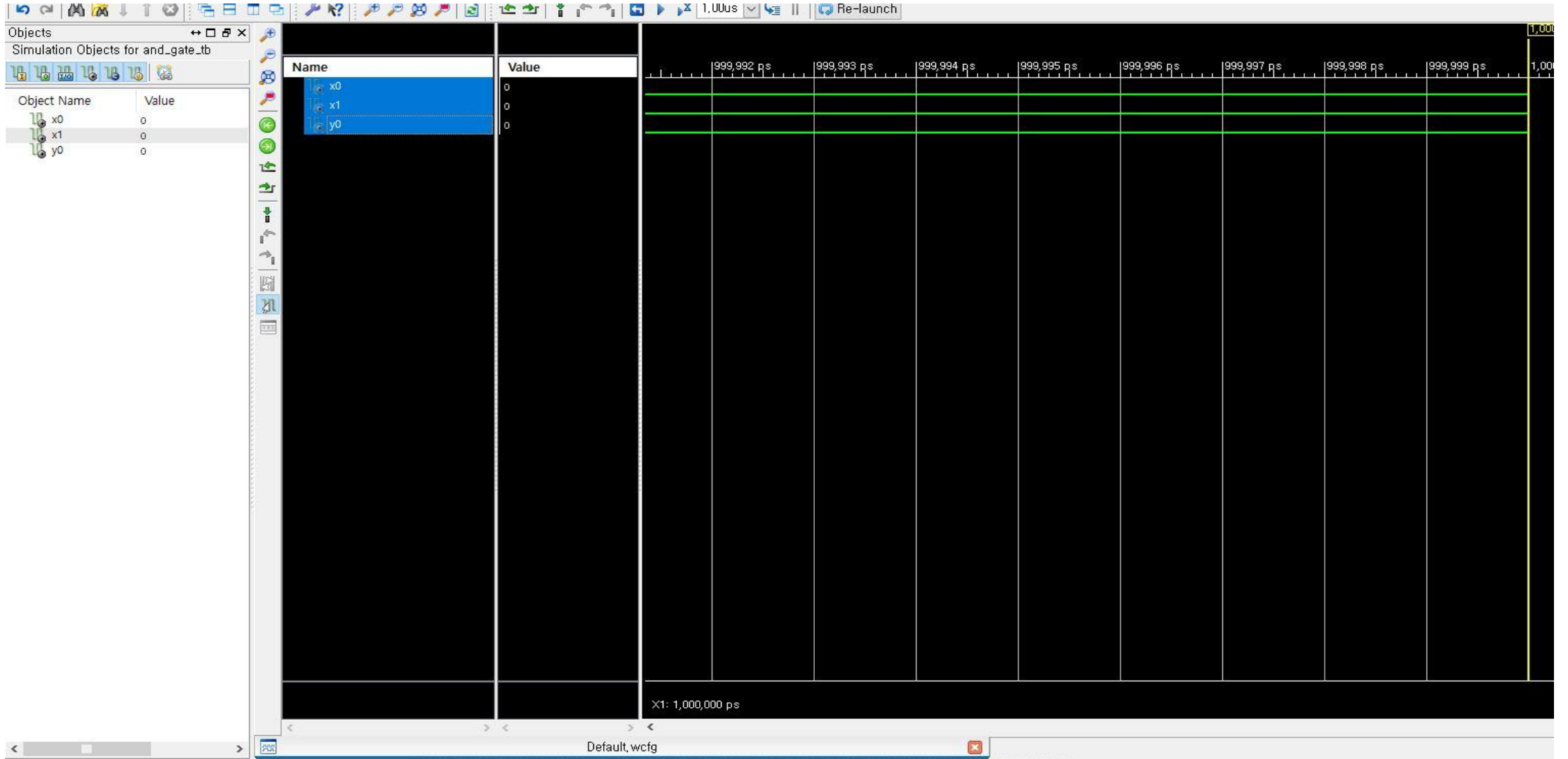
실습2 AND게이트 (동작테이블)

입력 출력 동작 Table		
X0 (스위치)	X1 (스위치)	Y0 (LED)
0	0	0
0	1	0
1	0	0
1	1	1

실습2 AND게이트 (가상회로, fpga 회로)



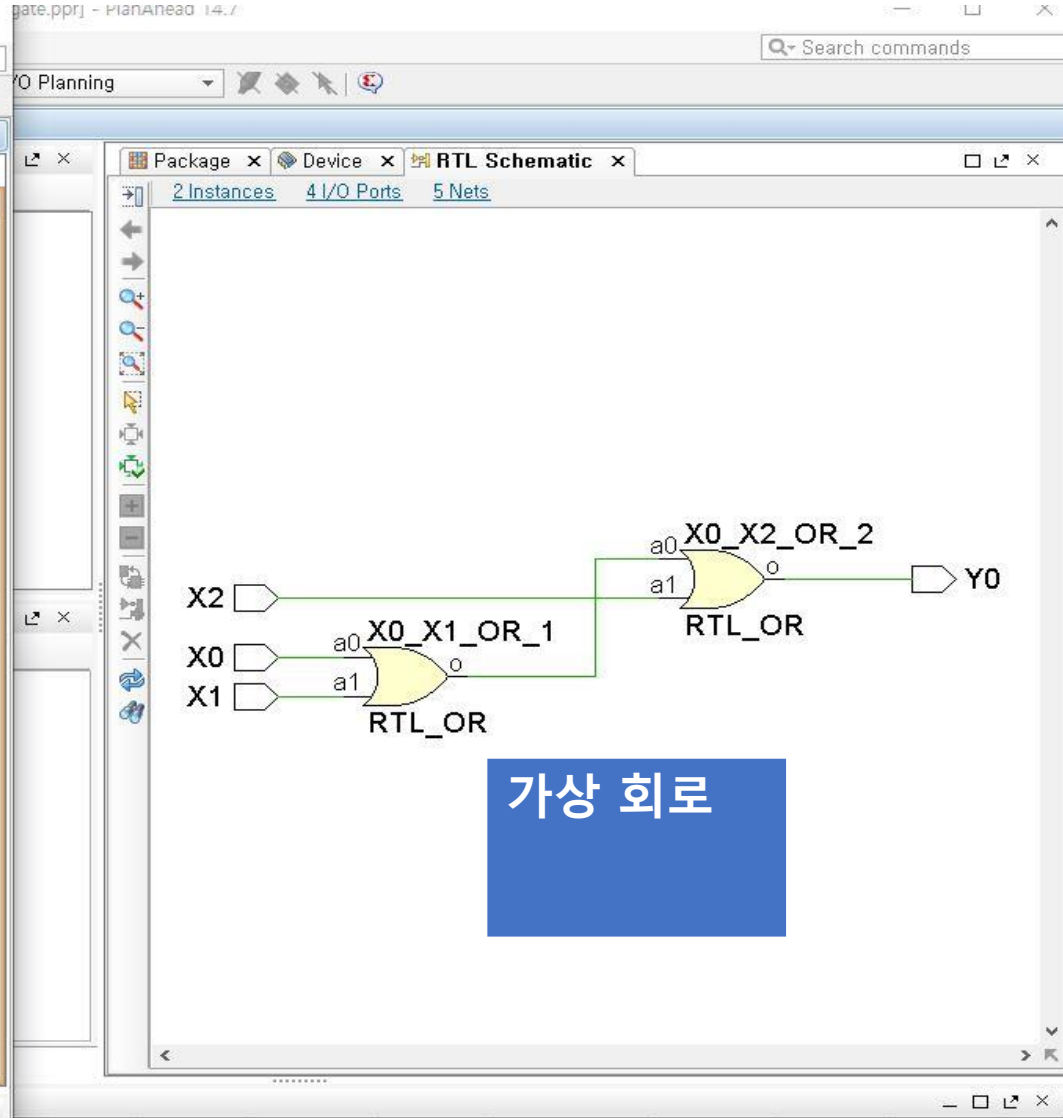
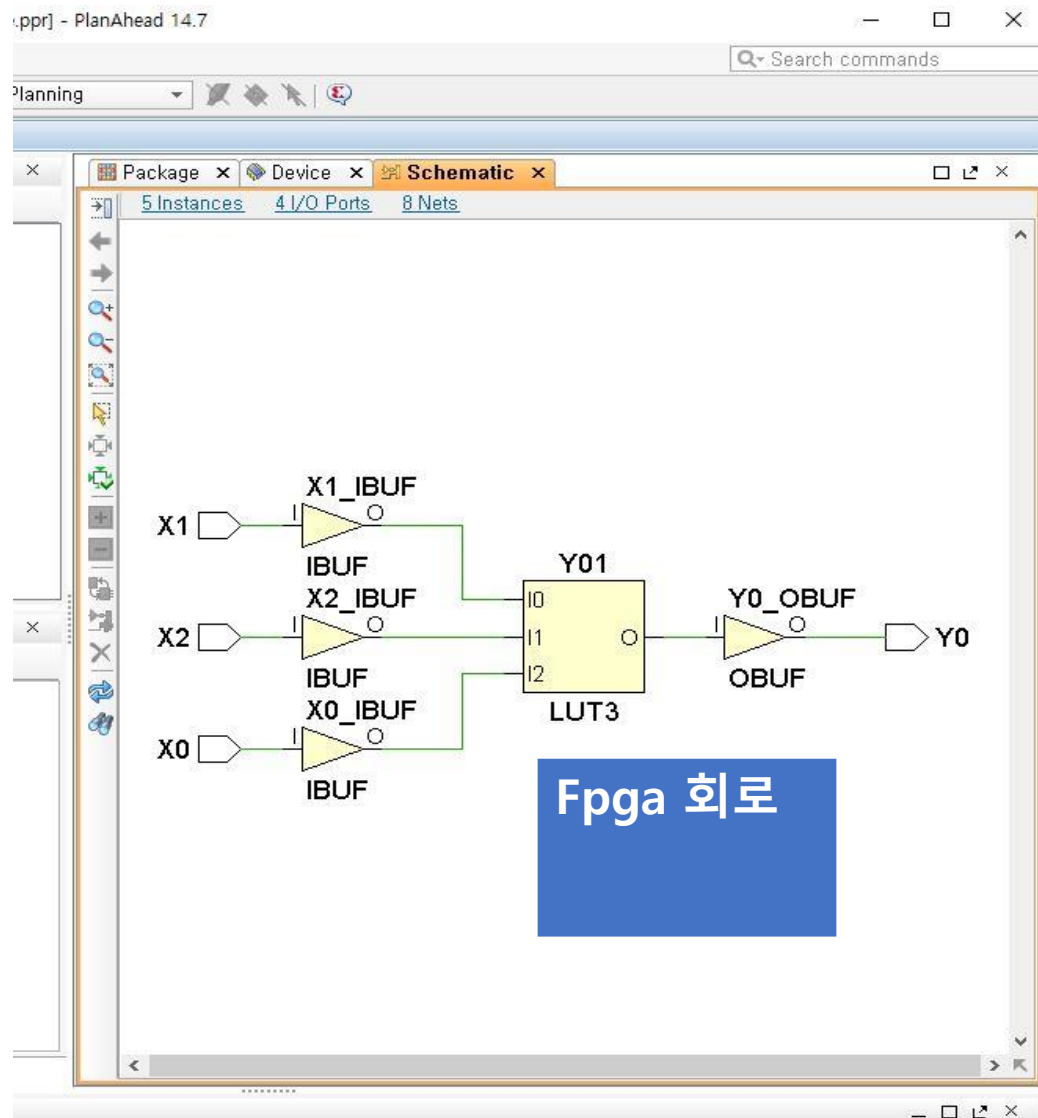
실습2 AND게이트 (시뮬레이션 화면)



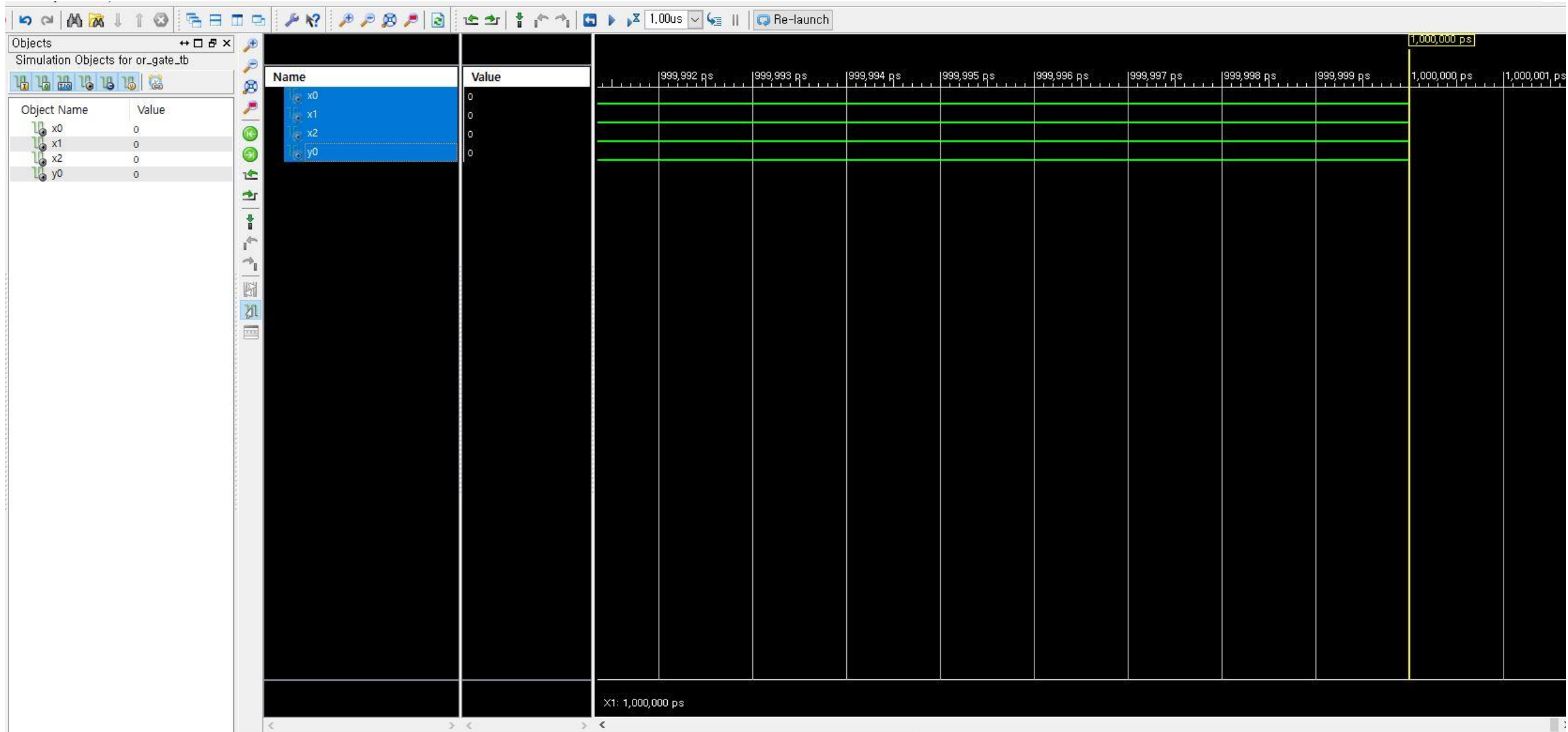
실습3 OR게이트 (동작테이블)

입력 출력 동작 Table			
X0 (스위치)	X1 (스위치)	X2 (스위치)	Y0 (LED)
0	0	0	0
1	0	0	1
0	1	0	1
0	0	1	1
1	1	0	1
0	1	1	1
1	0	1	1
1	1	1	1

실습3 OR게이트 (가상회로, fpga 회로)



실습3 OR게이트 (시뮬레이션 화면)

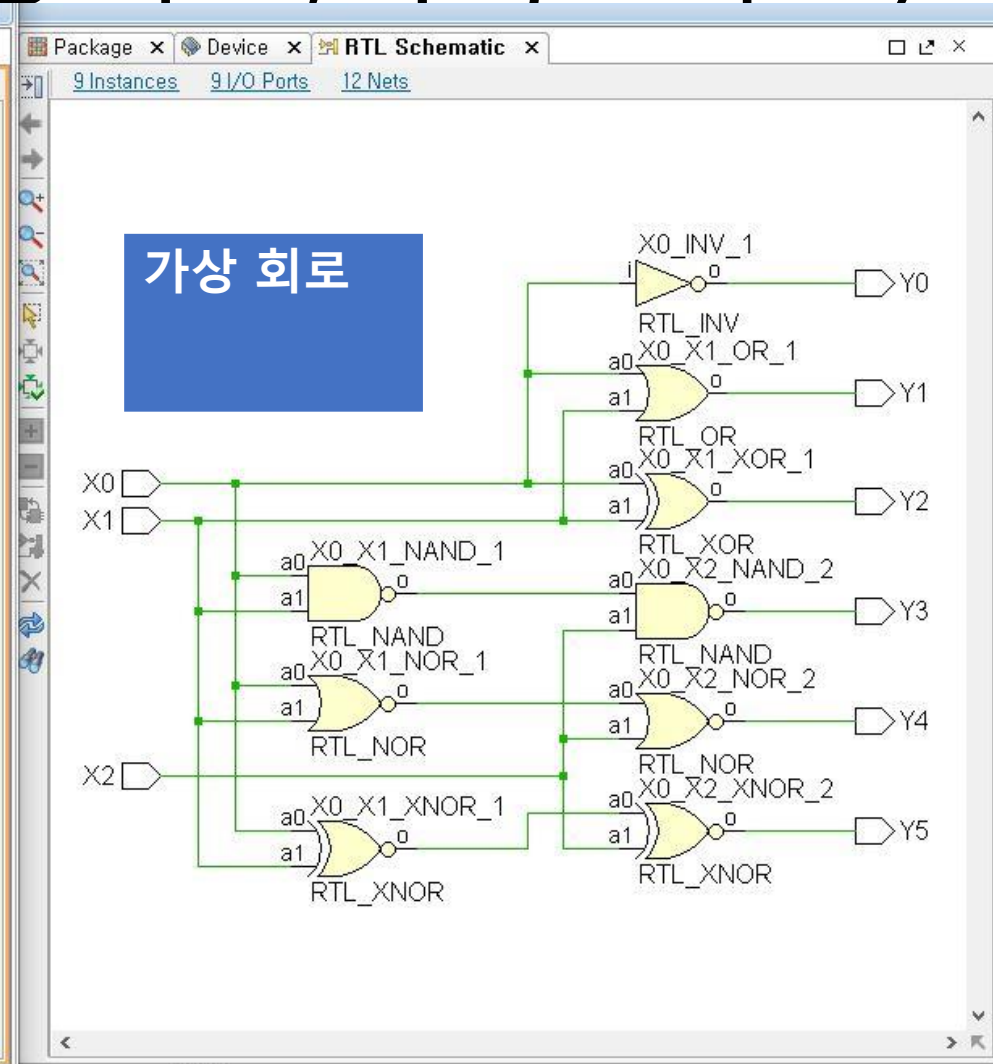
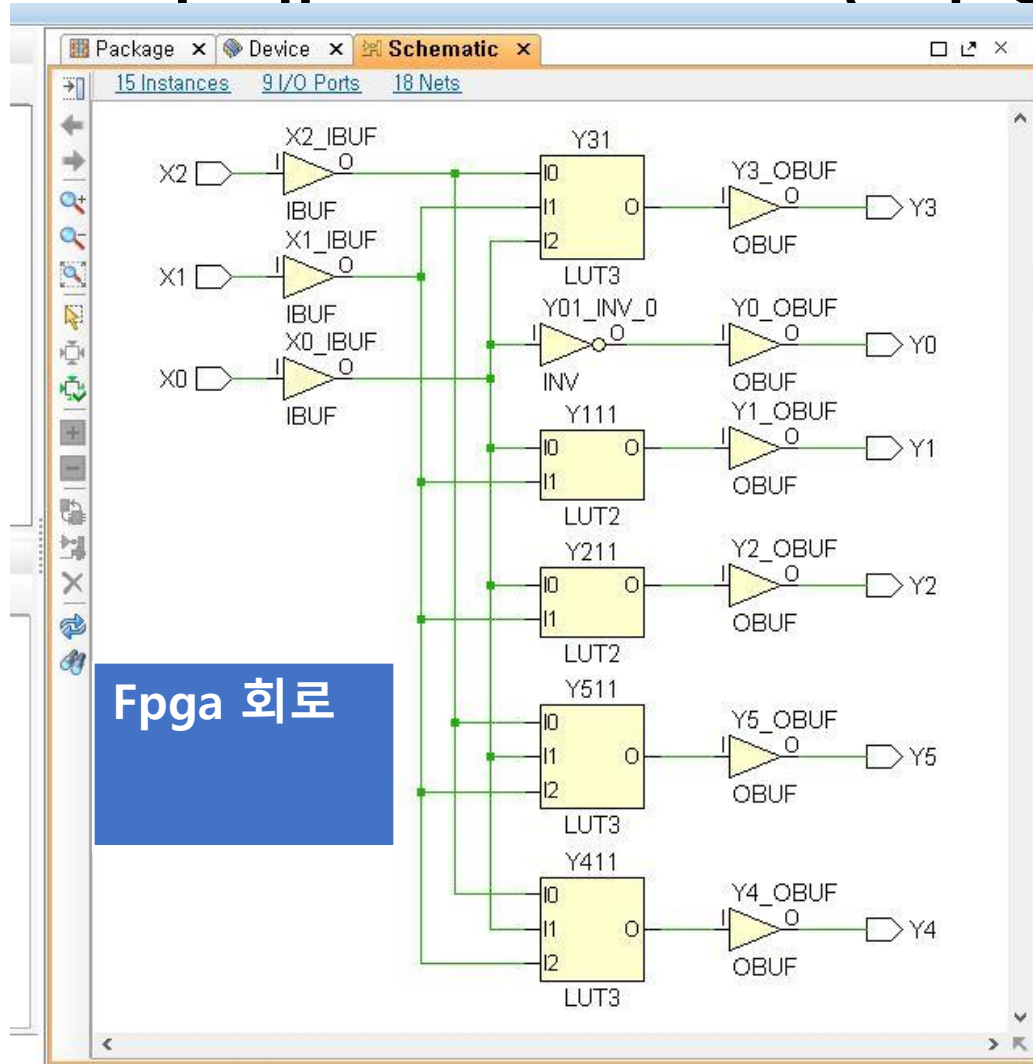


과제 All Gate 동작테이블

1입력 출력 동작 Table		2입력 출력 동작 Table			
X0 (스위치)	Y0 (LED)	X0 (스위치)	X1 (스위치)	Y1 (LED) OR 게이트	Y2 (LED) XOR 게이트
0	1	0	0	0	0
1	0	0	1	1	1
		1	0	1	1
		1	1	1	0

3입력 출력 동작 Table (X0, X1 우선입력)					
X0 (스위치)	X1 (스위치)	X2 (스위치)	Y3(LED) NAND게이트	Y4(LED) NOR게이트	Y5(LED) XNOR게이트
0	0	0	1	0	0
1	0	0	1	1	1
0	1	0	1	1	1
0	0	1	0	0	1
1	1	0	1	1	0
0	1	1	0	0	0
1	0	1	0	0	0
1	1	1	1	0	1

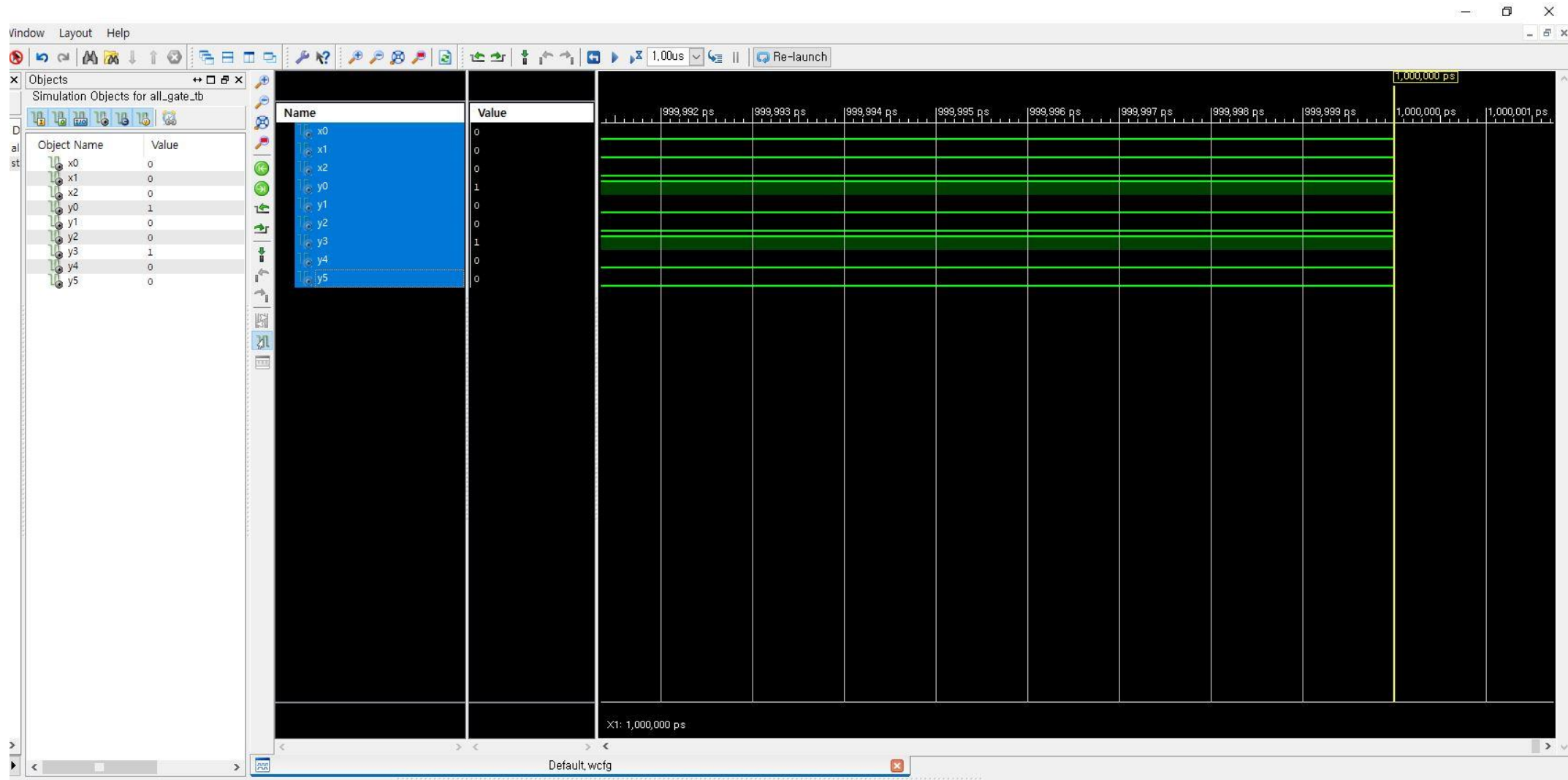
과제 All Gate (가상회로, fpga 회로)



Pair	Site	Fixed	Bank	I/O Std	Vcco	Vref	Drive Str...	Slew
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Site	Fixed	Bank	I/O Std	Vcco	Vref	Drive Str...	Slew
------	-------	------	---------	------	------	--------------	------

과제 All Gate (시뮬레이션 화면)



week5

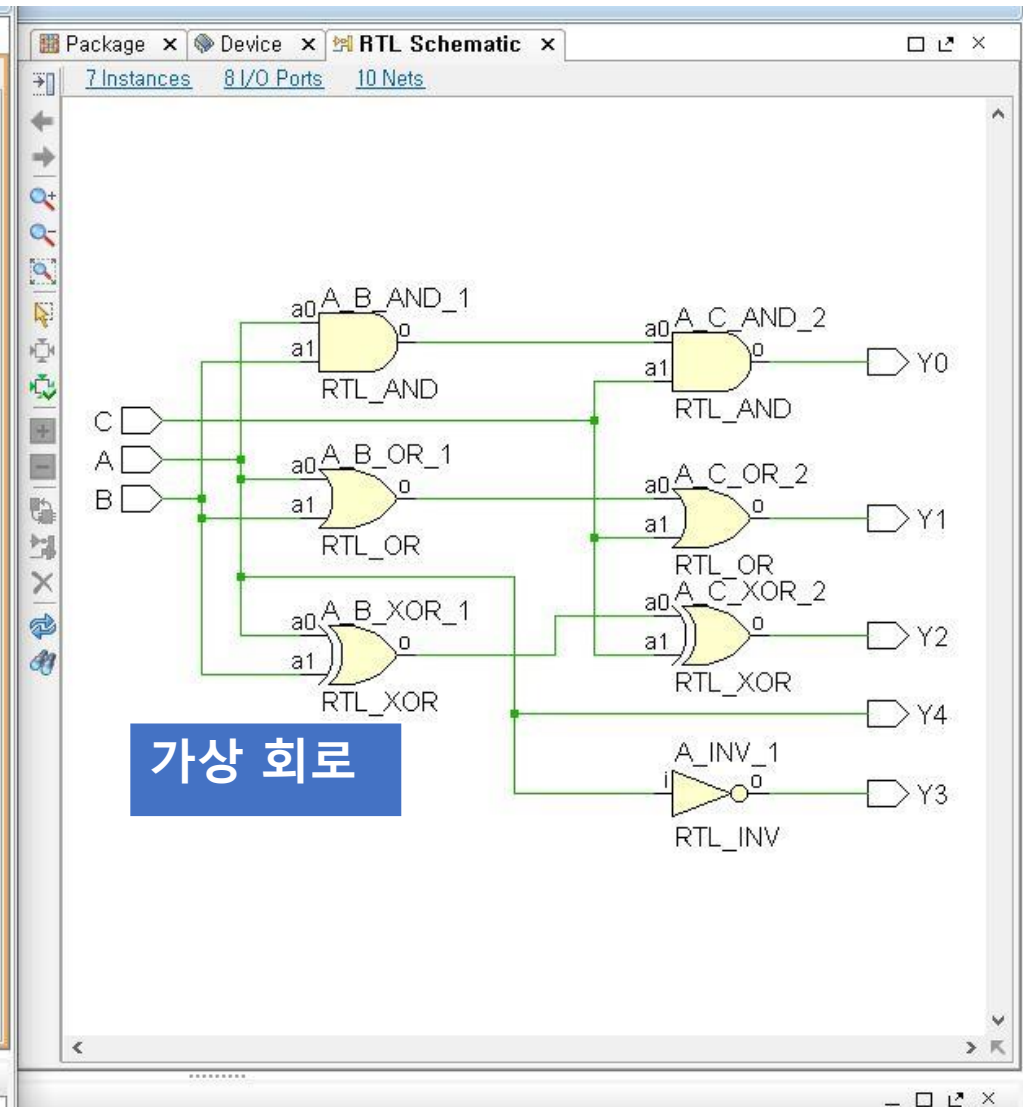
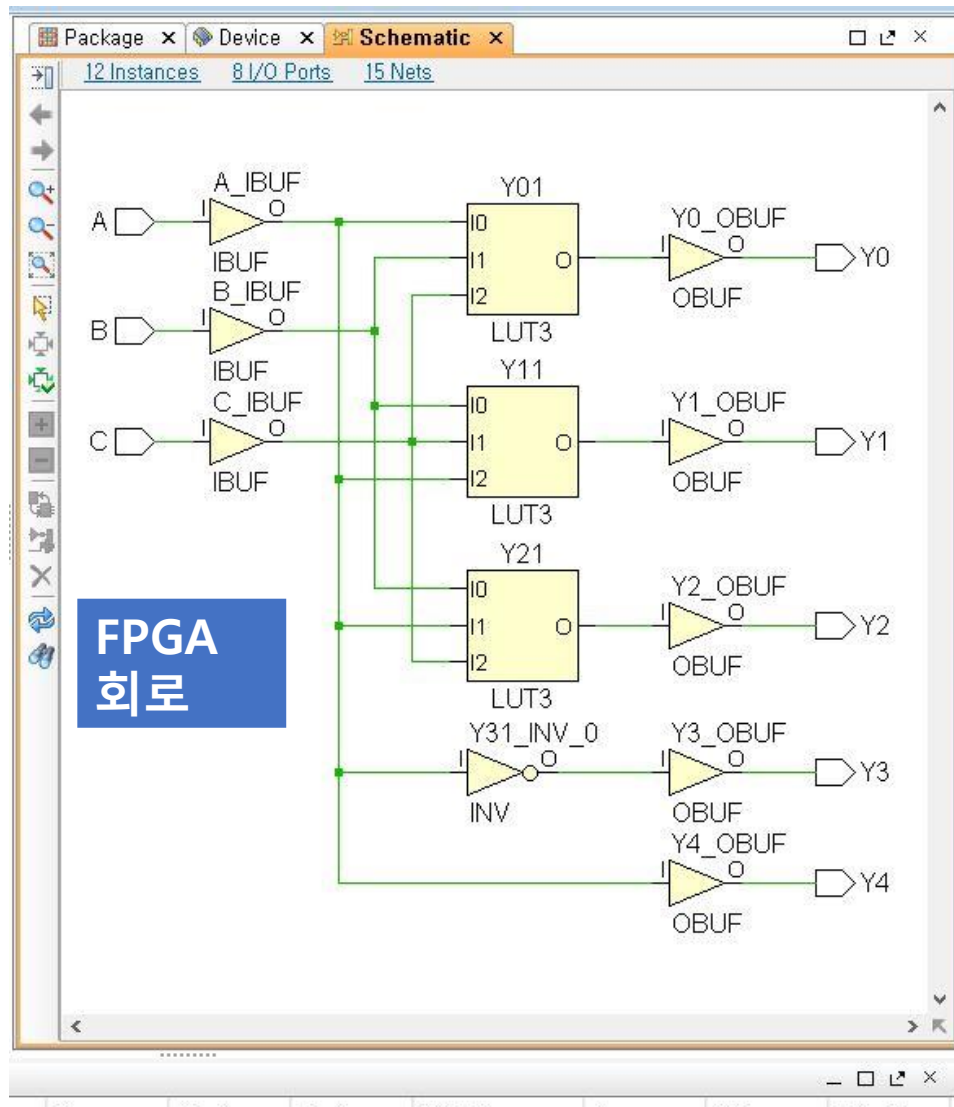
2학년_2반_2021041089_최태민

유튜브 영상 주소 : <https://www.youtube.com/watch?v=l2-7cLv8yA>

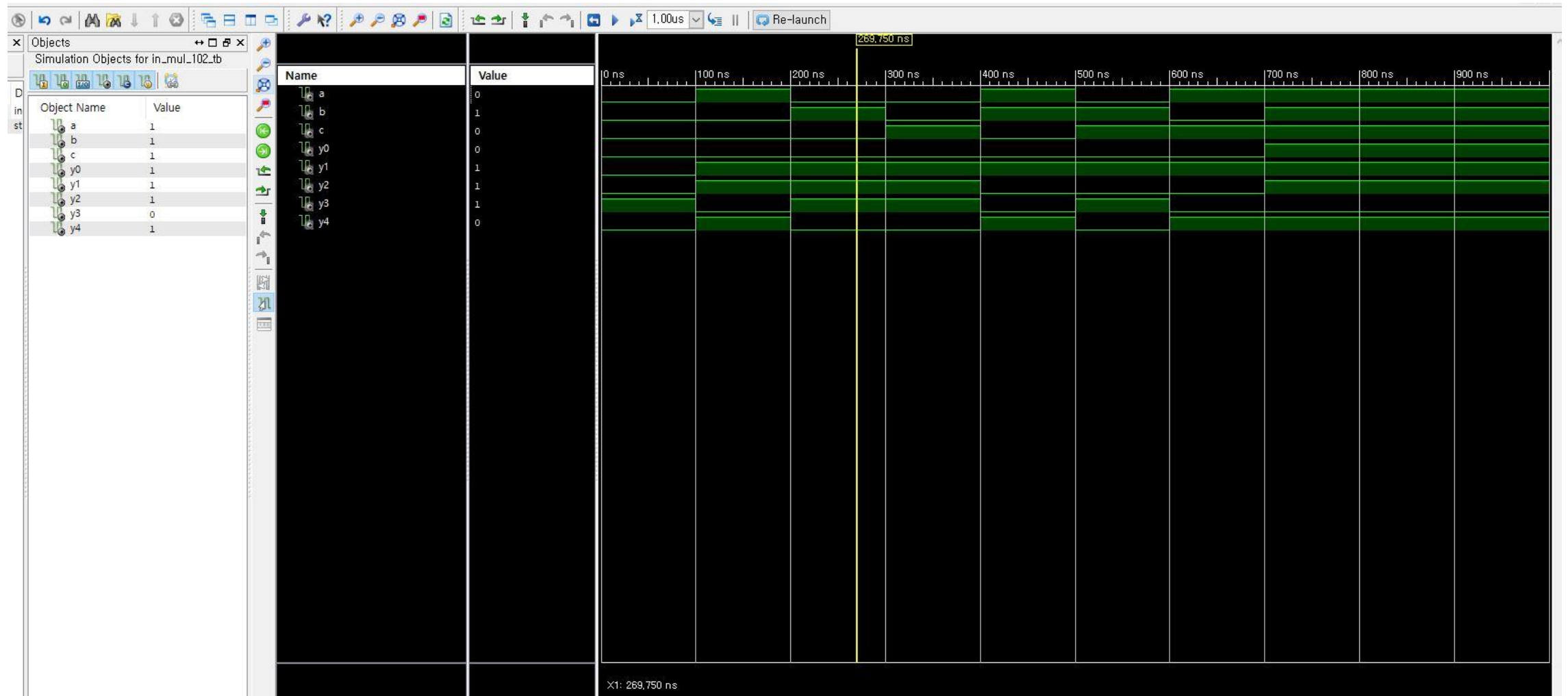
과제 : 동작테이블

A (입력)	B (입력)	C (입력)	Y0 (3입력AND) (A x B) C = Y0	Y1 (3입력OR) (A + B) + C = Y0	Y2 (3입력XOR) (A ⊕ B) ⊕ C = Y1	Y3 (1입력A) (invertor) $\overline{A} = Y4$	Y4 (1입력A) (Buffer) A = Y4
0	0	0	0	0	0	1	0
1	0	0	0	1	1	0	1
0	1	0	0	1	1		
0	0	1	0	1	1		
1	1	0	0	1	0		
0	1	1	0	1	0		
1	0	1	0	1	0		
1	1	1	1	1	1		

과제 : 가상회로, FPGA 회로



과제 : 시뮬레이션 화면



Week6

2학년 2반 2021041089 최태민

유튜브 영상 주소

실습 1 : <https://youtu.be/2qOY2NzABxk>

실습 2 : <https://youtu.be/-r-CvaiYMWE>

과제 1 : <https://youtu.be/sDVg5UrgNhM>

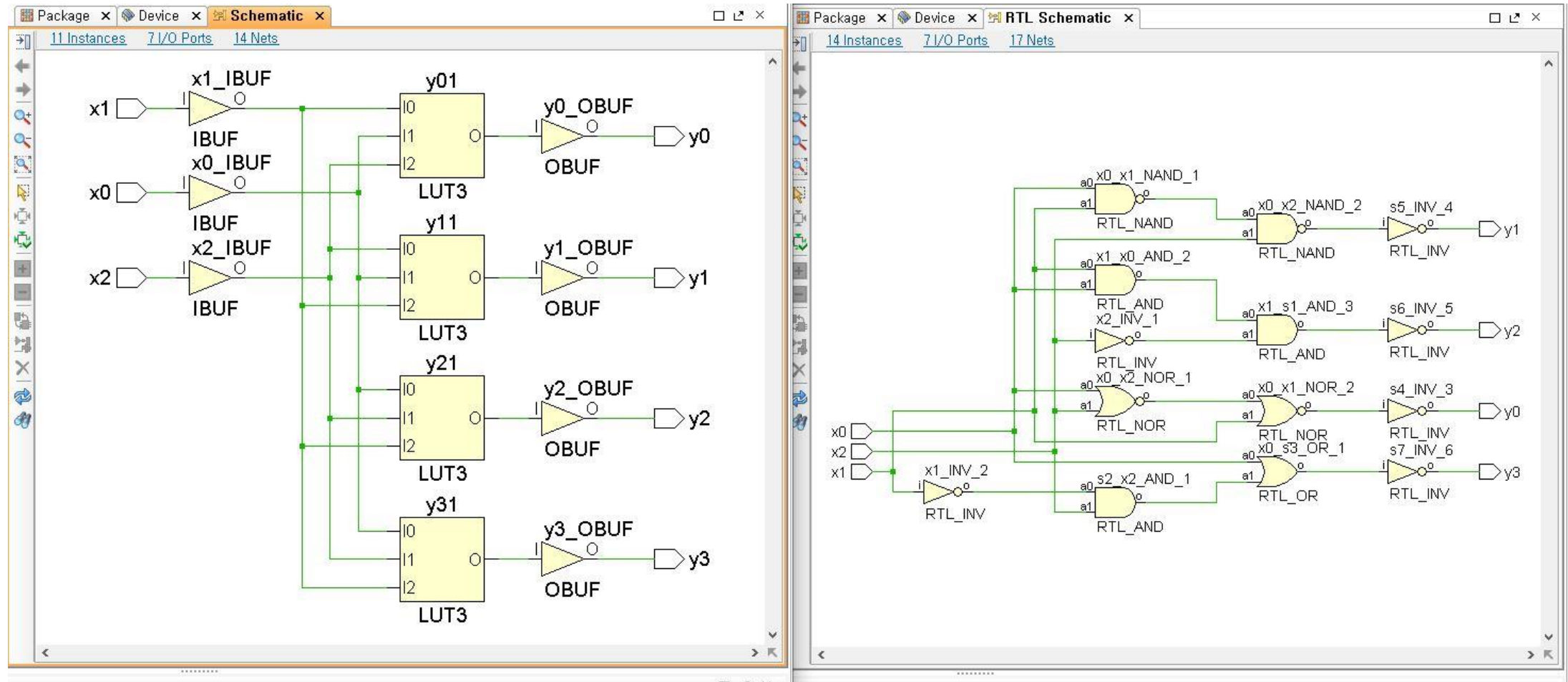
과제 2 : <https://youtu.be/YLviLQWIdIU>

실습 1 : 동작표

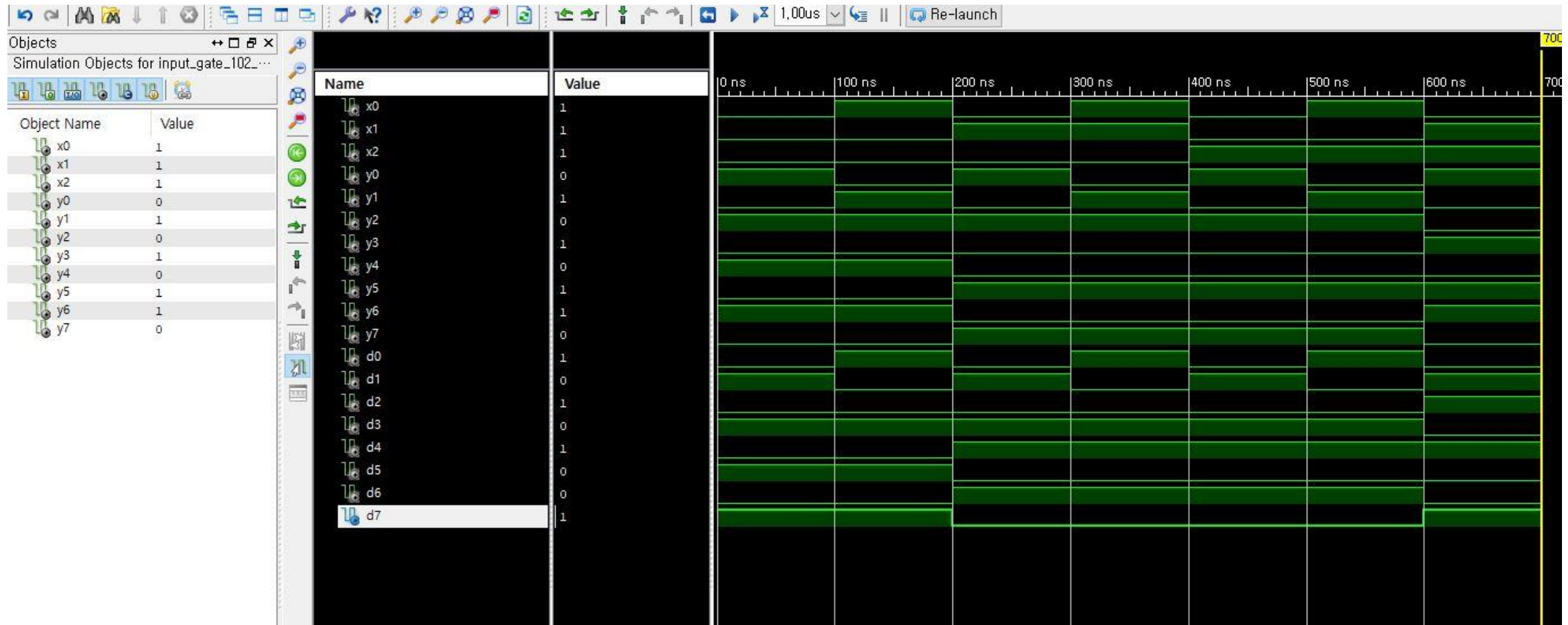
입력 출력 동작 Table								
X1 (스위치)	X2 (스위치)		Y2	Y3	Y4	Y5	Y6	Y7
0	0		1	0	1	0	1	0
0	1		1	0	0	1	0	1
1	0		1	0	0	1	0	1
1	1		0	1	0	1	1	0

입력 출력 동작 Table		
입력 스위치 X0	출력 LED Y0	출력 LED Y1
0	1	0
1	0	1

실습 1 : 가상회로, fpga 회로



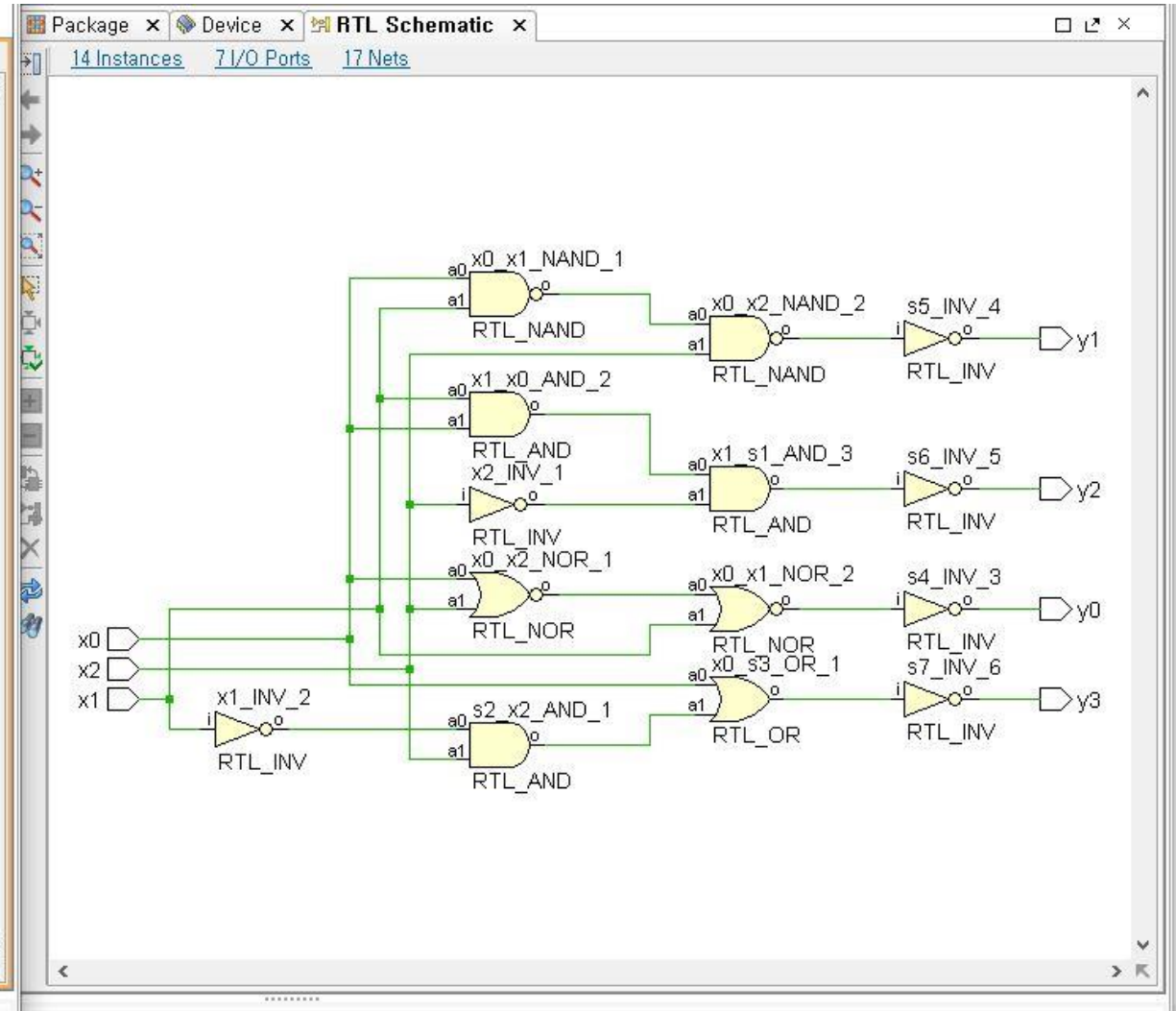
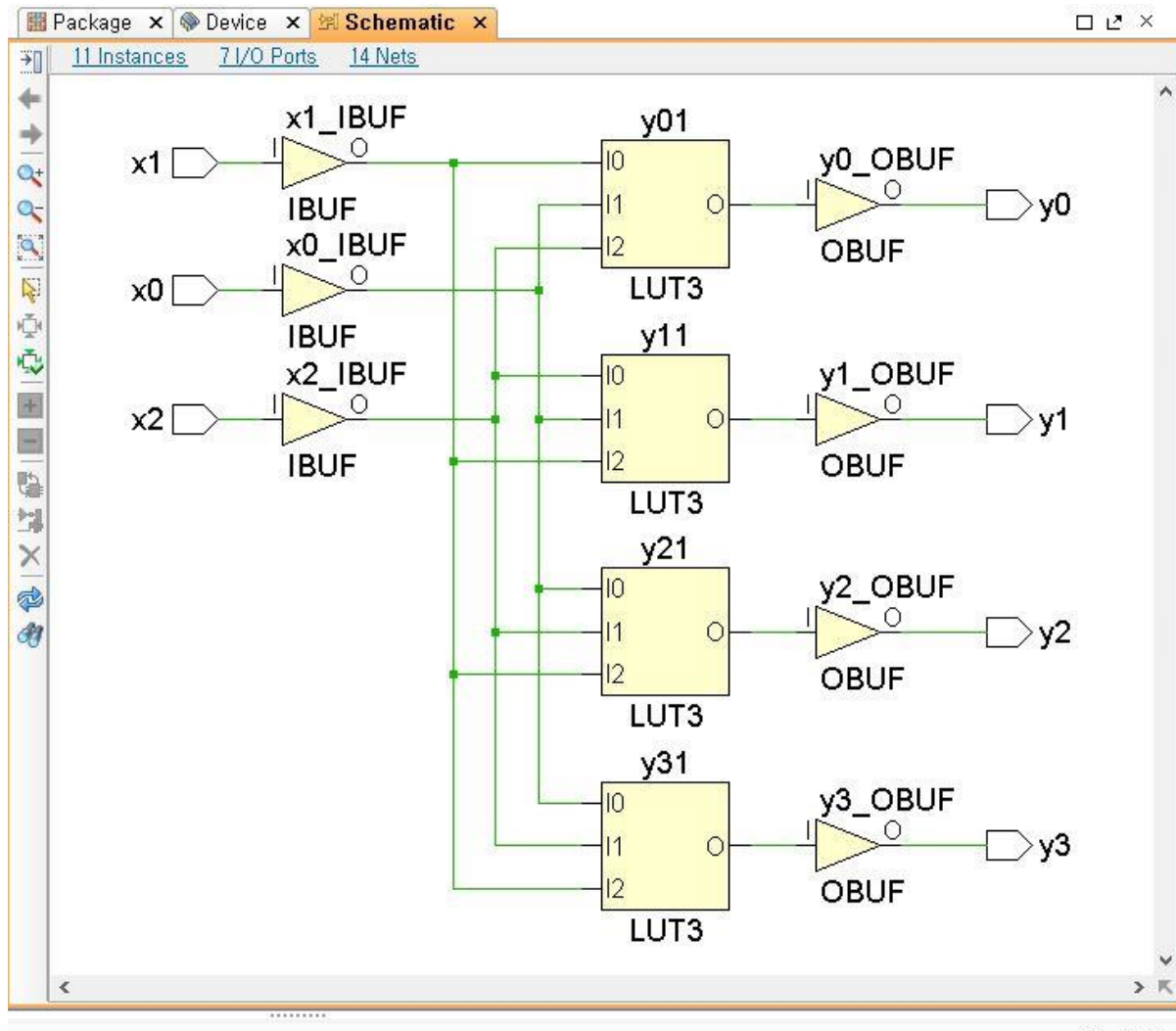
실습 1 : 시뮬레이션 화면



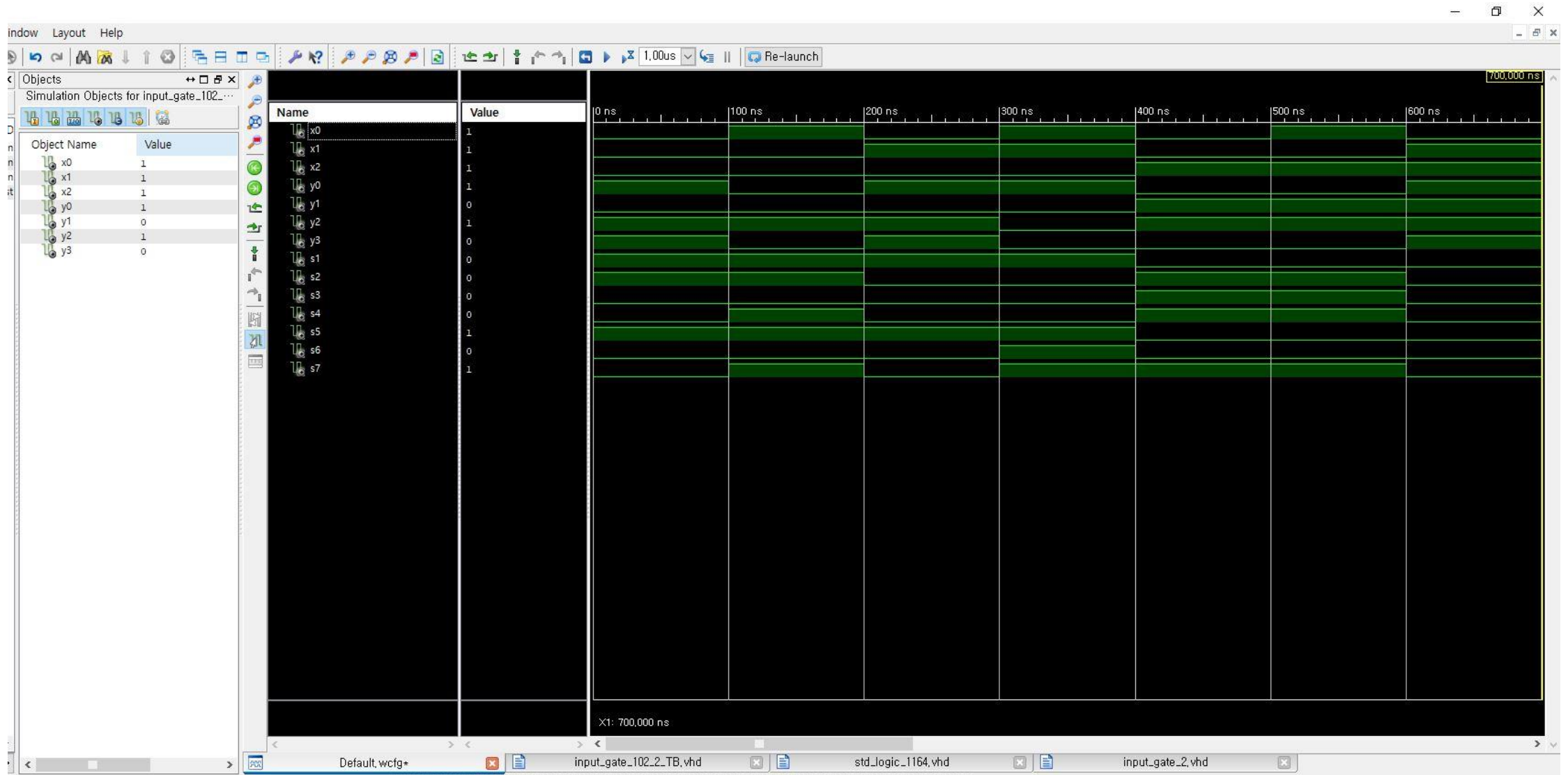
실습 2 : 동작표

input			output			
x2	x1	x0	y0	y1	y2	y3
0	0	0	1	0	1	1
0	0	1	0	0	1	0
0	1	0	1	0	1	1
0	1	1	1	0	0	0
1	0	0	0	1	1	0
1	0	1	0	1	1	0
1	1	0	1	1	1	1
1	1	1	1	0	1	0

실습 2 : 가상회로, fpga 회로



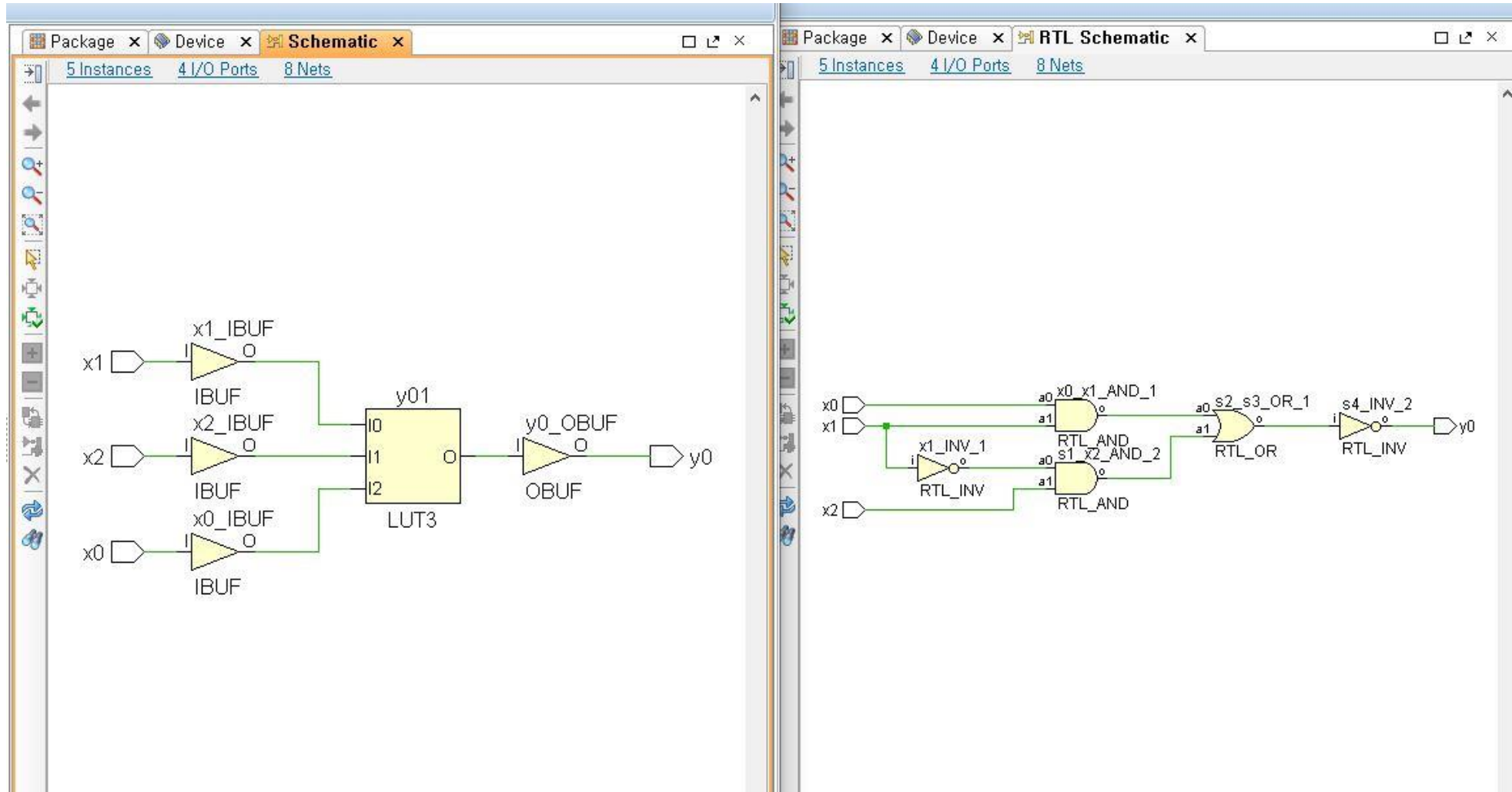
실습 2 : 시뮬레이션 화면



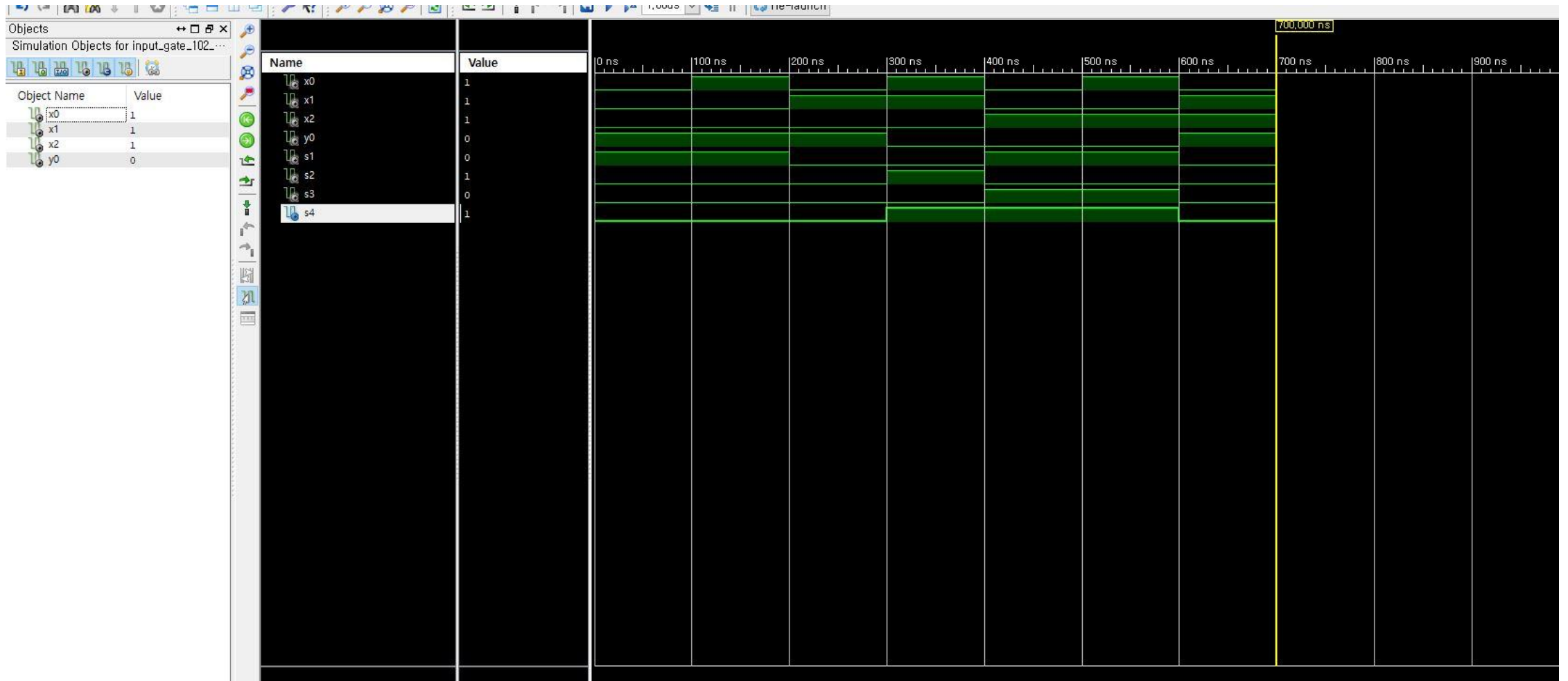
과제 1 : 동작표

input			output
x2	x1	x0	y0
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

과제 1 : 가상회로, fpga 회로



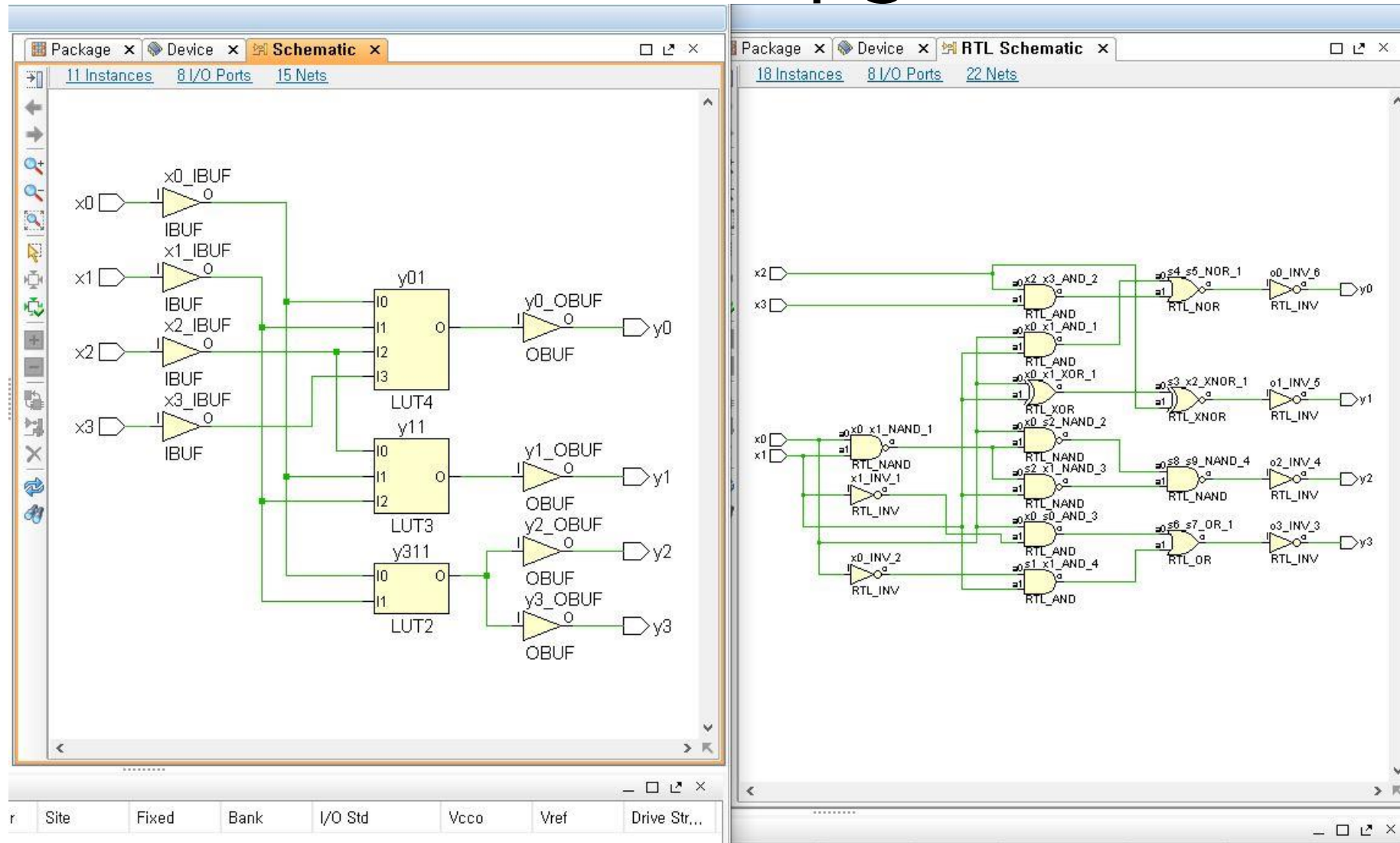
과제 1 : 시뮬레이션 화면



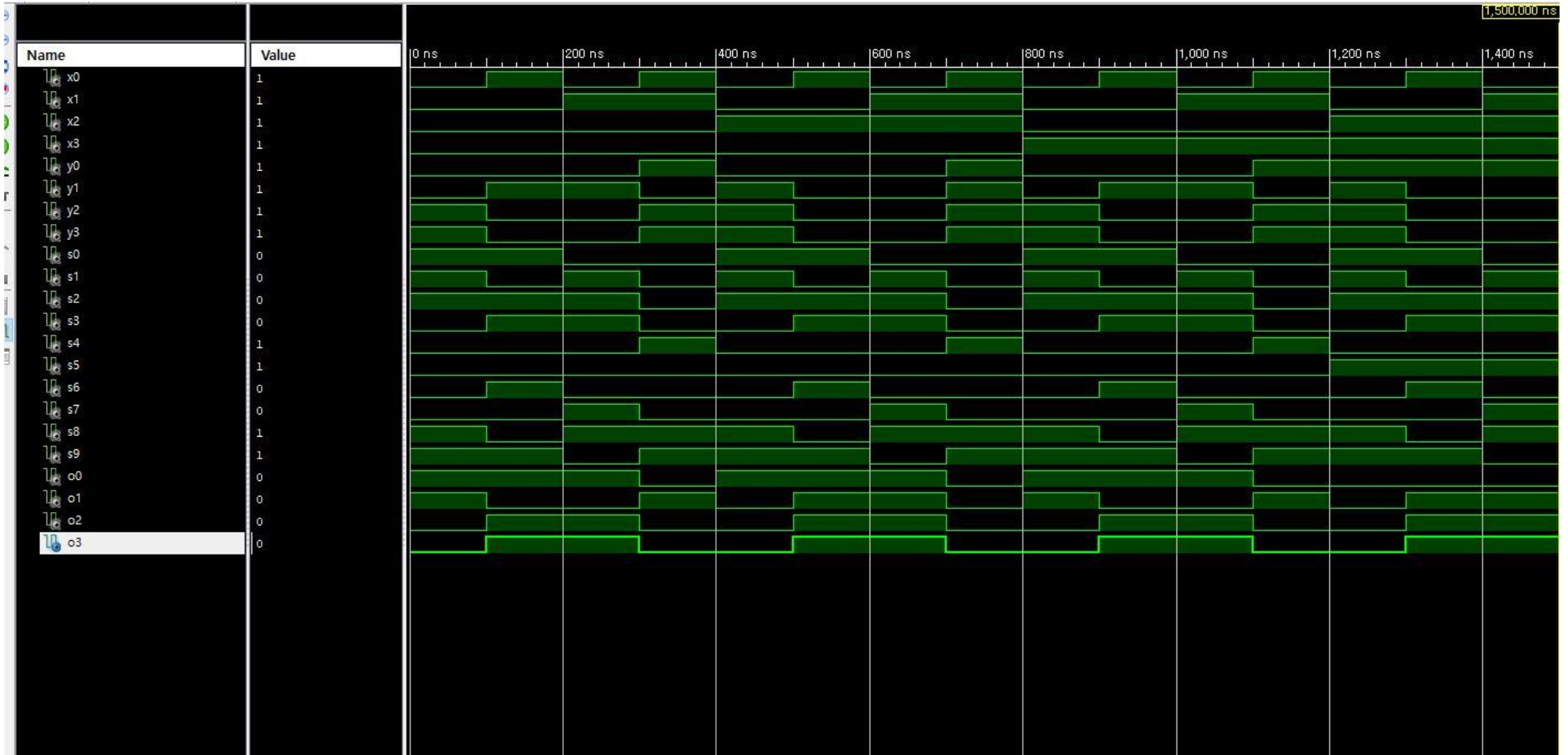
과제 2 : 동작표

input				output			
x0	x1	x2	x3	y0	y1	y2	y3
0	0	0	0	0	0	1	1
1	0	0	0	0	1	0	0
0	1	0	0	0	1	0	0
1	1	0	0	1	0	1	1
0	0	1	0	0	1	1	1
1	0	1	0	0	0	0	0
1	1	1	0	0	0	0	0
0	0	0	1	1	1	1	1
1	0	0	1	0	0	1	1
0	1	0	1	0	1	0	0
1	1	0	1	0	1	0	0
0	0	1	1	1	0	1	1
1	0	1	1	1	1	1	1
1	1	1	1	1	0	0	0

과제 2 : 가상회로, fpga 회로



과제 2 : 시뮬레이션 화면



임베디드 FPGA week7 실습,과제

2019041066 송기준 2021041089 최태민

실습1 1입력 IN_X7_OUT_Y7 영상주소 : https://youtu.be/A_xMtIQAUy8

실습2 2입력 AND_gate 영상주소 : <https://youtu.be/JCM5srvIBGA>

실습3 3입력 OR_gate 영상주소 : <https://youtu.be/iflAiSAXOr0>

과제 1, 2, 3 입력 게이트 영상주소 : <https://youtu.be/fq0-KQaXvw0>

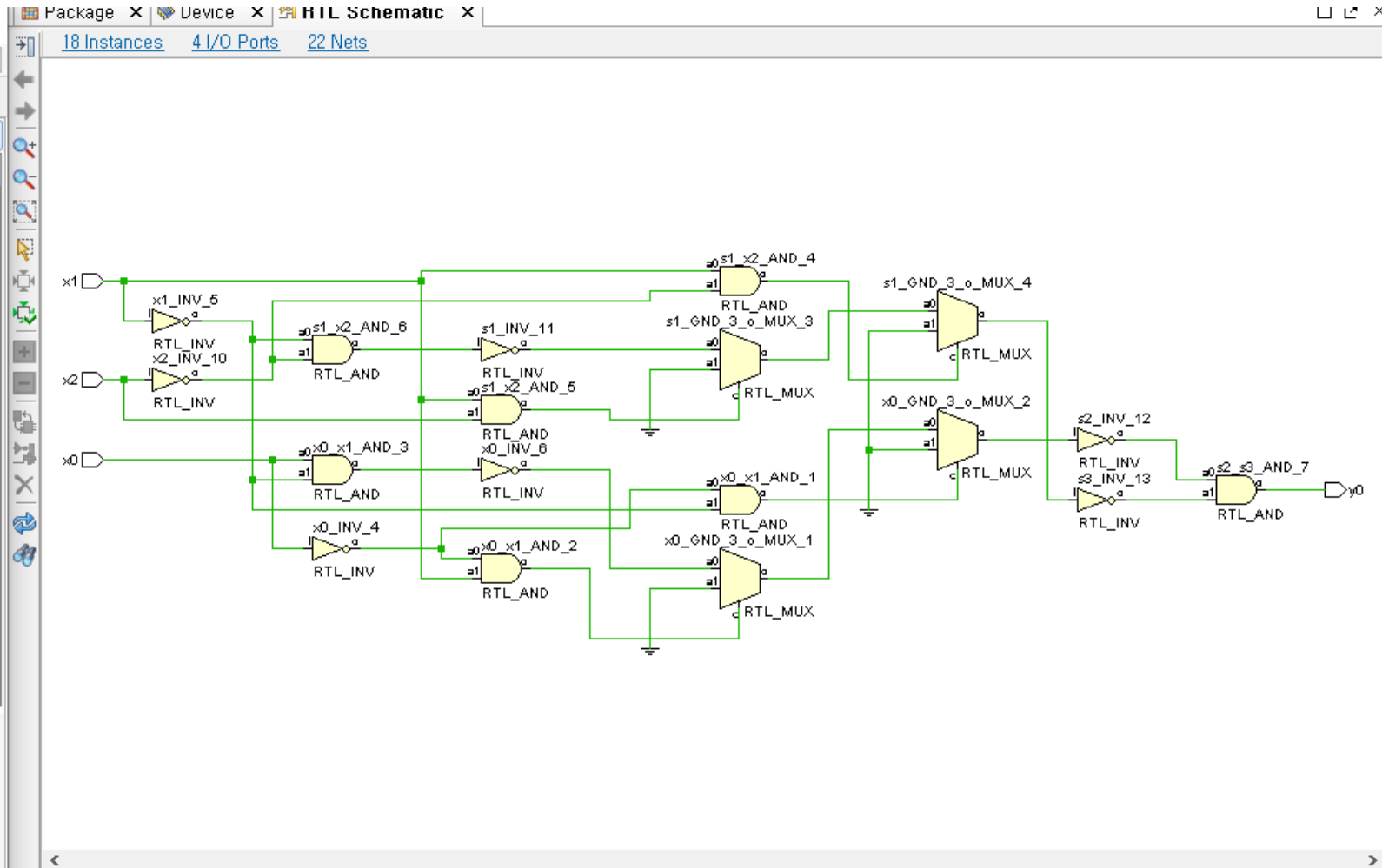
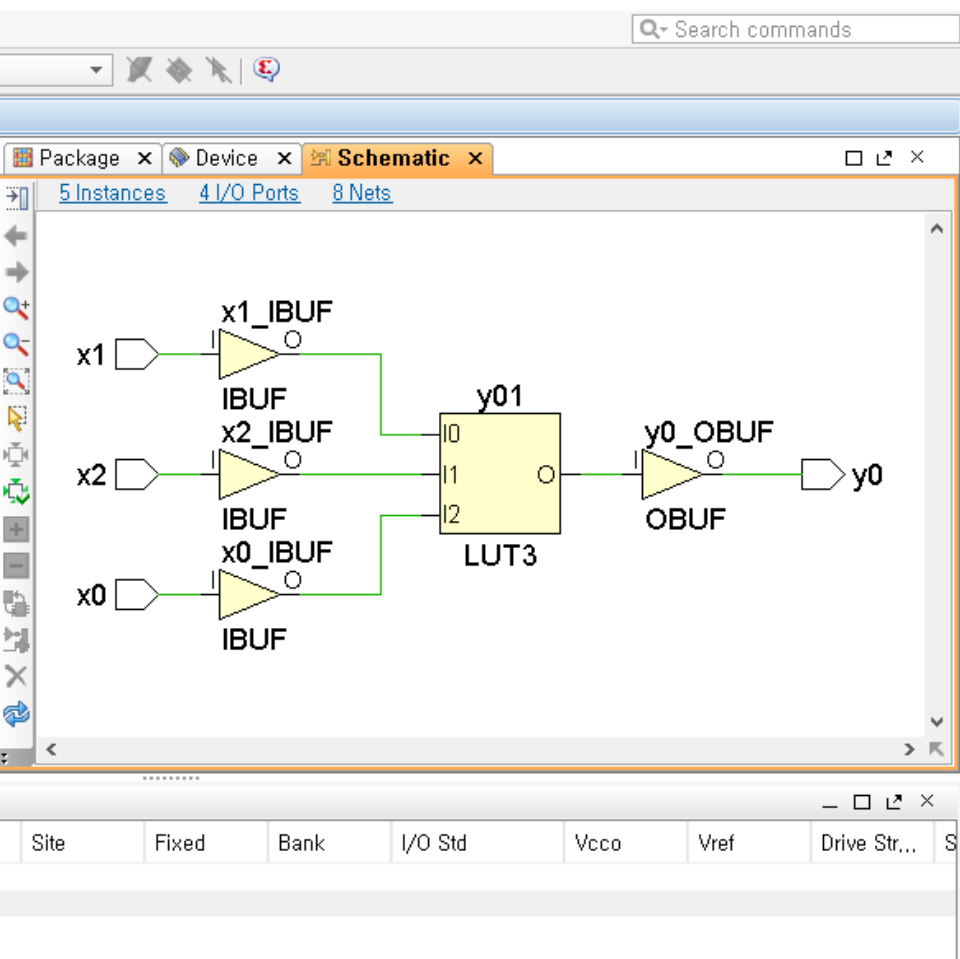
과제 1) std_logic 동작테이블

입력			출력
X2	X1	X0	Y0
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

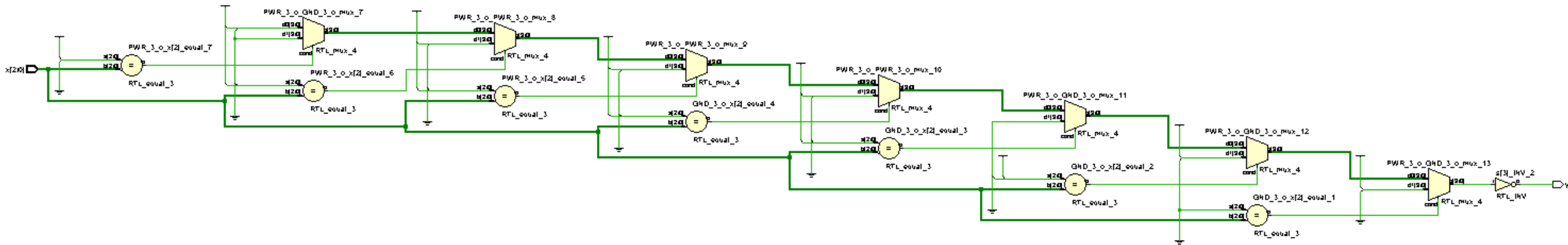
과제 1) FPGA 회로

과제 1) 가상회로

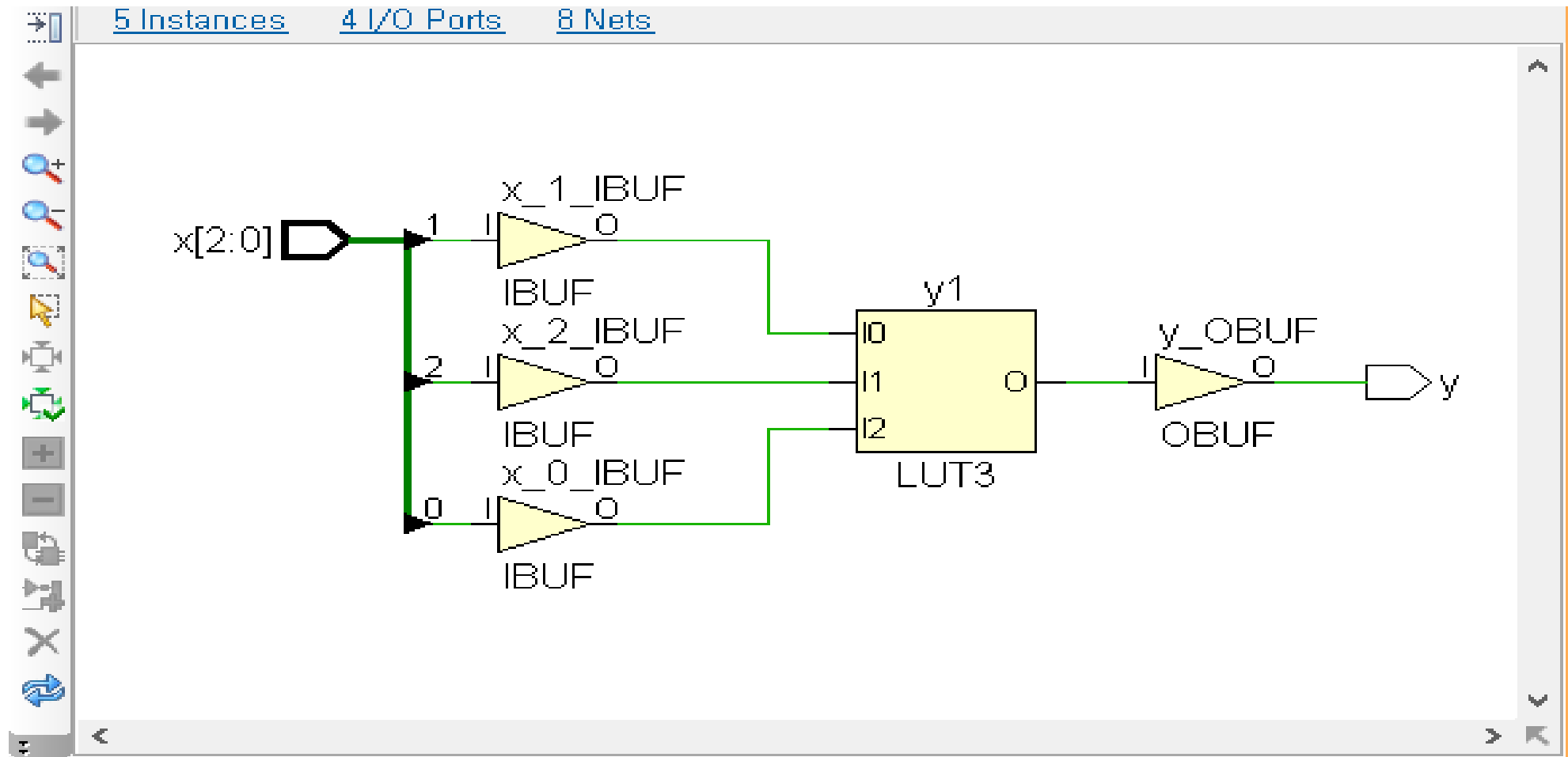
ut_gate_102_ex1.ppr] - PlanAhead 14.7



과제1) VECTOR_내부회로



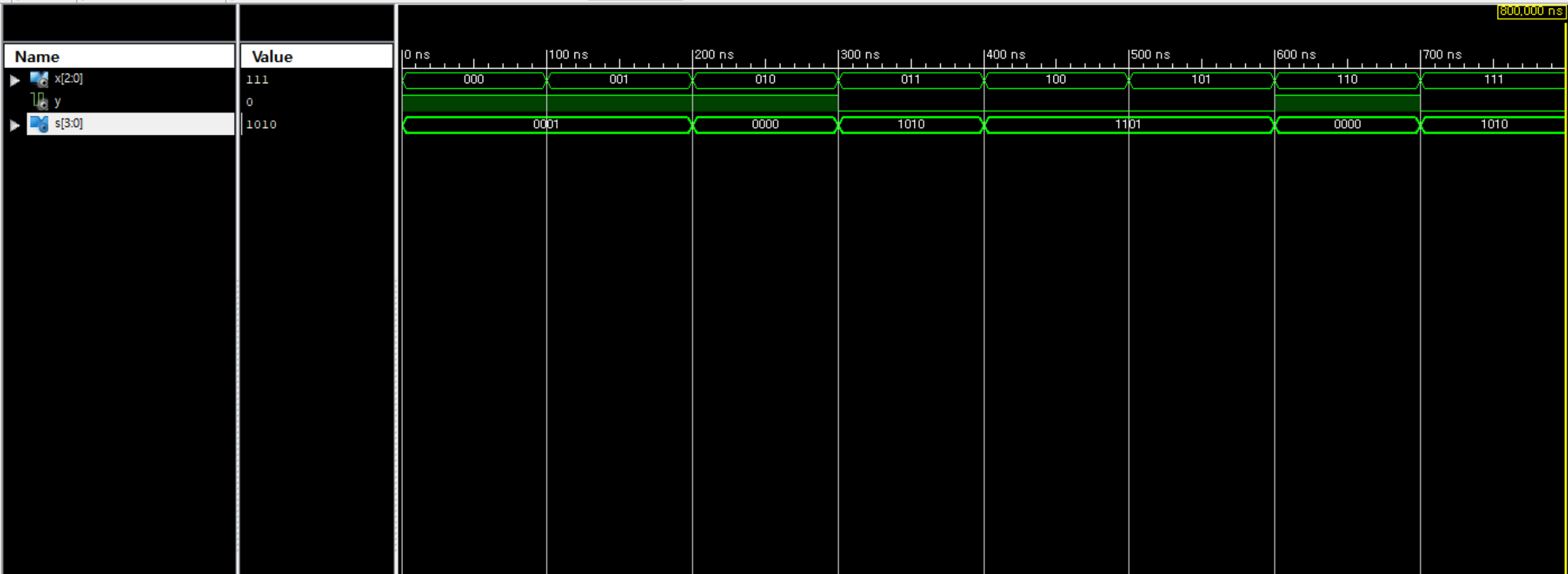
과제1) VECTOR_FPGA회로



과제1) LOGIC_시뮬레이션



과제1) VECTOR_시뮬레이션

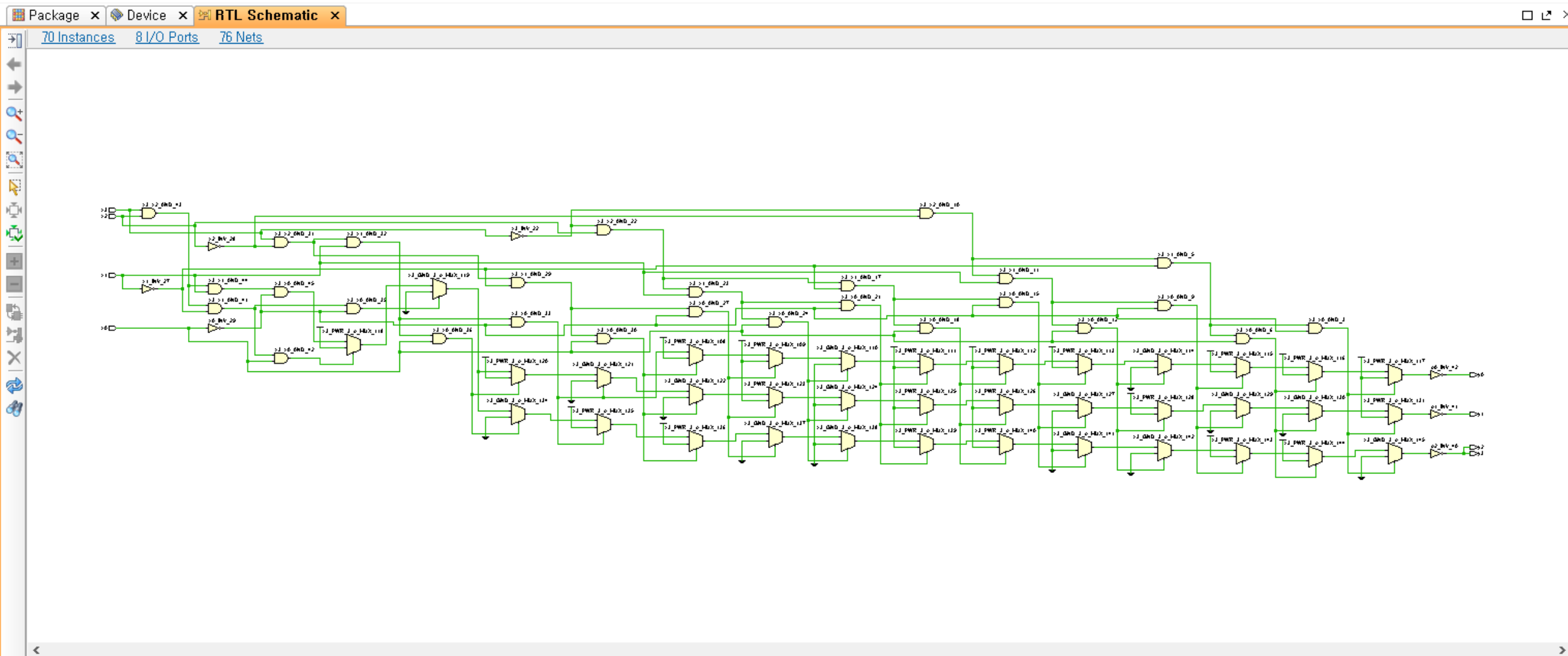


과제2) std_logic 동작테이블

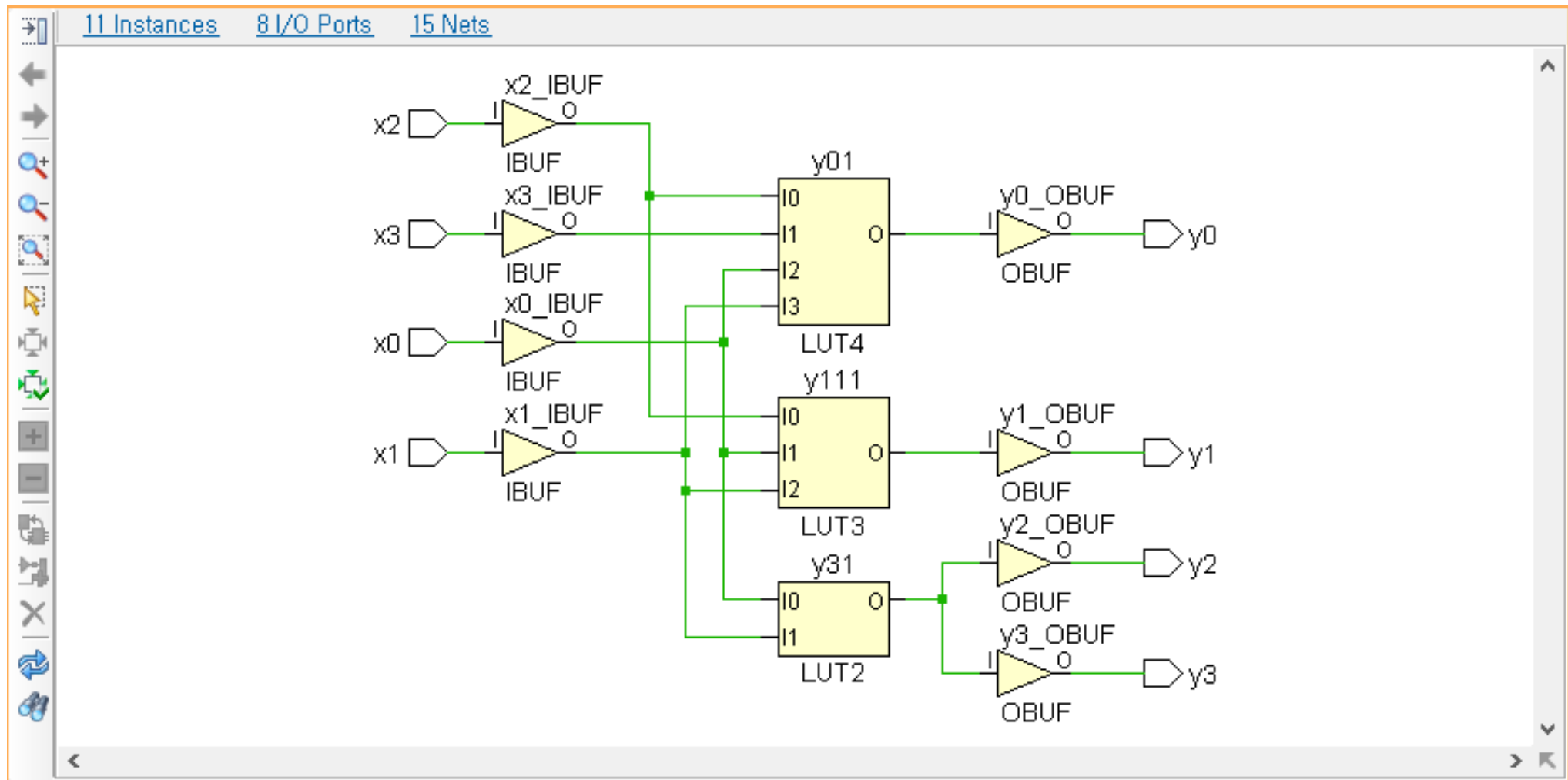
입력				출력			
X3	X2	X1	X0	Y3	Y2	Y1	Y0
0	0	0	0	1	1	0	0
0	0	0	1	0	0	1	0
0	0	1	0	0	0	1	0
0	0	1	1	1	1	0	1
0	1	0	0	1	1	1	0
0	1	0	1	0	0	0	0
0	1	1	0	0	0	0	0
0	1	1	1	1	1	1	1

입력				출력			
X3	X2	X1	X0	Y3	Y2	Y1	Y0
1	0	0	0	1	1	0	0
1	0	0	1	0	0	1	0
1	0	1	0	0	0	1	0
1	0	1	1	1	1	0	1
1	1	0	0	1	1	1	1
1	1	0	1	0	0	0	1
1	1	1	0	0	0	0	1
1	1	1	1	1	1	1	1

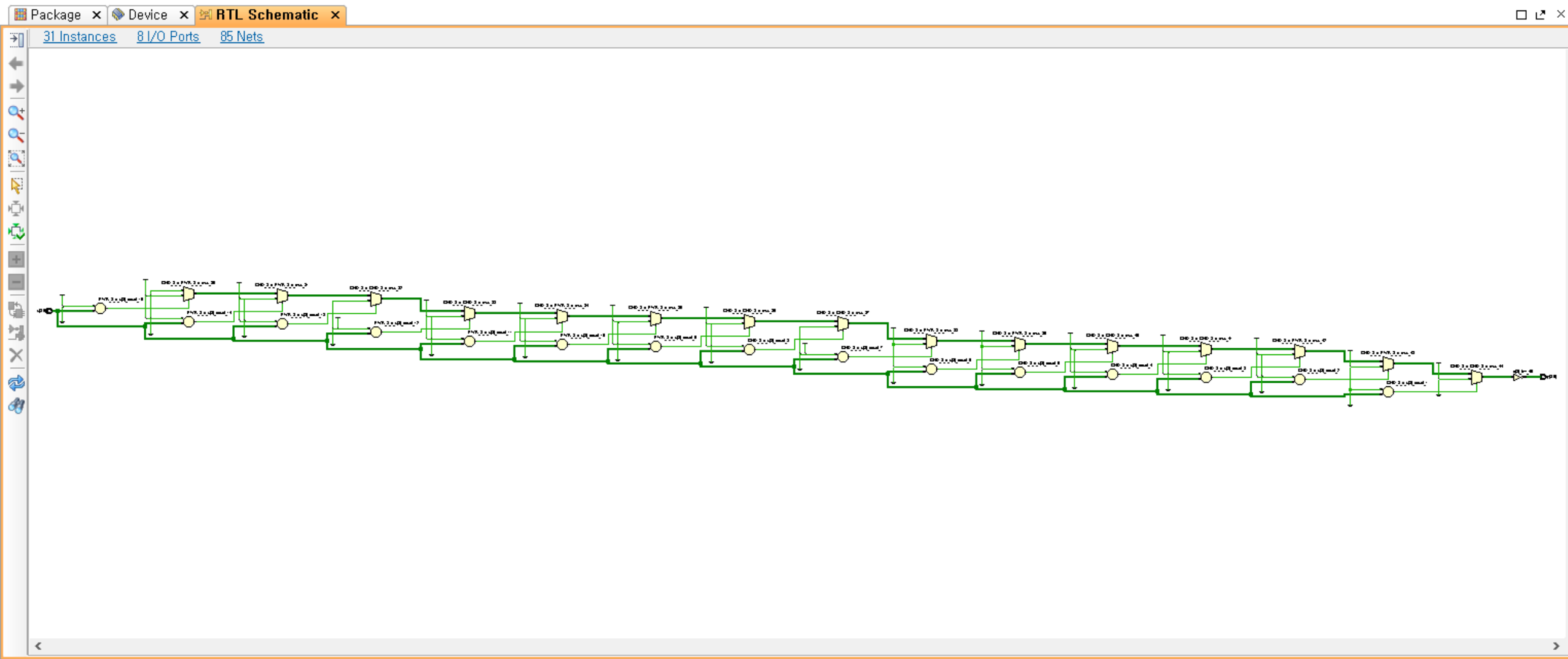
과제2) std_logic 내부회로



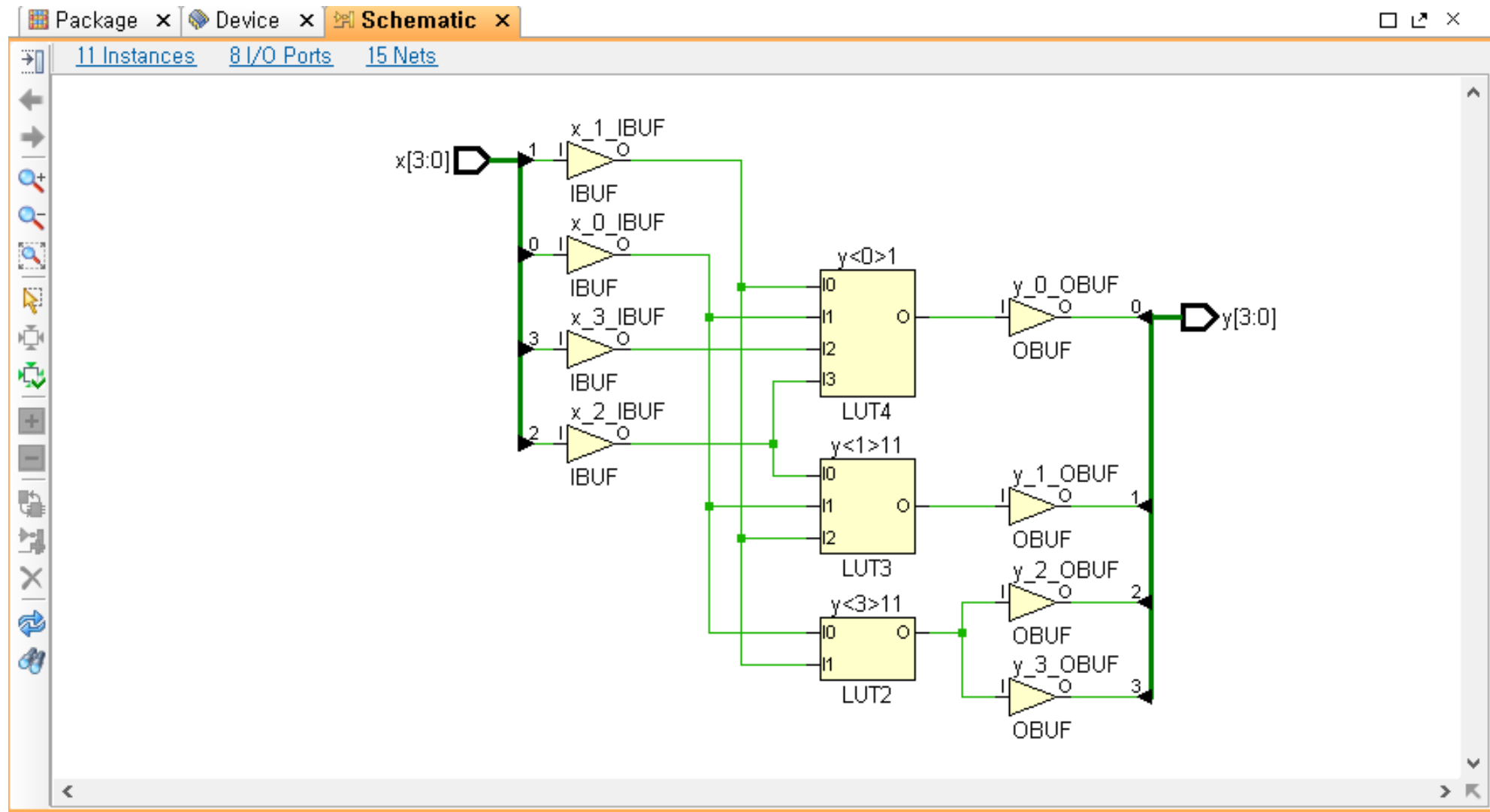
과제2) FPGA회로



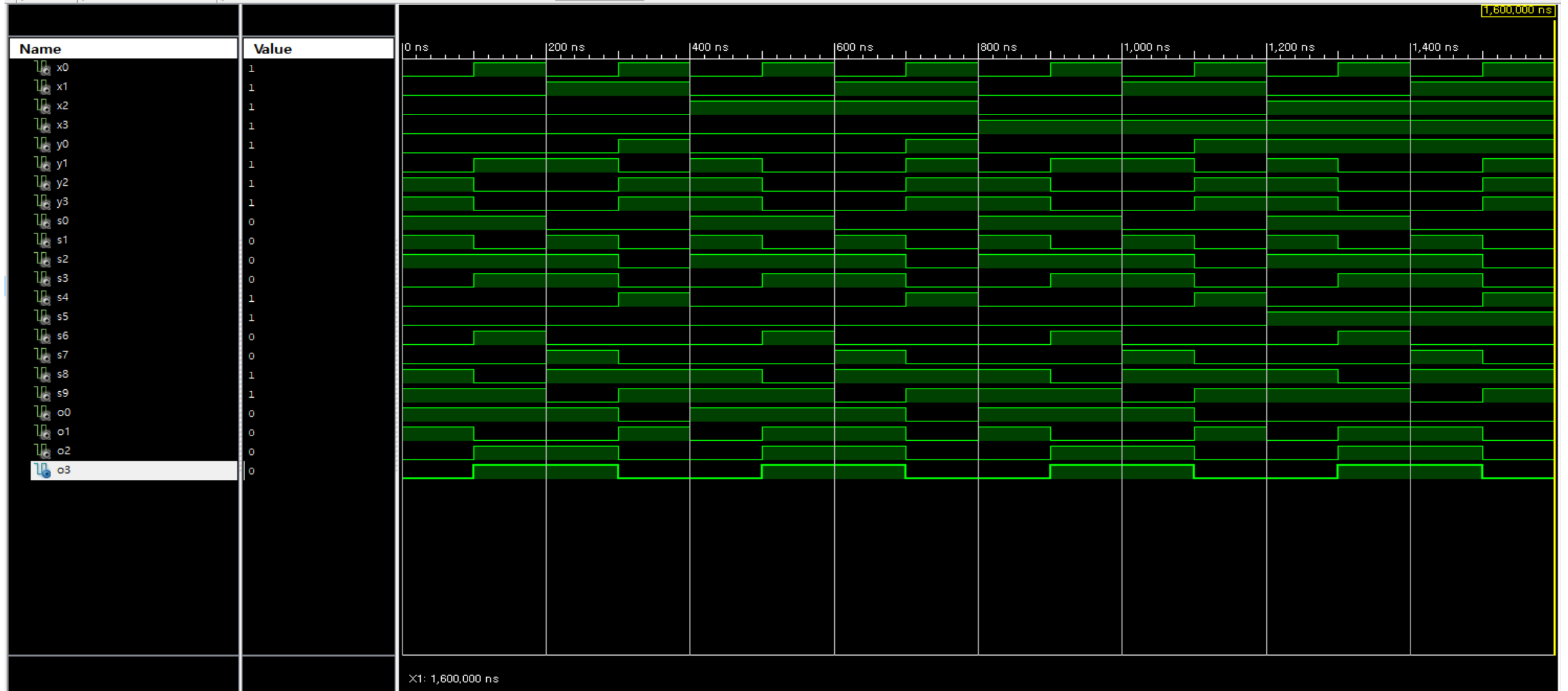
과제2) VECTOR_내부회로



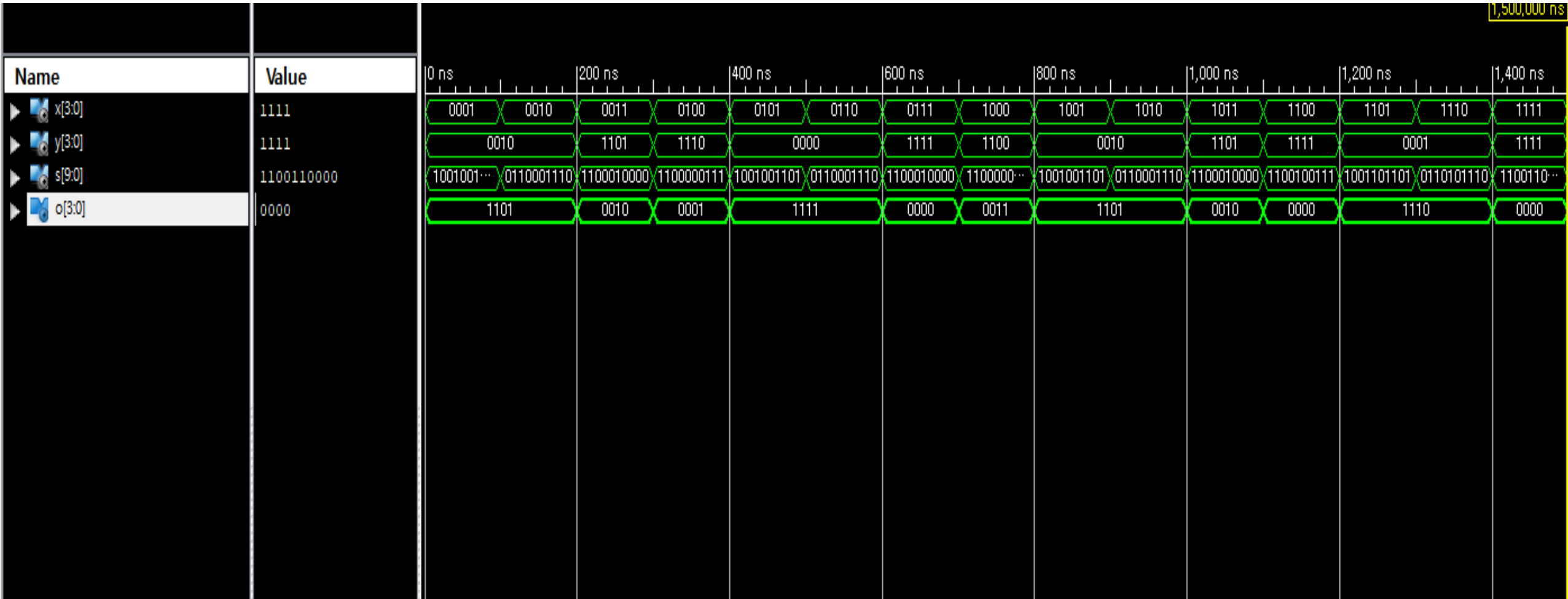
과제2) VECTOR_FPGA회로



과제2) STD_LOGIC 시뮬레이션



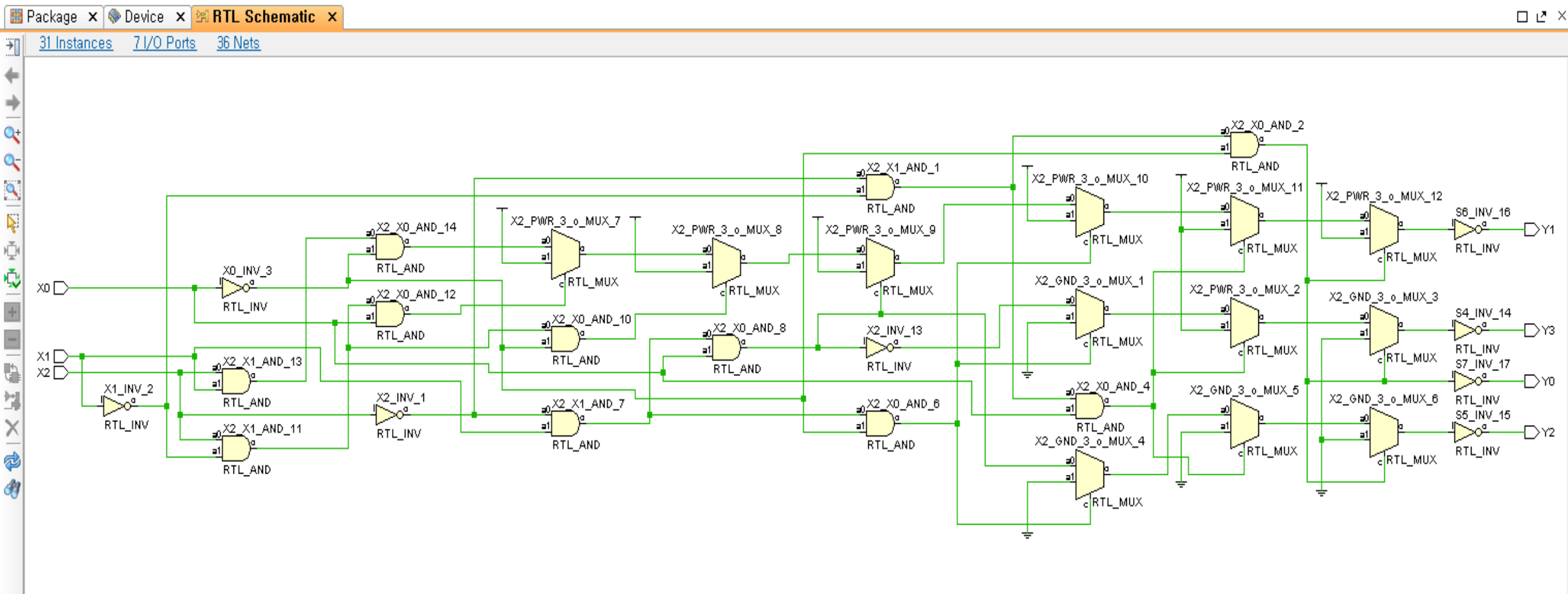
과제2) VECTOR 시뮬레이션



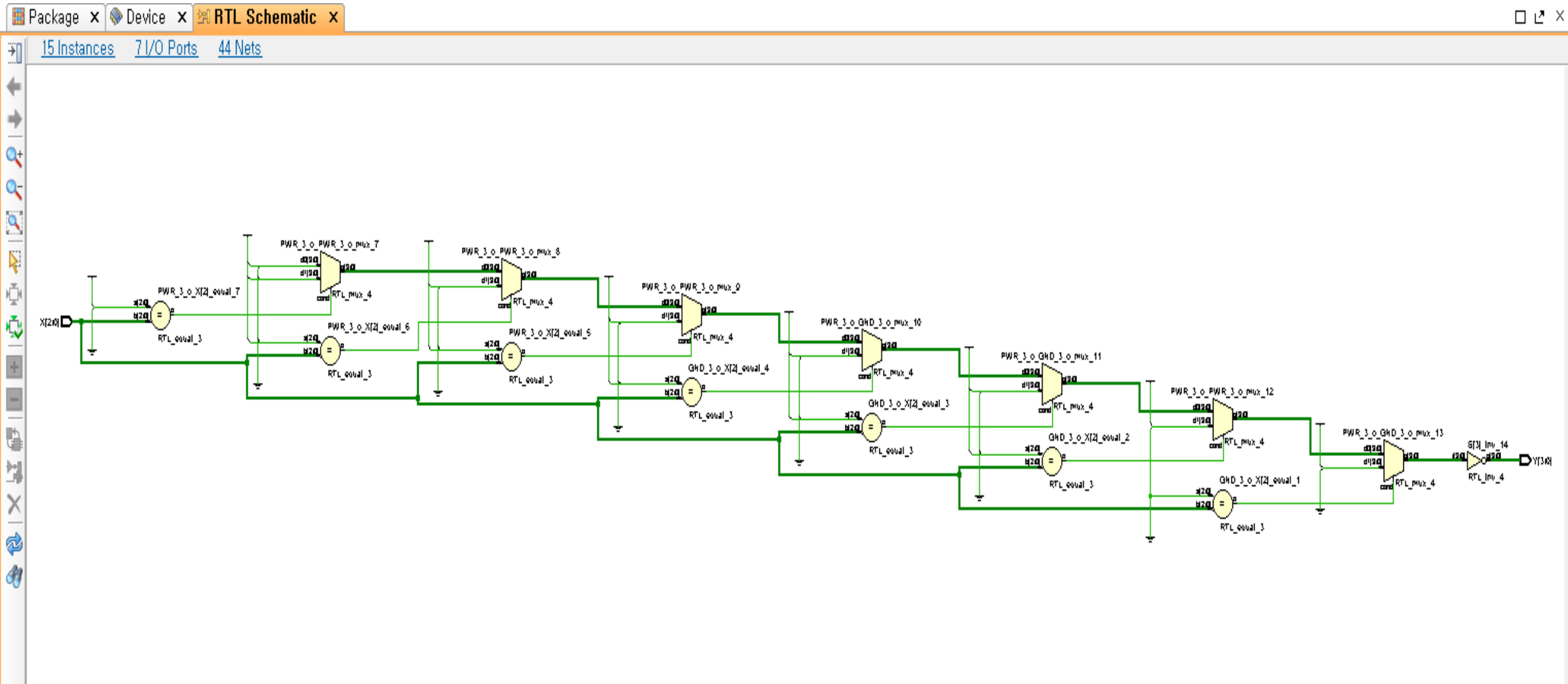
과제3) std_logic 동작테이블

입력			출력			
X2	X1	X0	Y3	Y2	Y1	Y0
0	0	0	1	1	0	0
0	0	1	0	1	0	1
0	1	0	1	1	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	1
1	0	1	0	1	0	1
1	1	0	0	1	0	1
1	1	1	0	1	1	1

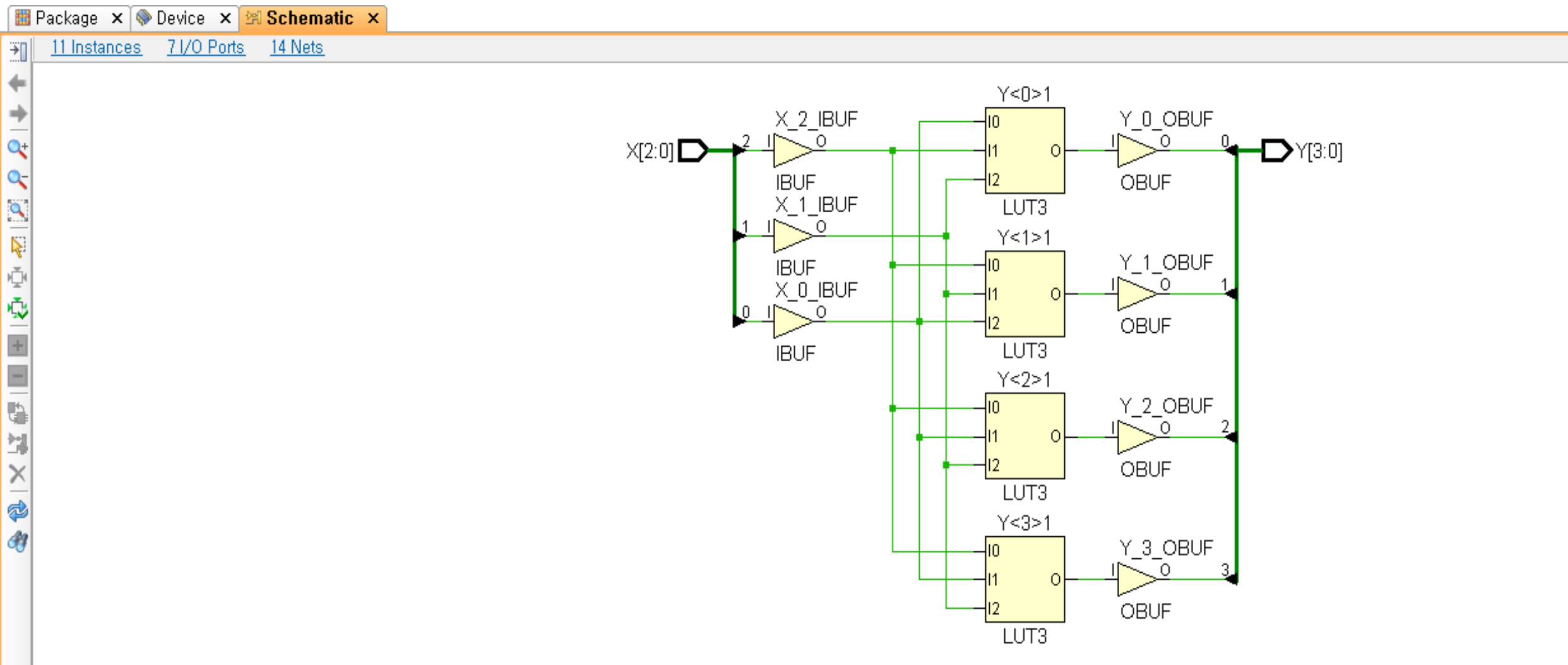
과제3) STD_LOGIC 가상회로



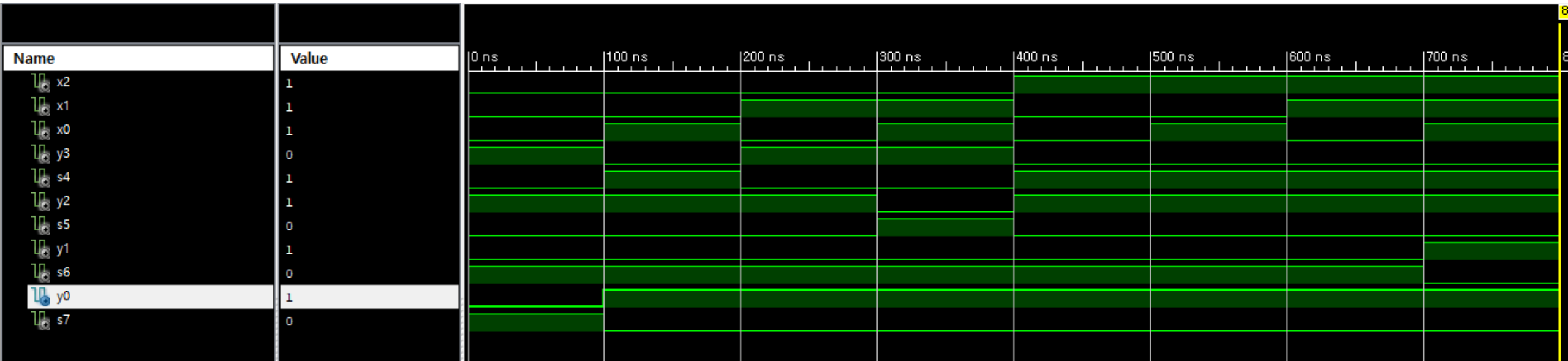
과제3) STD_LOGIC_VECTOR 가상회로



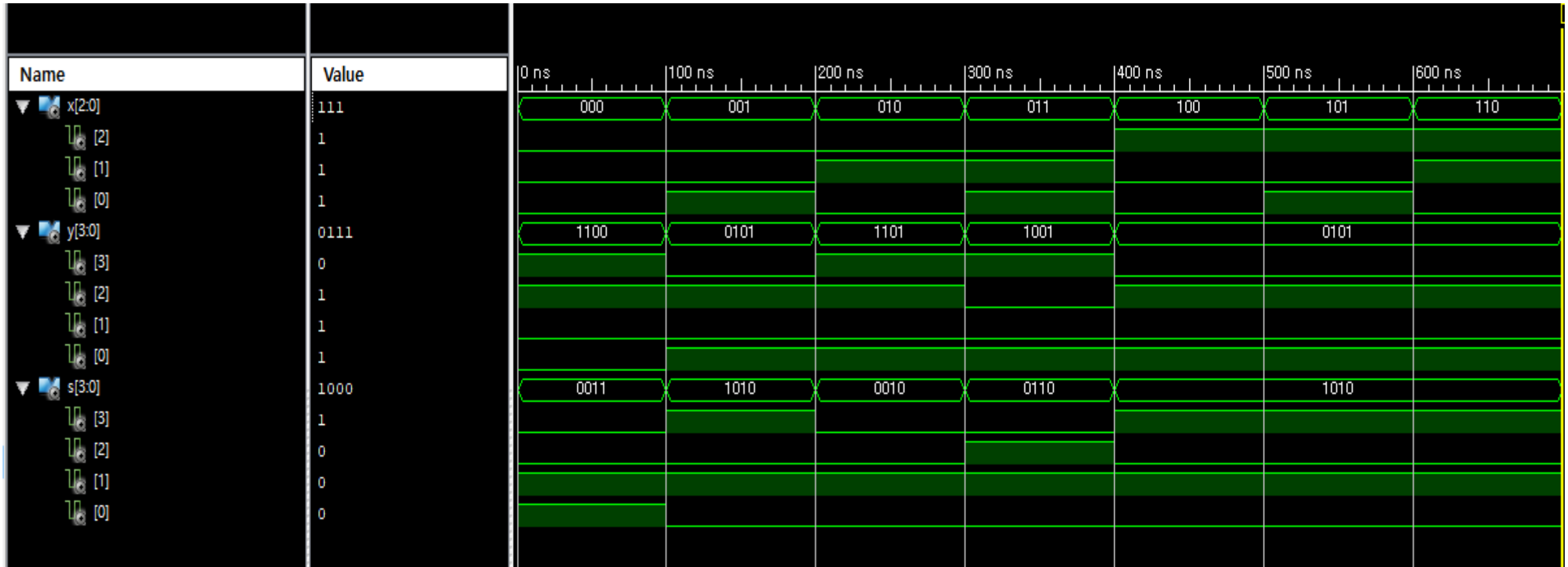
FPGA 회로



과제3) STD_LOGIC 시뮬레이션



과제3) STD_LOGIC_VECTOR 시뮬레이션



Week10 team과제

2학년 2반 2021041089 최태민. 2학년 2반 2021041043 서재권

연습문제2번 동영상 주소:<https://youtu.be/NeBVun9NQzI>

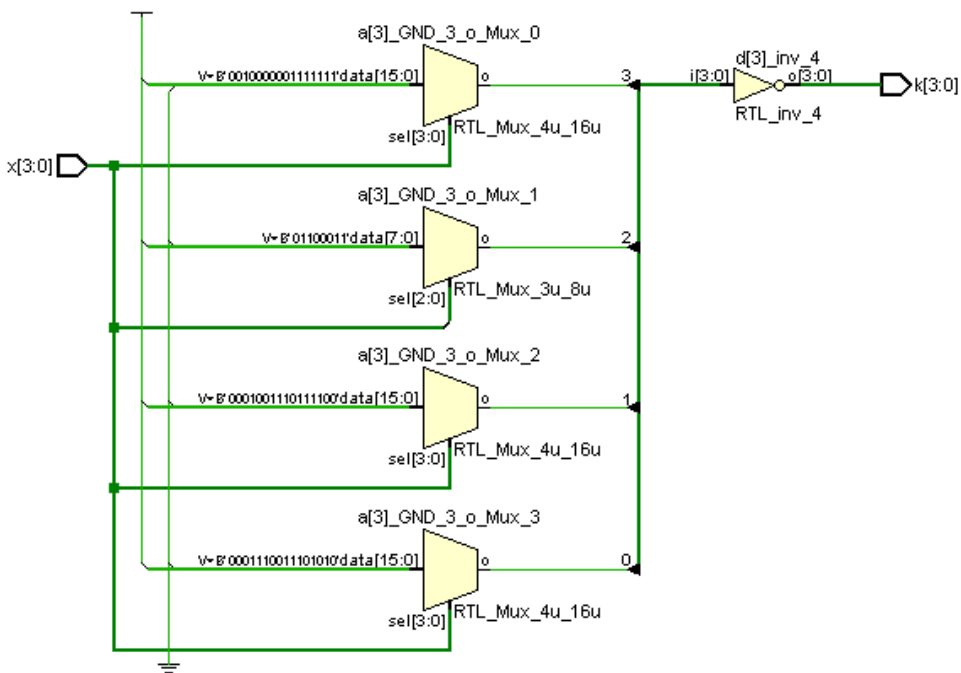
연습문제6번 동영상 주소:<https://youtu.be/v3drQ2zxbi4>

문제 2 : 동작테이블

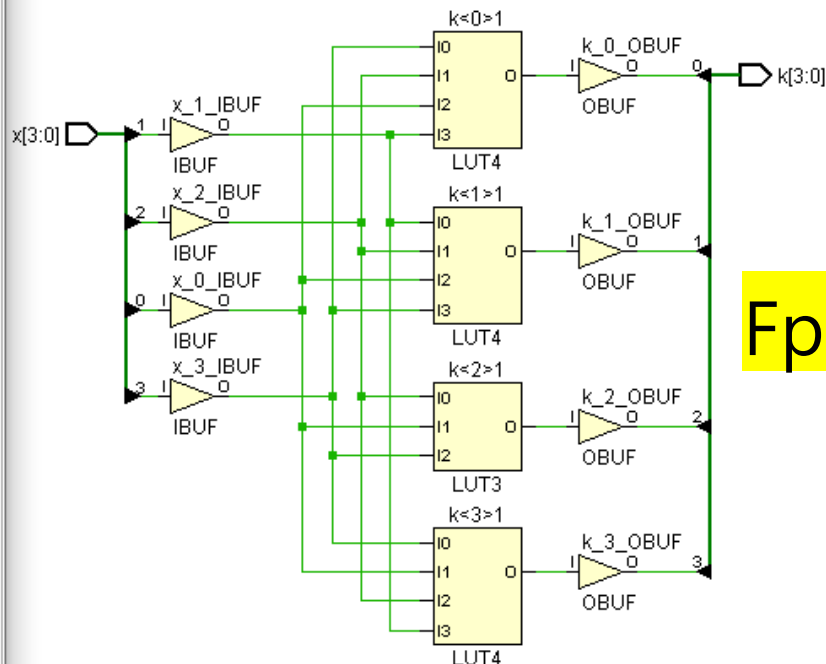
X3 스위치	X2 스위치	X1 스위치	X0 스위치	D3	D2	D1	D0	LED K3	LED K2	LED K1	LED K0	
0	0	0	0	1	1	0	0	0	0	1	1	
0	0	0	1	1	1	0	1	0	0	1	0	
0	0	1	0	1	1	1	0	0	0	0	1	
0	0	1	1	1	1	1	1	0	0	0	0	
0	1	0	0	1	0	1	0	0	1	0	1	
0	1	0	1	1	0	1	1	0	1	0	0	
0	1	1	0	1	0	0	1	0	1	1	0	
0	1	1	1	0	0	1	1	1	1	0	0	
1	0	0	0	0	0	1	0	1	1	0	1	
1	0	0	1	0	1	1	0	1	0	0	1	
1	0	1	0	0	0	0	1	1	1	1	0	
1	0	1	1	0	1	0	1	1	0	1	0	
1	1	0	0	0	1	1	1	1	0	0	0	
1	1	0	1	1	0	0	0	0	1	1	1	
1	1	1	0	0	1	0	0	1	0	1	1	
1	1	1	1	0	0	0	0	1	1	1	1	

문제2 조건처리문(case) : 가상회로 및 fpga회로

가상회로



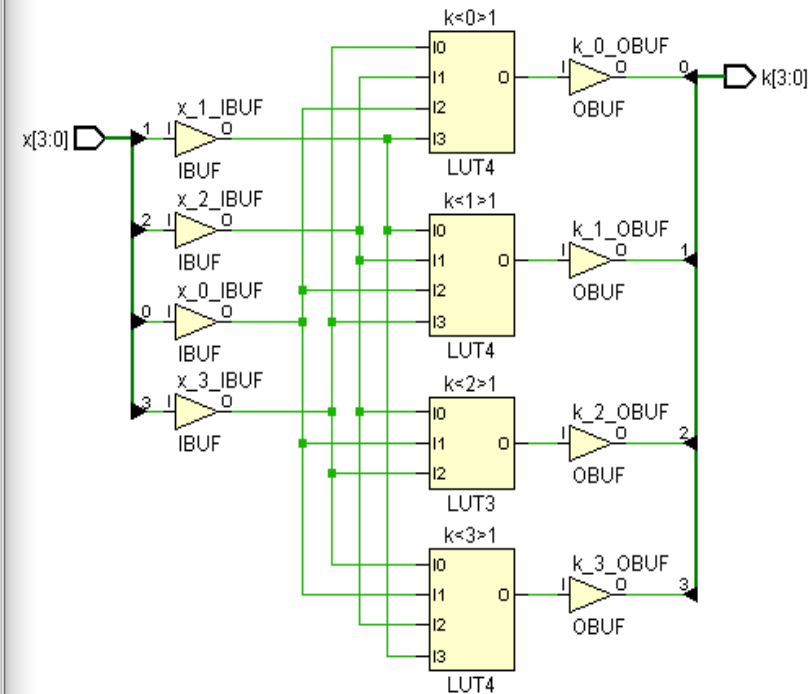
Fpga 회로



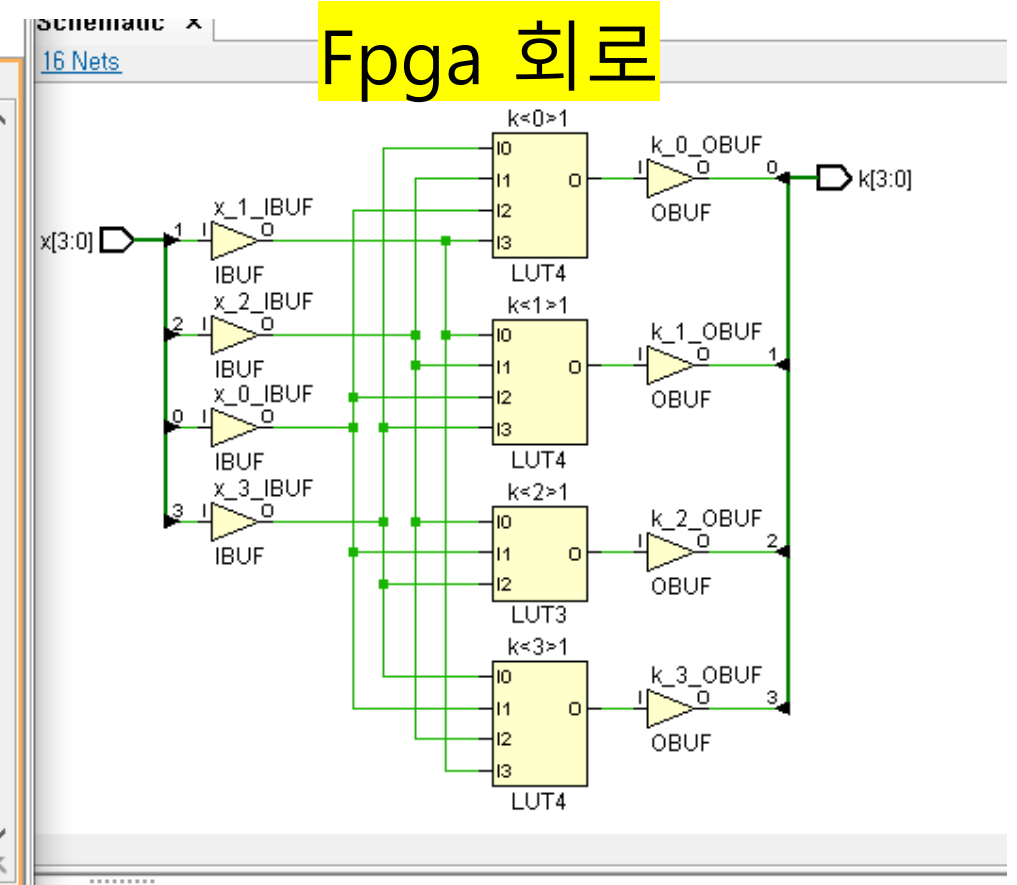
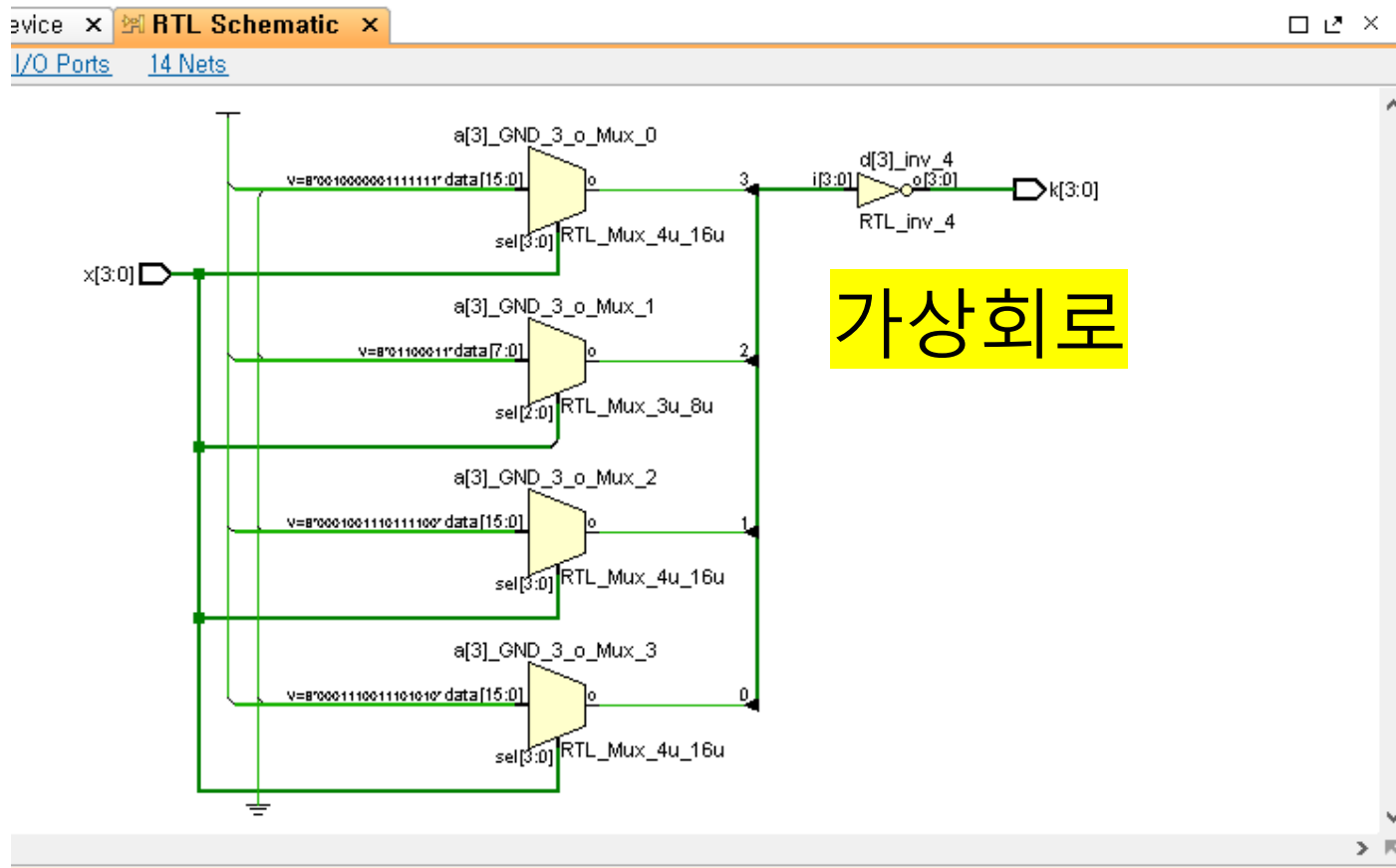
문제2 조건처리문(if~then~else) : 가상회로 및 fpga회로

가상회로

Fpga 회로



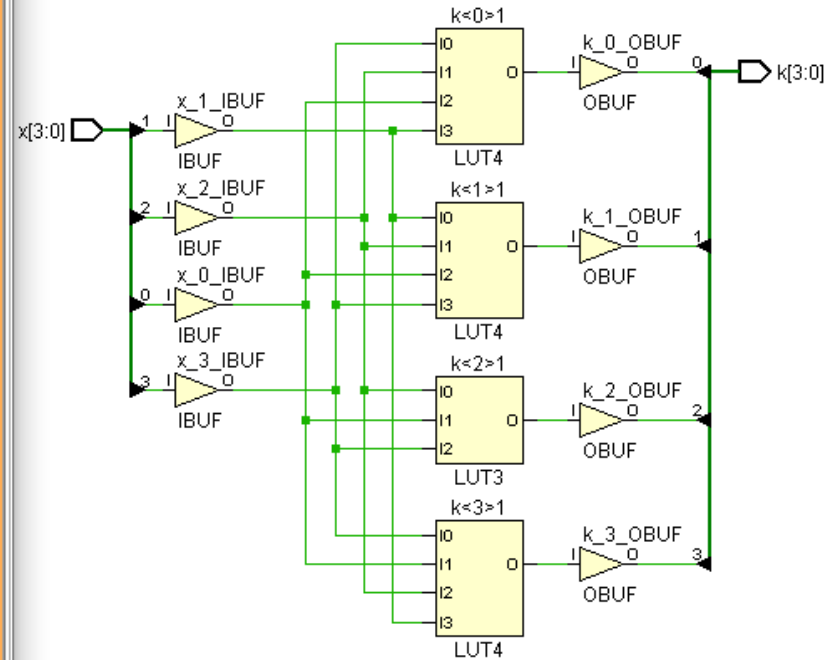
문제2 병행처리문(with~select~when) : 가상회로 및 fpga회로



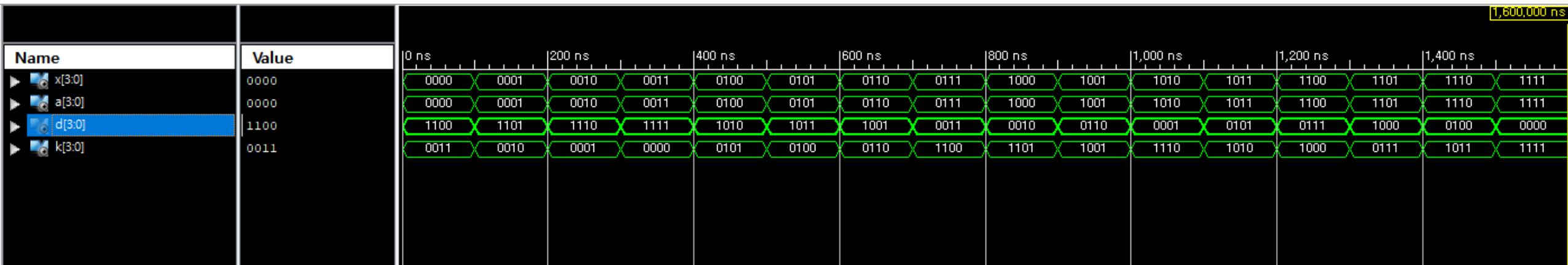
문제2 병행처리문(when~else) : 가상회로 및 fpga회로

가상회로

Fpga 회로



문제2 : 실행화면



문제 6 : 동작테이블 (X9 스위치와 X8의 스위치가 0일 때)

X7 스위치	X6 스위치	X5 스위치	X4 스위치	X3 스위치	X2 스위치	X1 스위치	X0 스위치	X9 스위치	X8 스위치	K3 LED	K2 LED	K1 LED	K0 LED
0	0	0	0	0	0	0	0	0	0	X	X	X	X
0	0	0	0	0	0	0	1	0	0	X	X	X	X
0	0	0	0	0	0	1	0	0	0	X	X	X	X
0	0	0	0	0	0	1	1	0	0	X	X	X	X
0	0	0	0	0	1	0	0	0	0	X	X	X	X
0	0	0	0	0	1	0	1	0	0	X	X	X	X
0	0	0	0	0	1	1	0	0	0	X	X	X	X
0	0	0	0	0	1	1	1	0	0	X	X	X	X
0	0	0	0	1	0	0	0	0	0	X	X	X	X
0	0	0	0	1	0	0	1	0	0	X	X	X	X
0	0	0	0	1	0	1	0	0	0	X	X	X	X
0	0	0	0	1	0	1	1	0	0	X	X	X	X
0	0	0	0	1	1	0	0	0	0	X	X	X	X
0	0	0	0	1	1	0	1	0	0	X	X	X	X
0	0	0	0	1	1	1	0	0	0	X	X	X	X
0	0	0	0	1	1	1	1	0	0	X	X	X	X
0	0	0	1	0	0	0	0	0	0	X	X	X	X
0	0	0	1	0	0	0	1	0	0	X	X	X	X

문제 6 : 동작테이블 (X9 스위치가 0, X8의 스위치가 1일 때)

X7 스위치	X6 스위치	X5 스위치	X4 스위치	X3 스위치	X2 스위치	X1 스위치	X0 스위치	X9 스위치	X8 스위치	K3 LED	K2 LED	K1 LED	K0 LED
0	0	0	0	0	0	0	0	0	1	1	1	1	1
0	0	0	0	0	0	0	1	0	1	1	1	1	0
0	0	0	0	0	0	1	0	0	1	1	1	0	1
0	0	0	0	0	0	1	1	0	1	1	0	1	1
0	0	0	0	0	1	0	0	0	1	1	0	1	1
0	0	0	0	0	1	0	1	0	1	1	0	1	0
0	0	0	0	0	1	1	0	0	1	1	0	0	1
0	0	0	0	0	1	1	1	0	1	1	0	0	0
0	0	0	0	1	0	0	0	0	1	0	1	1	1
0	0	0	0	1	0	0	1	0	1	0	1	1	0
0	0	0	0	1	0	1	0	0	1	0	1	0	1
0	0	0	0	1	0	1	1	0	1	0	1	0	0
0	0	0	0	1	1	0	0	0	1	0	0	1	1
0	0	0	0	1	1	0	1	0	1	0	0	1	0
0	0	0	0	1	1	1	0	0	1	0	0	0	1
0	0	0	0	1	1	1	1	0	1	0	0	0	0
0	0	0	1	0	0	0	0	0	1	1	1	1	1
0	0	0	1	0	0	0	1	0	1	1	1	1	0

문제 6 : 동작테이블 (X9 스위치가 1, X8의 스위치가 0일 때)

X7 스위치	X6 스위치	X5 스위치	X4 스위치	X3 스위치	X2 스위치	X1 스위치	X0 스위치	X9 스위치	X8 스위치	K3 LED	K2 LED	K1 LED	K0 LED
0	0	0	0	0	0	0	0	1	0	1	1	1	1
0	0	0	0	0	0	0	1	1	0	1	1	1	1
0	0	0	0	0	0	1	0	1	0	1	1	1	1
0	0	0	0	0	0	1	1	1	0	1	1	1	1
0	0	0	0	0	1	0	0	1	0	1	1	1	1
0	0	0	0	0	1	0	1	1	0	1	1	1	1
0	0	0	0	0	1	1	0	1	0	1	1	1	1
0	0	0	0	0	1	1	1	1	0	1	1	1	1
0	0	0	0	1	0	0	0	1	0	1	1	1	1
0	0	0	0	1	0	0	1	1	0	1	1	1	1
0	0	0	0	1	0	1	0	1	0	1	1	1	1
0	0	0	0	1	0	1	1	1	0	1	1	1	1
0	0	0	0	1	1	0	0	1	0	1	1	1	1
0	0	0	0	1	1	0	1	1	0	1	1	1	1
0	0	0	0	1	1	1	0	1	0	1	1	1	1
0	0	0	0	1	1	1	1	1	0	1	1	1	1
0	0	0	1	0	0	0	0	1	0	1	1	1	0
0	0	0	1	0	0	0	1	1	0	1	1	1	0

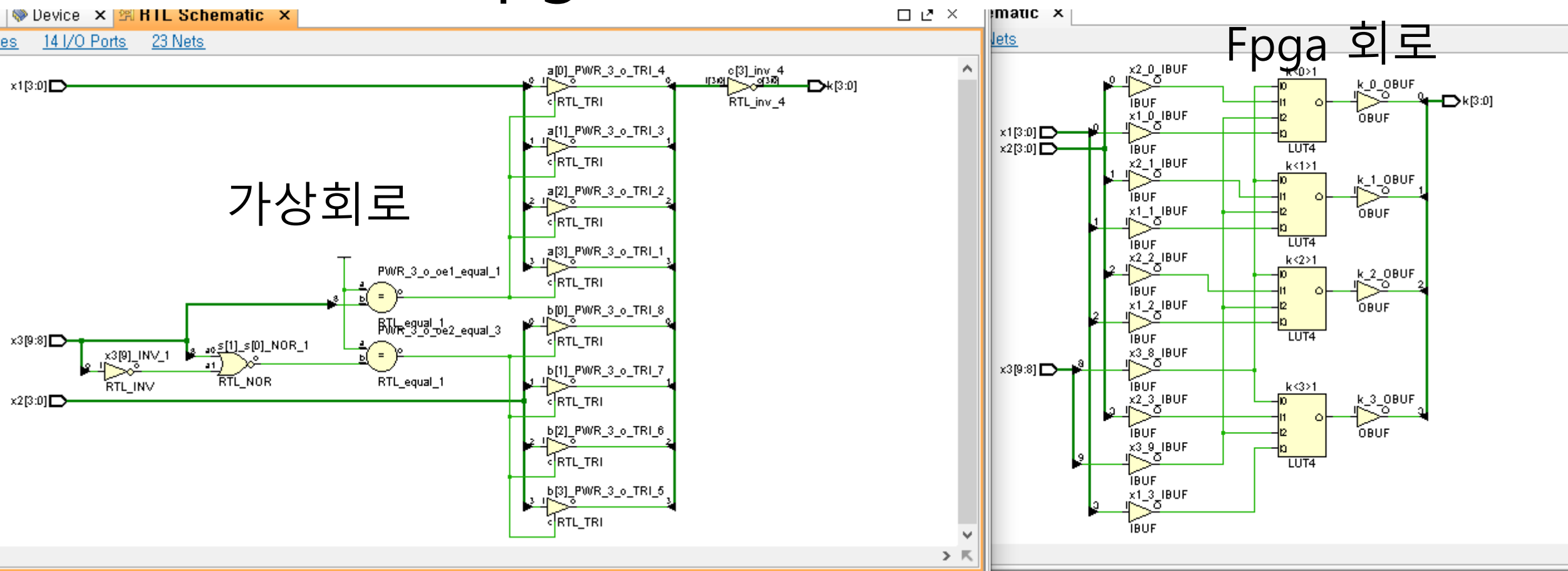
문제 6 : 동작테이블 (X9 스위치와 X8의 스위치가 1일 때)

X7 스위치	X6 스위치	X5 스위치	X4 스위치	X3 스위치	X2 스위치	X1 스위치	X0 스위치	X9 스위치	X8 스위치	K3 LED	K2 LED	K1 LED	K0 LED
0	0	0	0	0	0	0	0	1	1	1	1	1	1
0	0	0	0	0	0	0	1	1	1	1	1	1	0
0	0	0	0	0	0	1	0	1	1	1	1	0	1
0	0	0	0	0	0	1	1	1	1	1	1	0	0
0	0	0	0	0	1	0	0	1	1	1	0	1	1
0	0	0	0	0	1	0	1	1	1	1	0	1	0
0	0	0	0	0	1	1	0	1	1	1	0	0	1
0	0	0	0	0	1	1	1	1	1	1	0	0	0
0	0	0	0	1	0	0	0	1	1	0	1	1	1
0	0	0	0	1	0	0	1	1	1	0	1	1	0
0	0	0	0	1	0	1	0	1	1	0	1	0	1
0	0	0	0	1	0	1	1	1	1	0	1	0	0
0	0	0	0	1	1	0	0	1	1	0	0	1	1
0	0	0	0	1	1	0	1	1	1	0	0	1	0
0	0	0	0	1	1	1	0	1	1	0	0	0	1
0	0	0	0	1	1	1	1	1	1	0	0	0	0
0	0	0	1	0	0	0	0	1	1	1	1	1	1
0	0	0	1	0	0	0	1	1	1	1	1	1	0

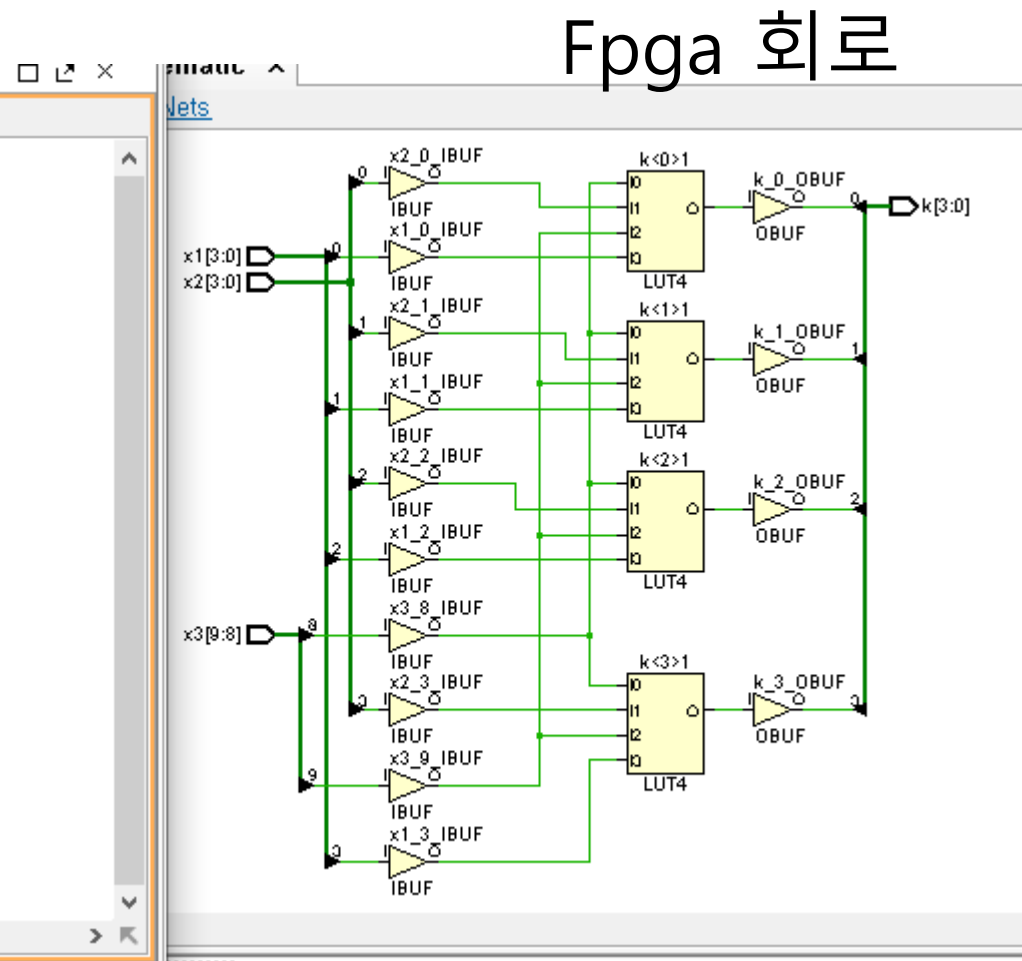
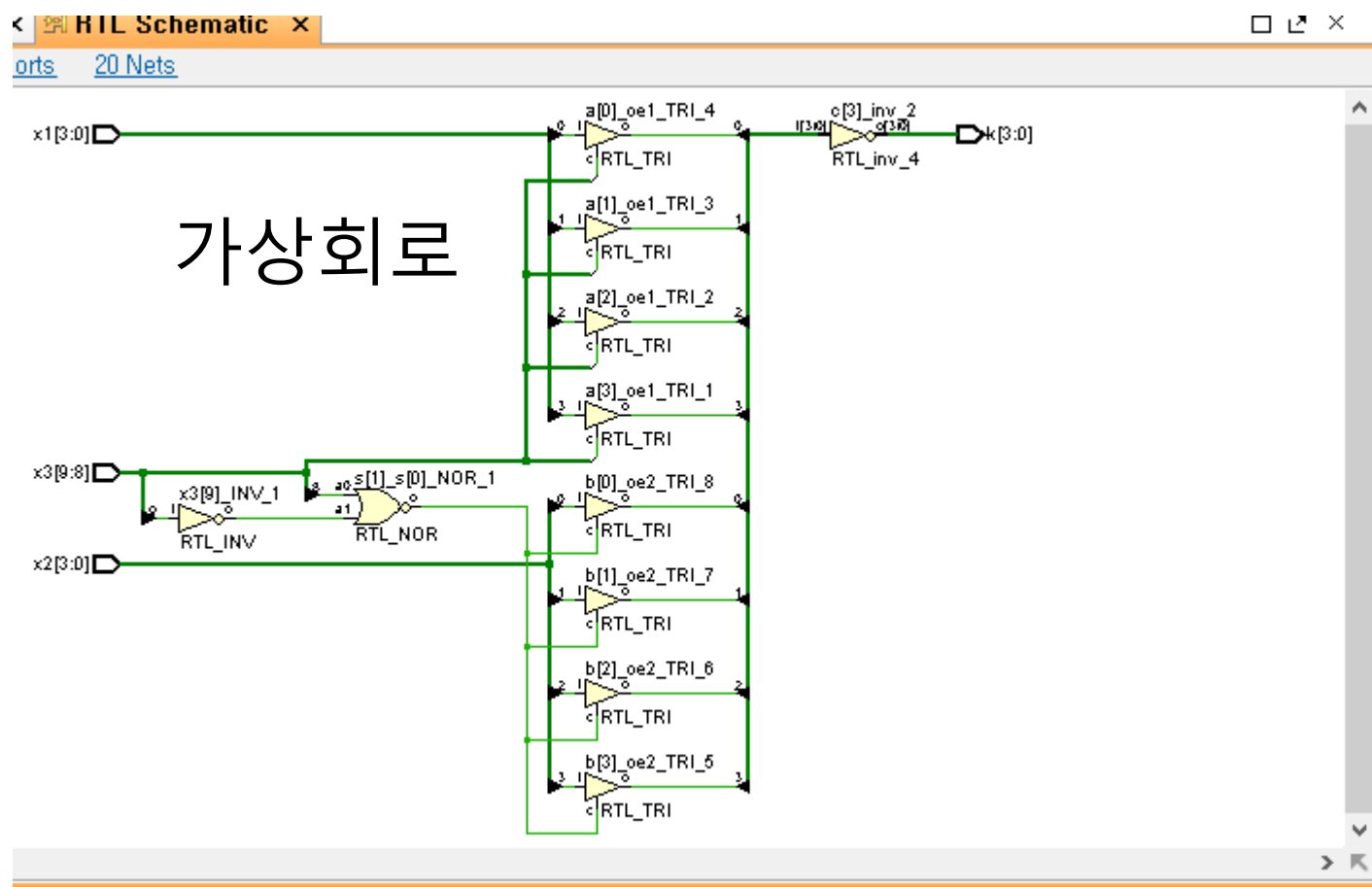
문제 6 : 동작테이블 해석

X8과 X9의 스위치가 둘 다 0일때 unknown(X)이 발생하고
X8과 X9의 스위치가 둘 다 1이거나 하나라도 1이면
unknown(X) 이 발생하지 않습니다.

문제6 조건처리문(case) : 가상회로 및 fpga회로

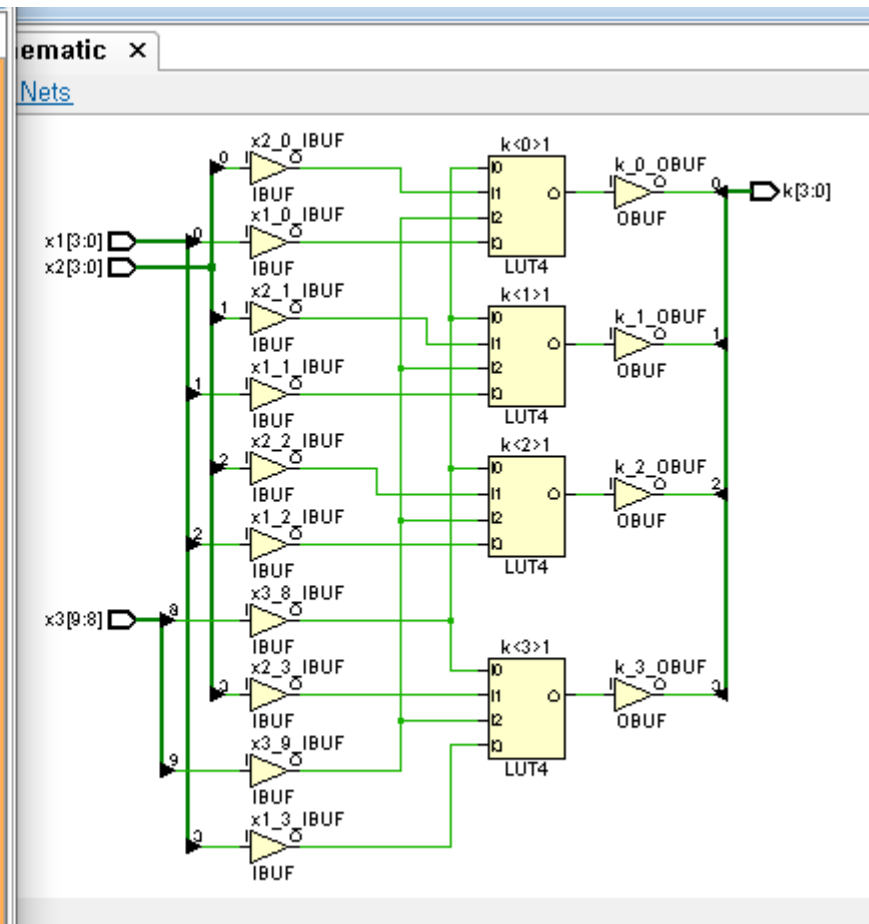
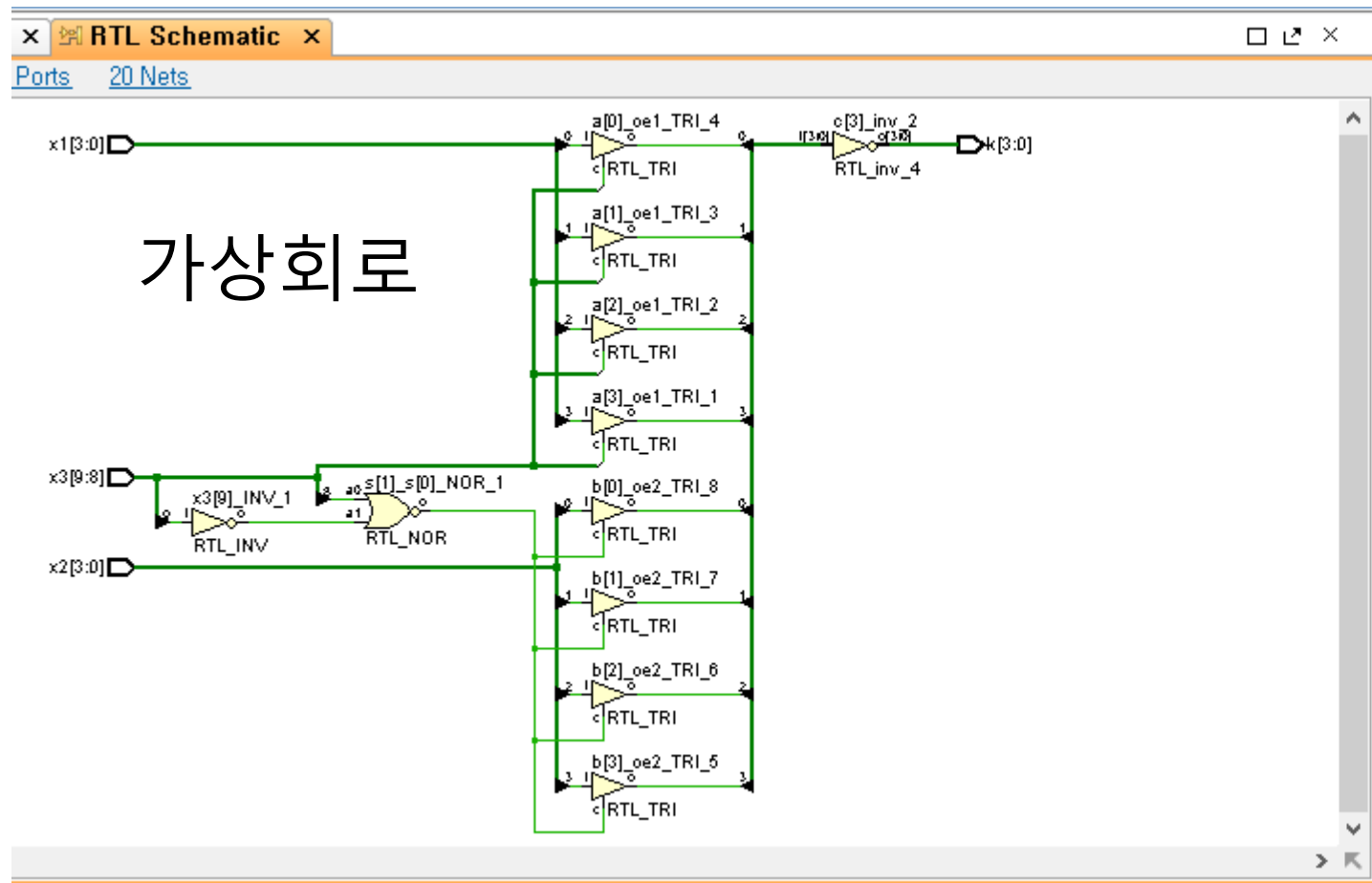


문제6 조건처리문(if~then~else) : 가상회로 및 fpga회로

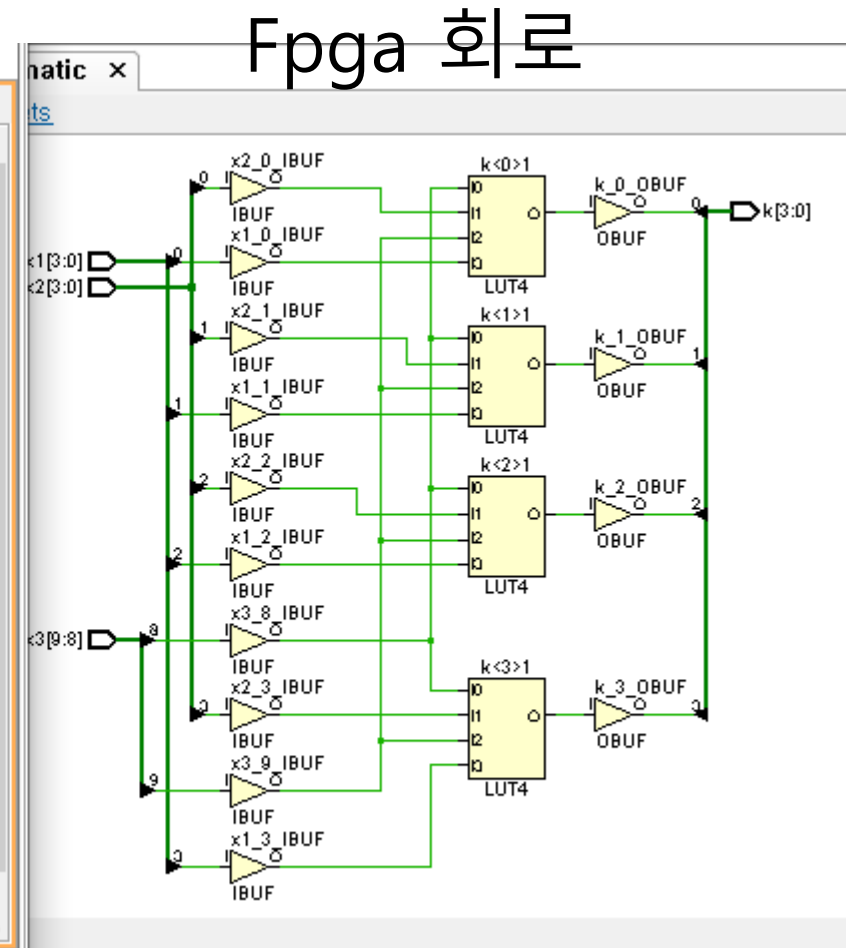
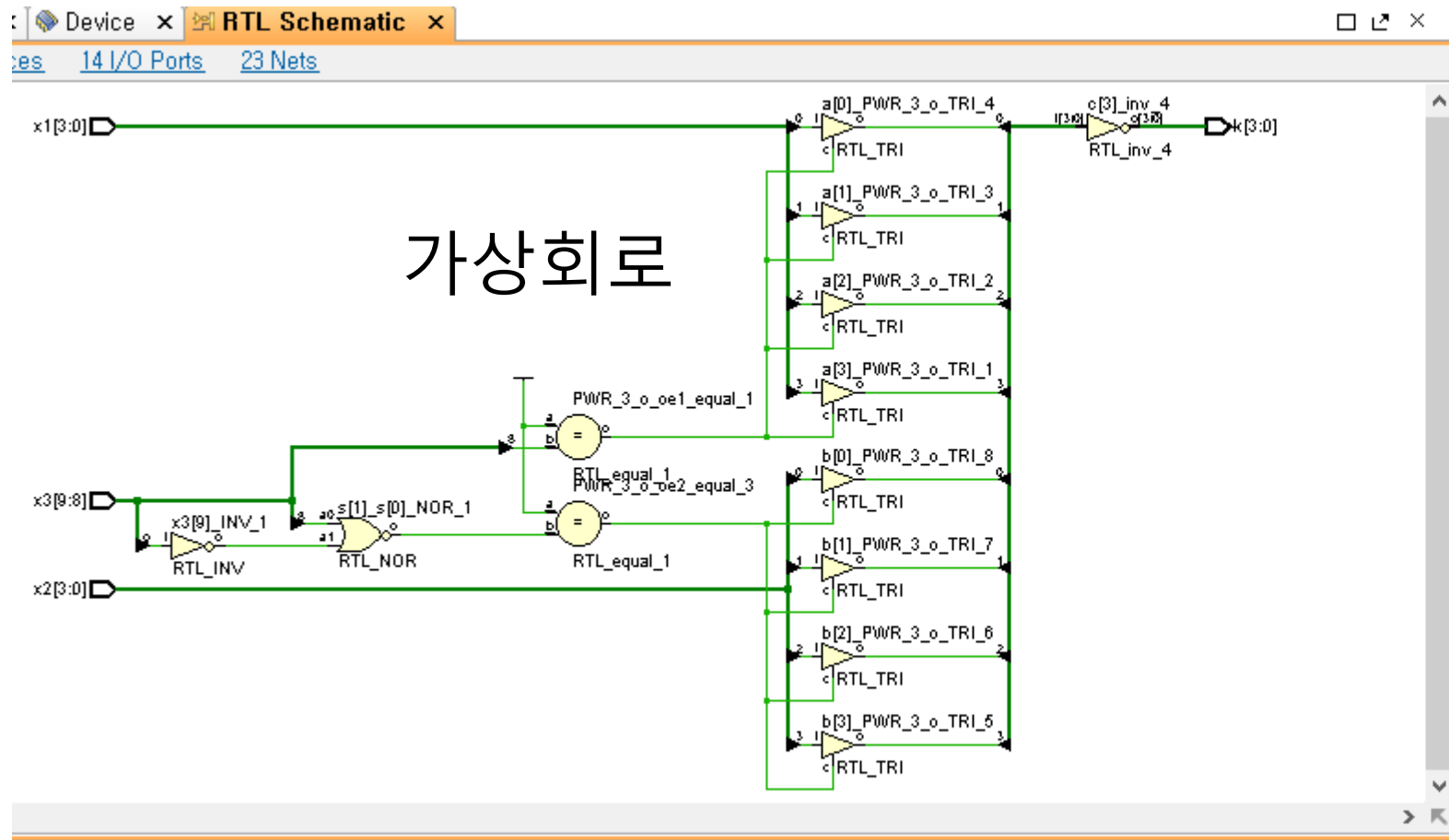


문제6 병행처리문(when~else) : 가상회로 및 fpga회로

Fpga 회로

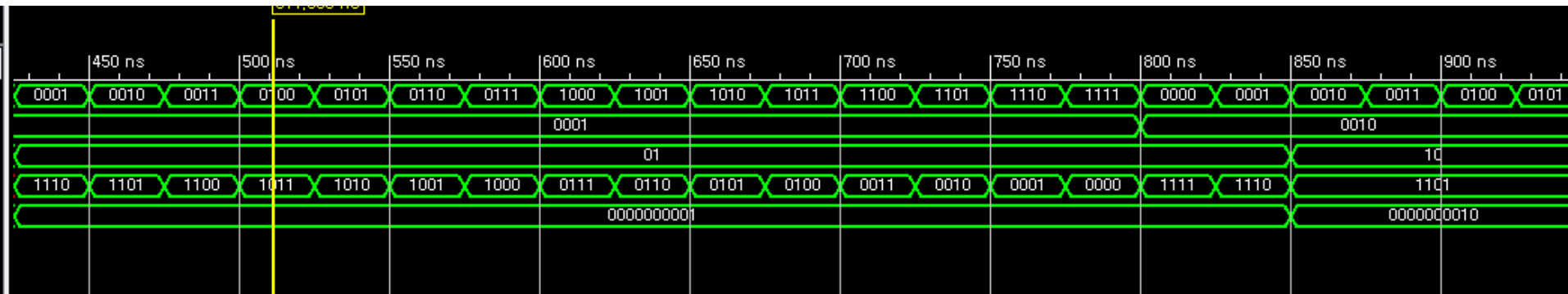


문제6 병행처리문(with~select~when) : 가상회로 및 fpga회로

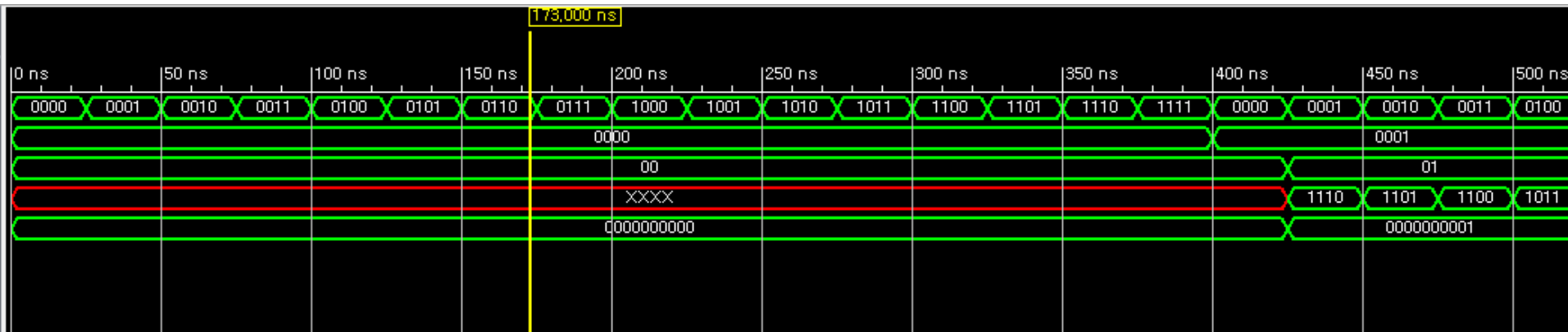


문제6 실행화면

Name	Value
x1[3:0]	0100
x2[3:0]	0001
x3[9:8]	01
k[3:0]	1011
count[9:0]	0000000001



Name	Value
x1[3:0]	0110
x2[3:0]	0000
x3[9:8]	00
k[3:0]	XXXX
count[9:0]	0000000000



문제6 실행화면

Name	Value	Timing Diagram (ns)																				
x1[3:0]	0101	0...	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111	0000	0001	0010	0011	0100	0101	0110
x2[3:0]	0011							0011										0100				
x3[9:8]	11	10								11											00	
k[3:0]	1010	1100	1011	1010	1001	1000	0111	0110	0101	0100	0011	0010	0001	0000	1111	1110	1101	1100	XXXX			
count[9:0]	0000000011	0...								0000000011										00000001	00	

Name	Value	850 ns	900 ns	936,500 ns	950 ns	1,000 ns	1,050 ns	1,100 ns	1,150 ns	1,200 ns	1,250 ns	1,300 ns	1,350 ns
x1[3:0]	0101	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101
x2[3:0]	0010					0010						0011	
x3[9:8]	10						10					11	
k[3:0]	1101					1101				1100		1011	1010
count[9:0]	0000000010						0000000010					0000000011	

week11 과제

2학년_2반_2021041089_최태민

연습문제2번 동영상 주소:<https://youtu.be/NeBVun9NQzI> 연
습문제6번 동영상 주소:<https://youtu.be/v3drQ2zxbi4>

과제 : 동작테이블

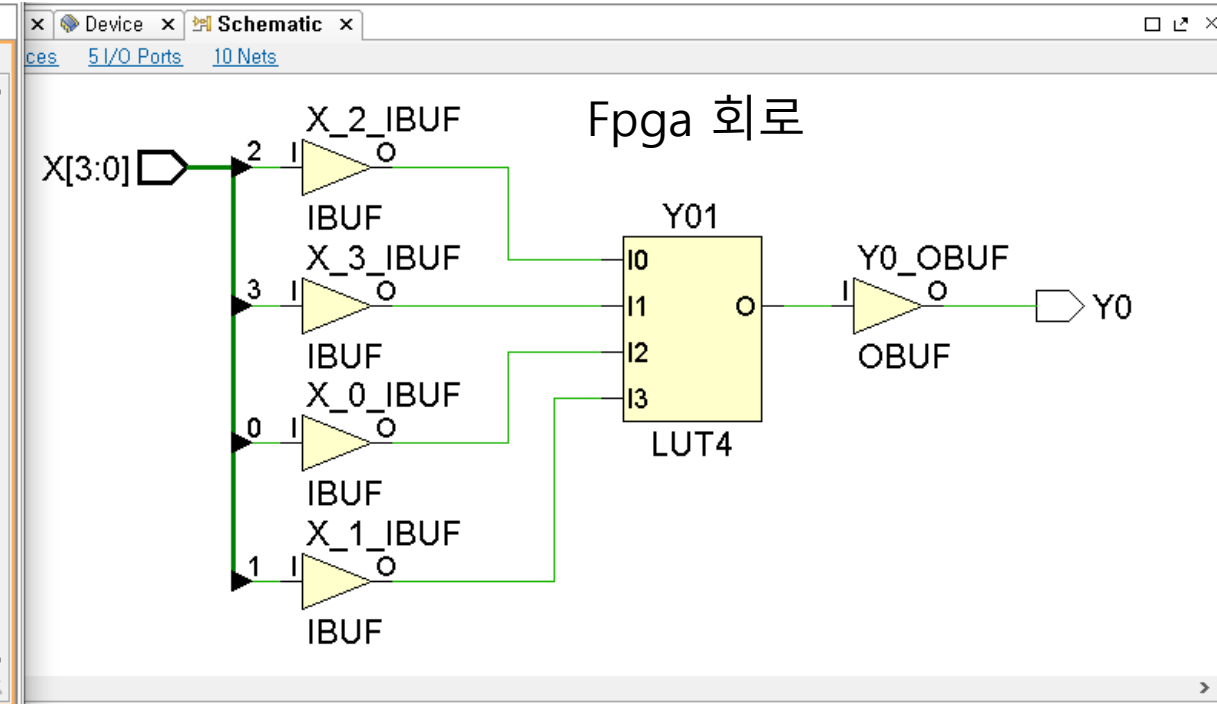
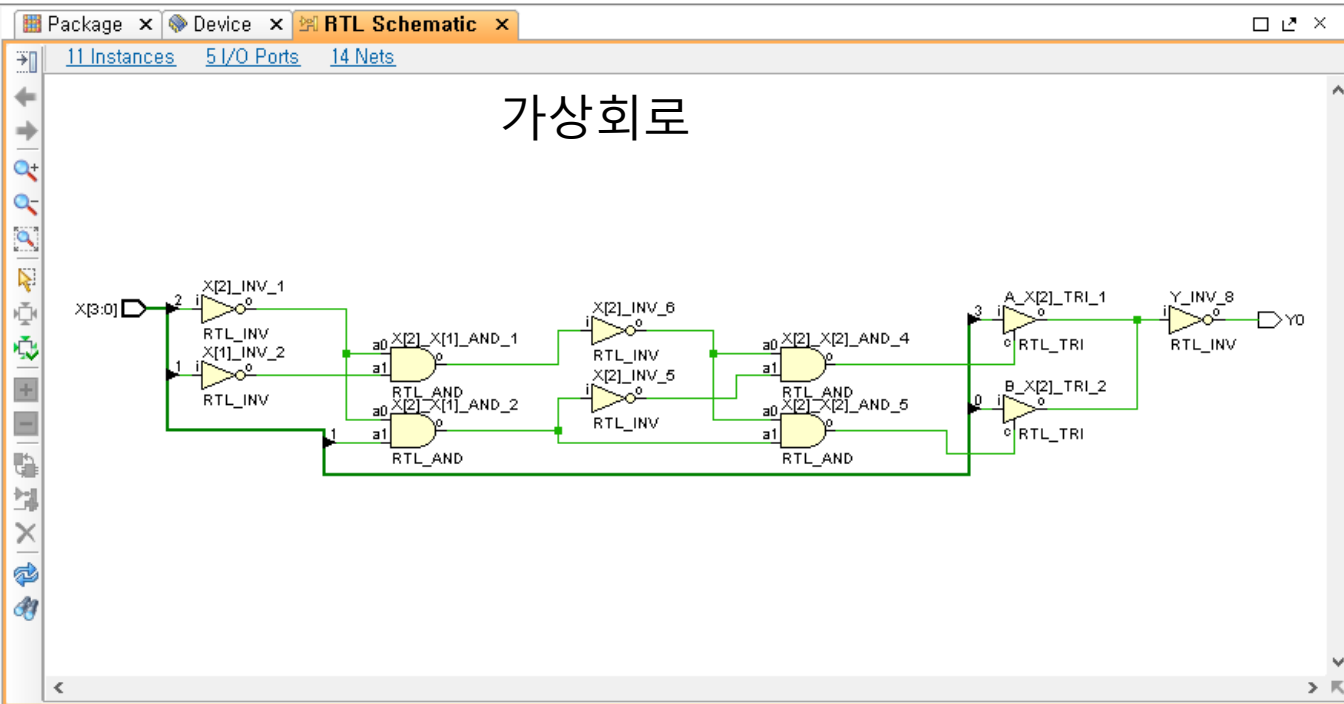
X2 = 0, X1 = 0 일 때				
X3	X0	X2	X1	Y0
0	0	0	0	X
0	1	0	0	X
1	0	0	0	X
1	1	0	0	X

X2 = 0, X1 = 1 일 때				
X3	X0	X2	X1	Y0
0	0	0	1	1
0	1	0	1	0
1	0	0	1	1
1	1	0	1	0

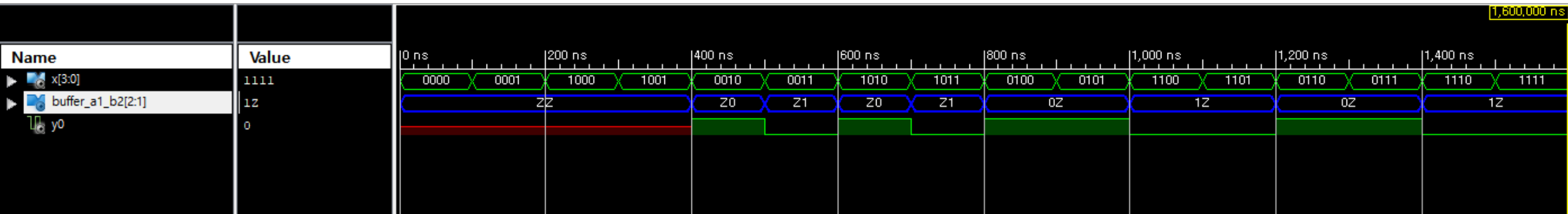
X2 = 1, X1 = 0 일 때				
X3	X0	X2	X1	Y0
0	0	1	0	1
0	1	1	0	1
1	0	1	0	0
1	1	1	0	0

X2 = 1, X1 = 1 일 때				
X3	X0	X2	X1	Y0
0	0	1	1	1
0	1	1	1	1
1	0	1	1	0
1	1	1	1	0

과제 : 가상회로, fpga 회로



과제 : 시뮬레이션화면



Fpga2 4주차 과제

2학년 3반 2021041089_최태민_2021041068_이한수

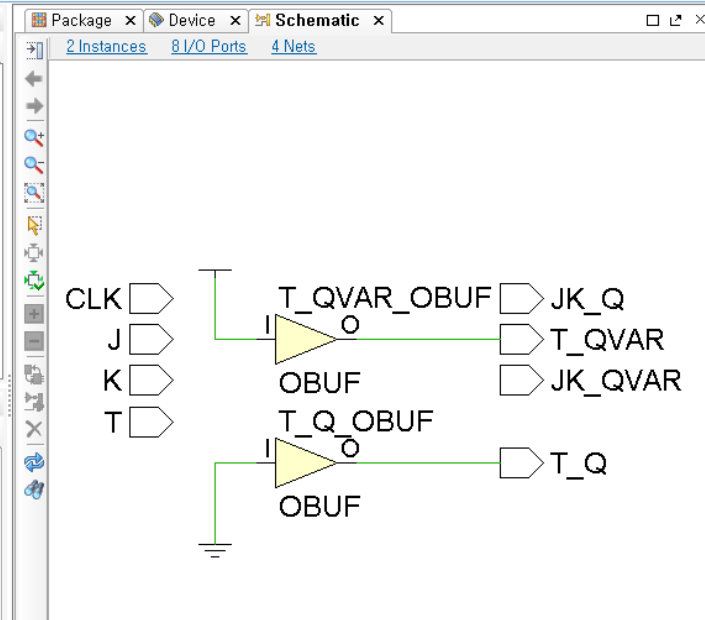
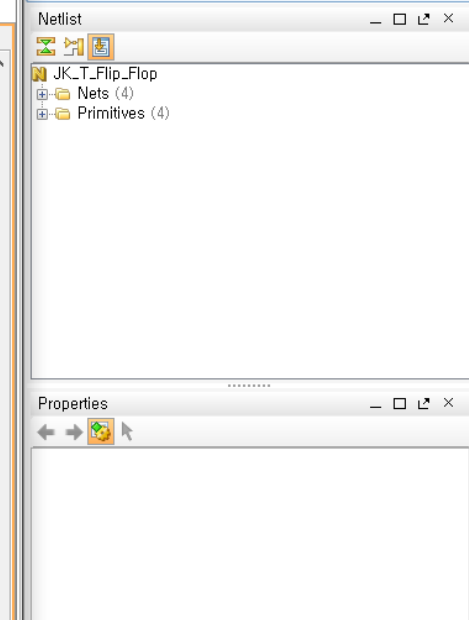
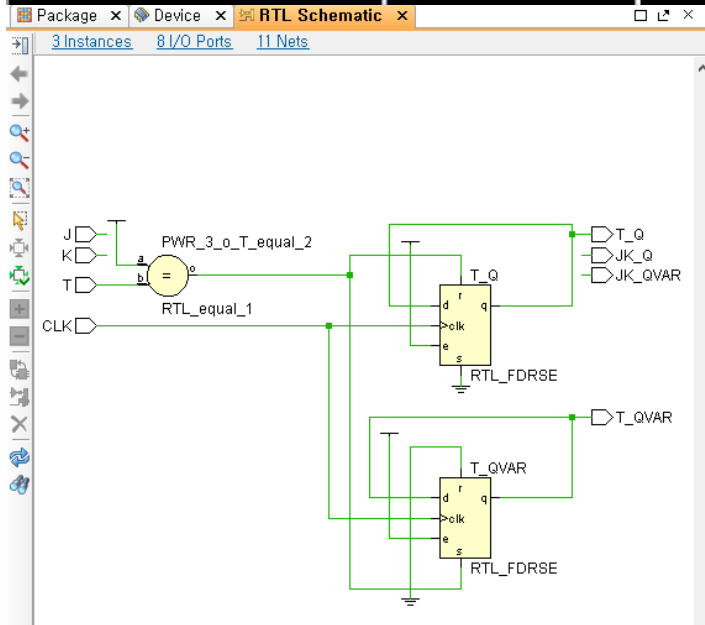
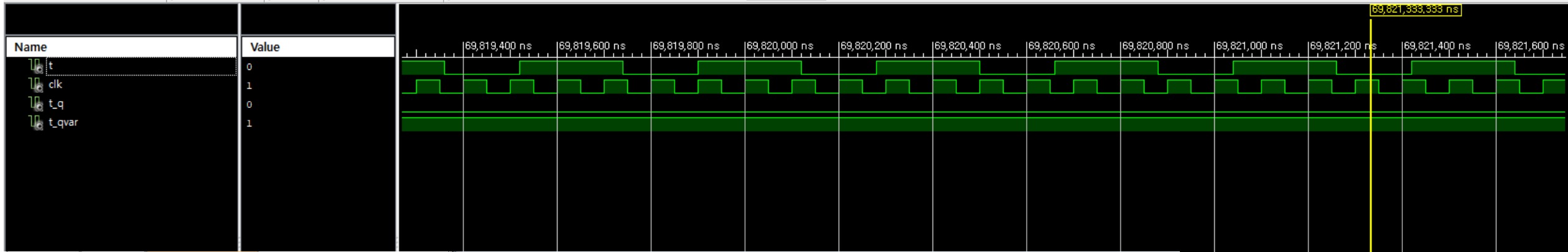
T-Flip-Flop 동작영상: <https://youtu.be/40fSx7eilFY>

JK-Flip-Flop 동작영상: <https://youtu.be/bDyMEUNakEw>

T Flip-Flop 동작 테이블

입력		출력		상태
CLOCK	T	Q	Qvar	
↓	X	-	-	전상태 기억
↑	0	-	-	전상태 기억
↑	1	0	1	Toggle

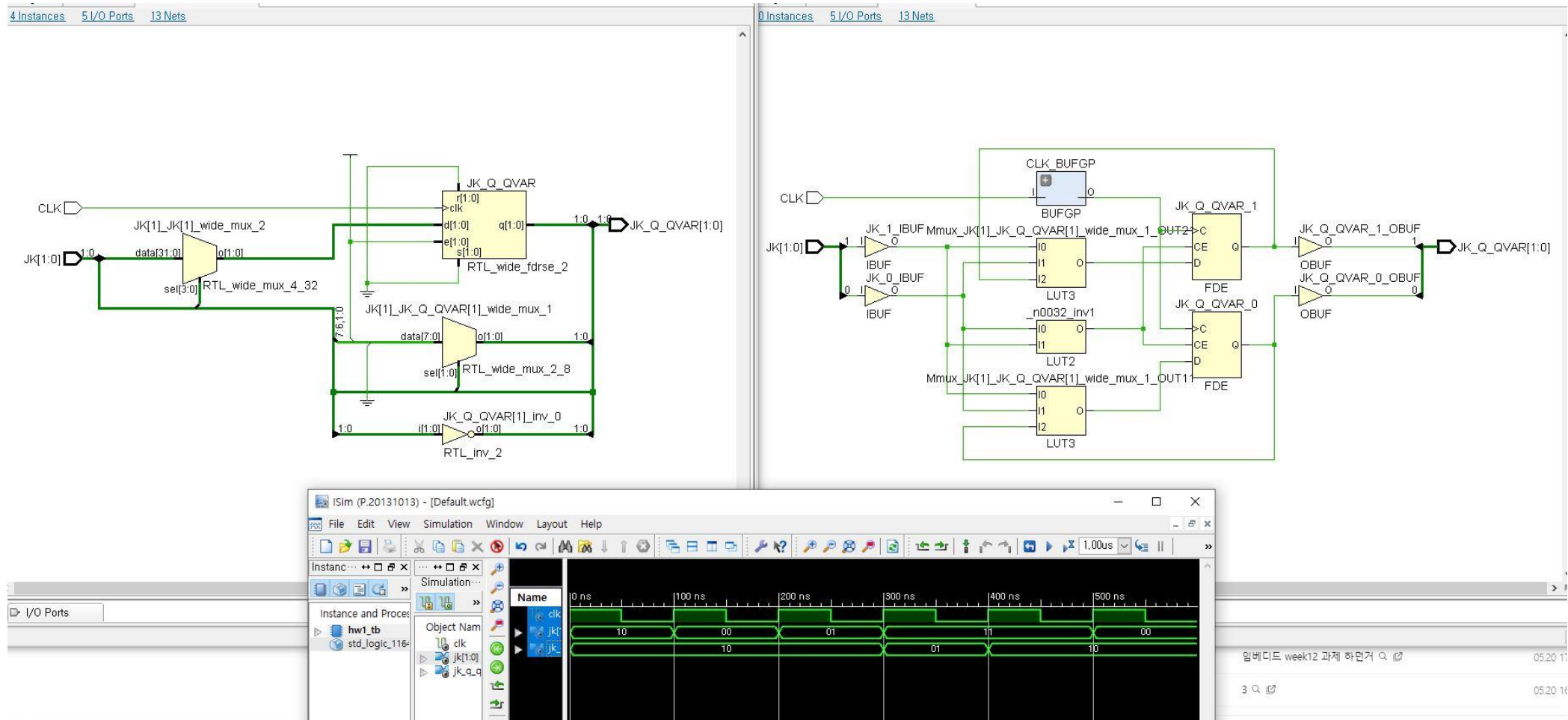
T Flip-Flop Simulate, 어헤드화면



JK Flip-Flop 동작테이블

입력			출력		상태
CLOCK	J	K	Q	Qvar	
↓	X	X	-	-	전상태 기억
↑	0	0	-	-	전상태 기억
↑	0	1	0	1	Reset
↑	1	0	1	0	Set
↑	1	1	Qvar	Q	Toggle

JK Flip-Flop Simulate, 어헤드화면



FPGA2 Week9 Team

2021041091_한규철, 2021041089_최태민

동작영상 유튜브 주소

존슨 카운터 영상 : https://youtu.be/vCX_CYNrquI?si=EhplESpn0wScBbYP

실습 응용 링카운터 영상 : <https://youtu.be/B00bfulwpys?si=VoyKOj2QwJHsFFvj>

실습 응용 리플카운터 영상 : <https://youtu.be/BZC5qBIQfkW?si=xXYTVWeW8OODsSJy>

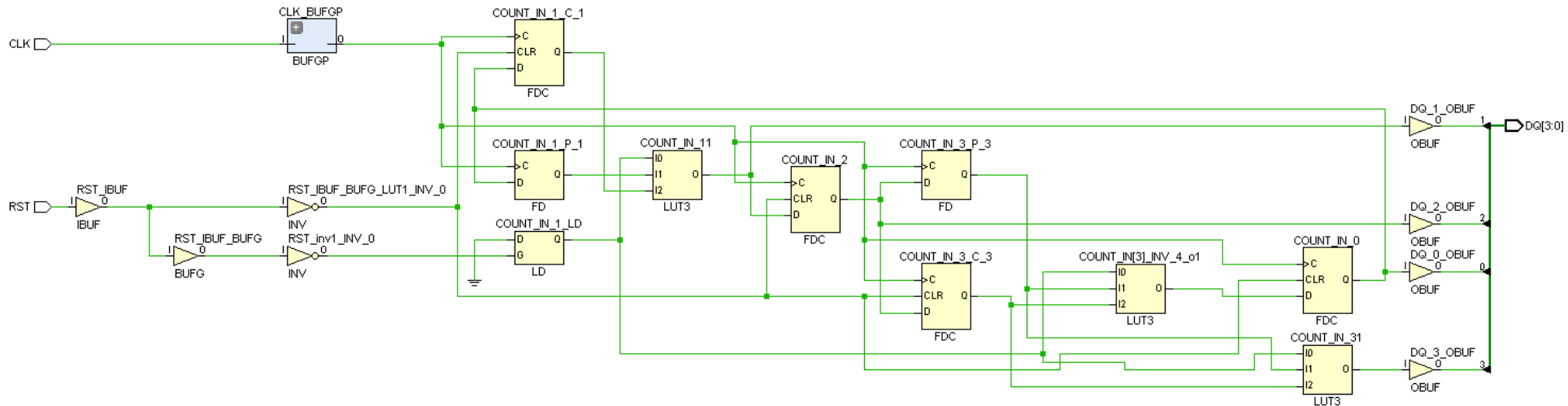
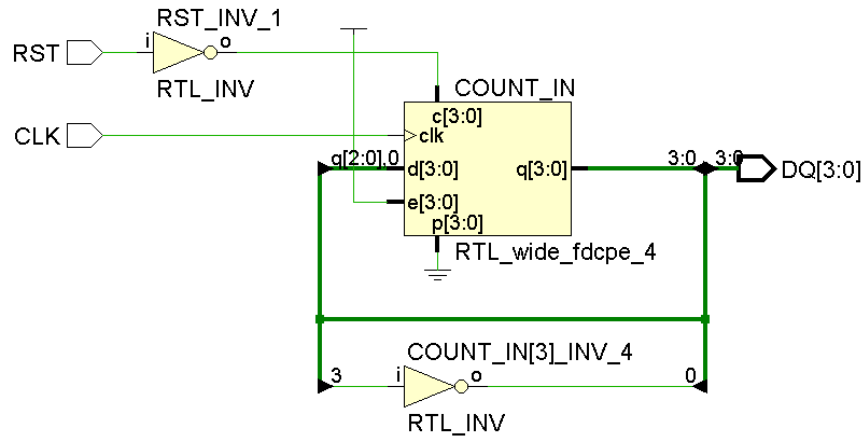
존슨 카운터 동작테이블

입력	Johnson COUNTER			
CLOCK	Q0	Q1	Q2	Q3
↑	0	0	0	0
	1	0	0	0
	1	1	0	0
	1	1	1	0
	1	1	1	1
	0	1	1	1
	0	0	1	1
	0	0	0	1
	0	0	0	0
	1	0	0	0
	1	1	0	0

존슨 카운터 시뮬레이션화면



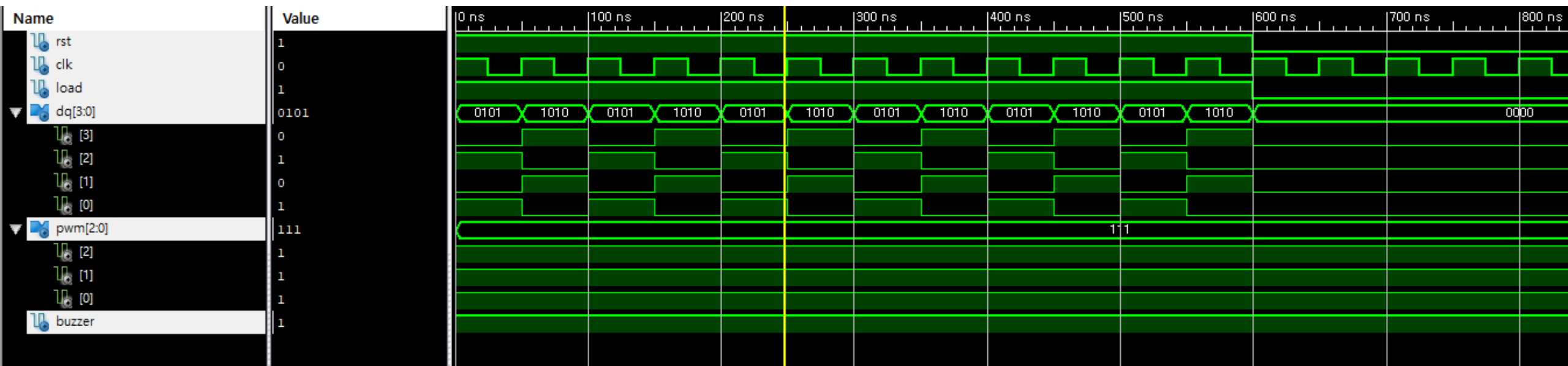
존슨 카운터 회로화면



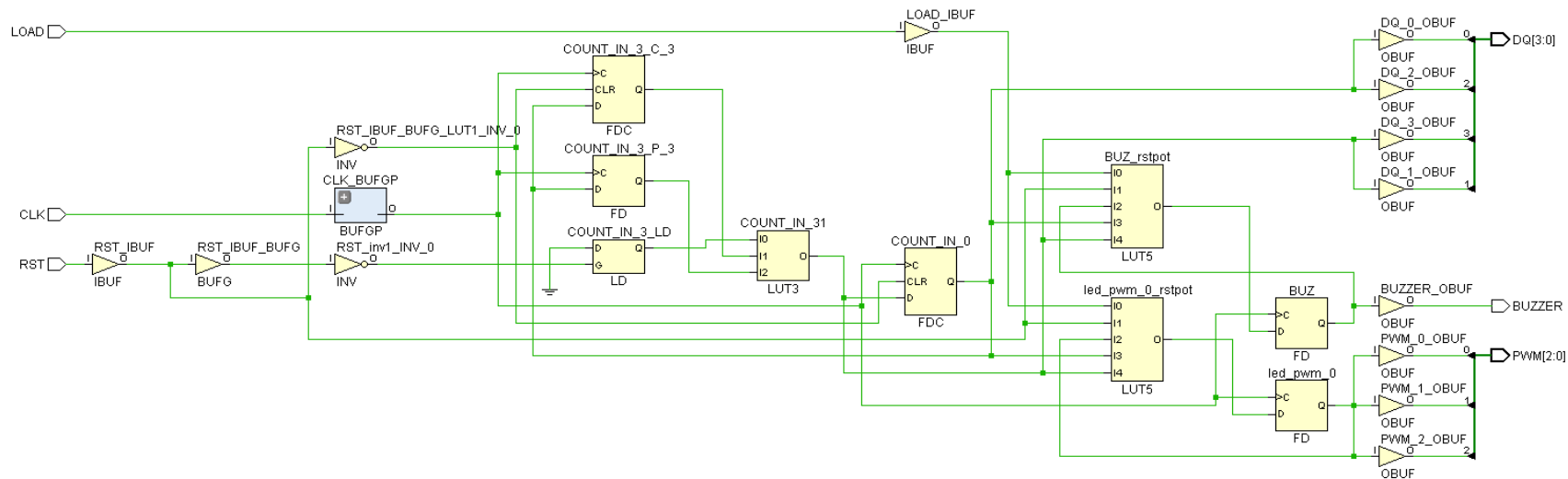
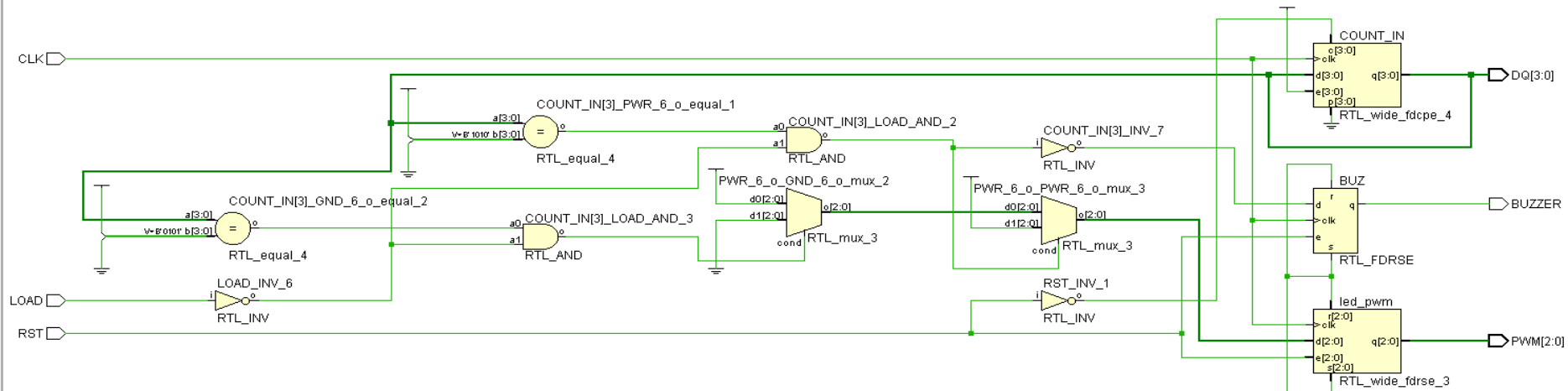
실습응용 링카운터 동작테이블

CLOCK	RESET	LOAD	Y0	Y1	Y2	Y3	BUZ	PWM
0	0	X	0	0	0	0	X	X
0	1	X	X	X	X	X	X	X
1	1	1	1	0	1	0	1	1
1		1	0	1	0	1	1	1
1		1	1	0	1	0	0	1
1		1	0	1	0	1	1	0

실습응용 링카운터 시뮬레이션화면



실습응용 링카운터 회로화면

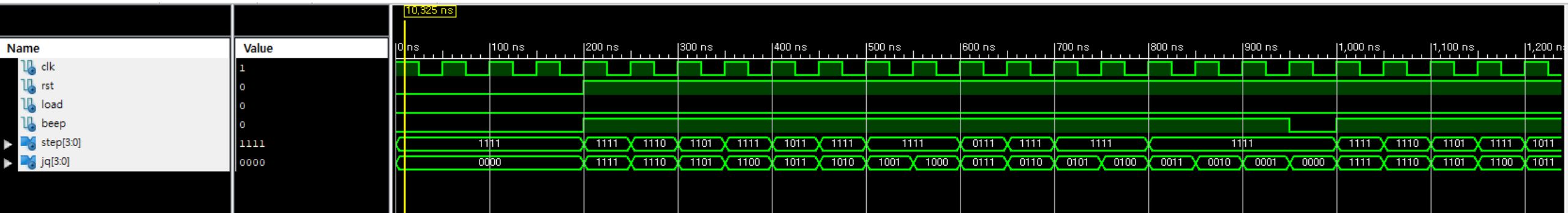


실습응용 리플카운터 동작테이블

입력	Johnson COUNTER							
CLOCK	Q0	Q1	Q2	Q3	Y4	Y5	Y6	Y7
↑	0	1	1	1	0	1	1	1
	1	0	1	1	1	0	1	1
	0	0	1	1	1	1	1	1
	1	1	0	1	1	1	0	1
	0	1	0	1	1	1	1	1
	1	0	0	1	1	1	1	1
	0	0	0	1	1	1	1	1
	1	1	1	0	1	1	1	0
	0	1	1	0	1	1	1	1
	1	0	1	0	1	1	1	1
	0	0	1	0	1	1	1	1
	1	1	0	0	1	1	1	1
	0	1	0	0	1	1	1	1
	1	0	0	0	1	1	1	1
	0	0	0	0	1	1	1	1

입력	Johnson COUNTER					
CLOCK	LOAD	Q0	Q1	Q2	Q3	Beep(부저)
X	1	0	1	1	1	1
	1	1	0	1	1	1
	1	0	0	1	1	1
	1	1	1	0	1	1
	1	0	1	0	1	1
	1	1	0	0	1	1
	1	0	0	0	1	1
	1	1	1	1	0	1
	1	0	1	1	0	1
	1	1	0	1	0	1
	1	0	0	1	0	1
	1	1	1	0	0	1
	1	0	1	0	0	1
	1	1	0	0	0	1
	1	0	0	0	0	1
	1	1	0	0	0	1
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0			

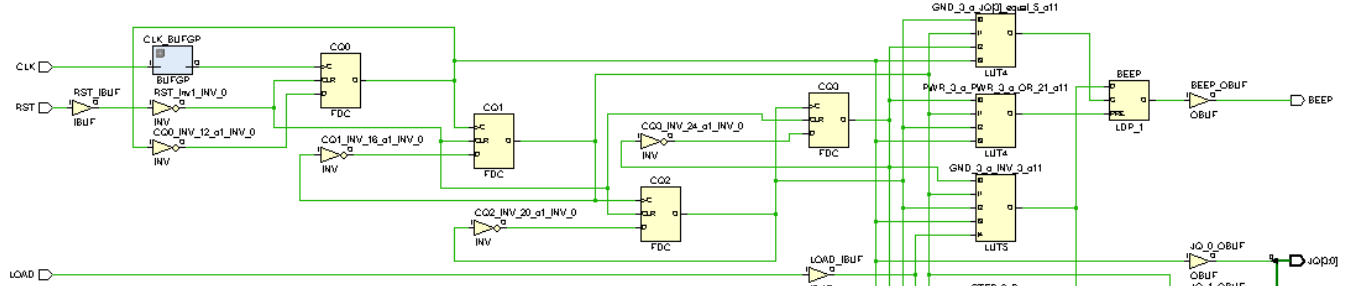
실습응용 리플카운터 시뮬레이션화면



실습응용 리플카운터 회로화면

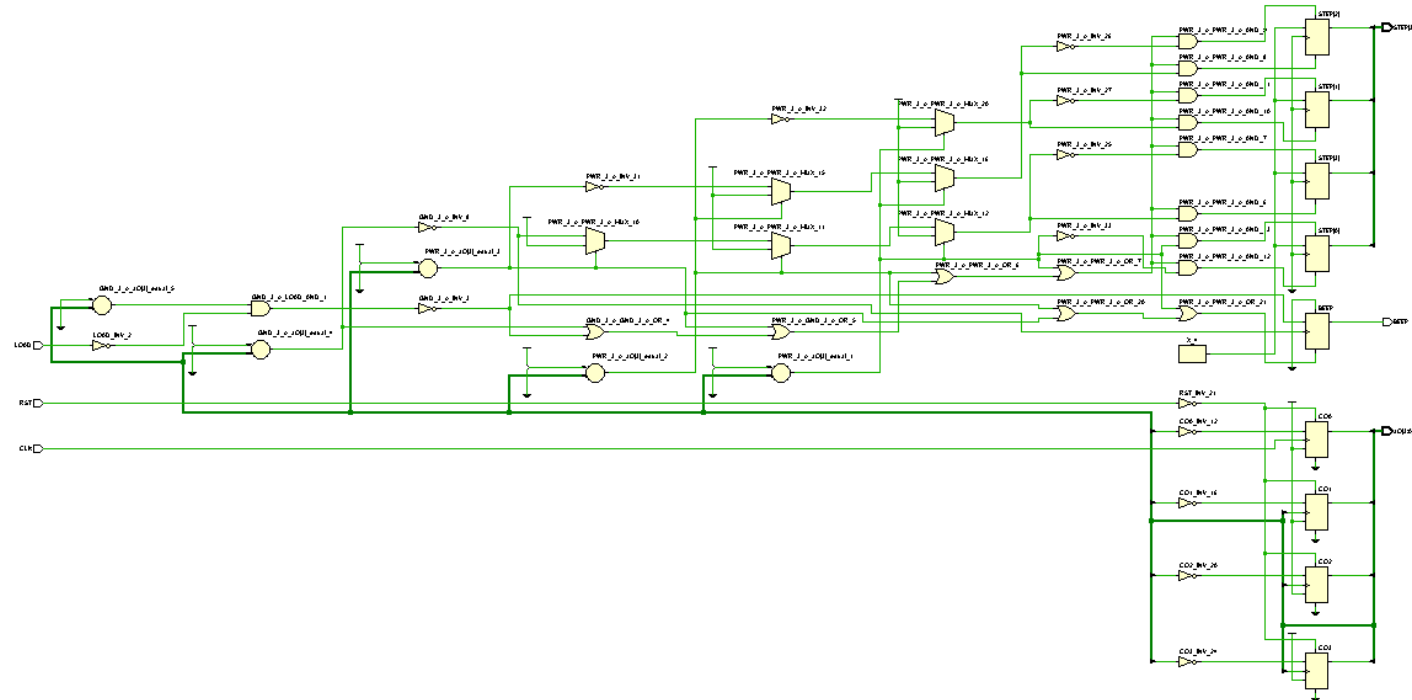
Schematic x

36 Nets



ice x RTL Schematic x

I/O Ports 55 Nets



FPGA2 Week10 Team3

2021041091_한규철, 2021041089_최태민

비동기식reset + 8진 segment counter + Beep 영상 : <https://youtu.be/DarWXztJbiU>

비동기식reset + 5진 segment counter + Beep 영상 : <https://youtu.be/IE1U33jugLM>

실습1 + 과제LOAD 기능추가: <https://youtu.be/h2nBaQFZ4as?si=Cy75qPw3K3OvoQFO>

실습2 + 과제LOAD 기능추가: <https://youtu.be/EpnJecLubIQ?si=l4l2YmURgBfpX1Tc>

동작영상 유튜브 주소

비동기식reset + 8진 segment counter + Beep 영상 : <https://youtu.be/DarWXztJbiU>

비동기식reset + 5진 segment counter + Beep 영상 : <https://youtu.be/IE1U33jugLM>

실습1 + 과제LOAD 기능추가: <https://youtu.be/h2nBaQFZ4as?si=Cy75qPw3K3OvoQFO>

실습2 + 과제LOAD 기능추가: <https://youtu.be/EpnJecLublQ?si=l4l2YmURgBfpX1Tc>

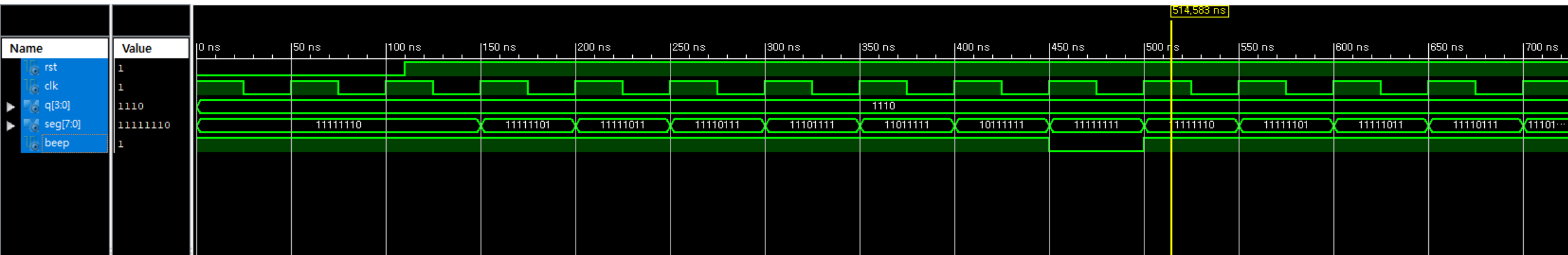
비동기식reset + 8진 BCD+ Beep 동작테이블

비동기식 Reset	Clock	beep	8진	INPUT-BCD		
EV3	CLK	Beep	내부 Count	C	B	A
J12	C9	E11		0	0	0
1	↑	1	0	0	0	0
1		1	0	0	0	0
1		1	1	0	0	1
1		1	2	0	1	0
1		1	3	0	1	1
1		1	4	1	0	0
1		1	5	1	0	1
1		1	6	1	1	0
1		1	7	1	1	1
0		0	0	0	0	0

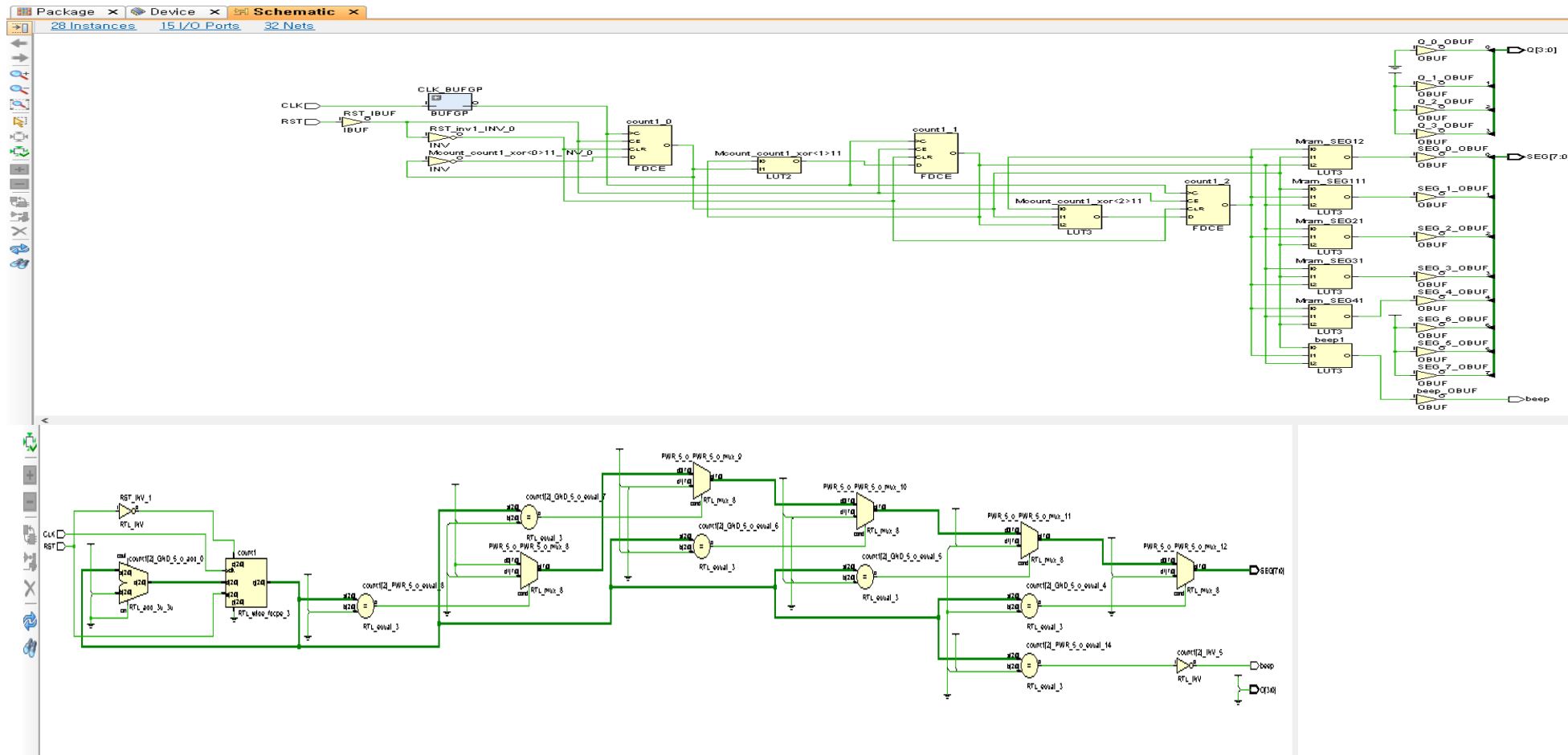
비동기식 reset + 8진 segment counter + Beep 동작테이블

16진 수	Common Anode 7-세그먼트LED 코드									Common Anode 7-세그먼트			
	dp	g	f	e	d	c	b	a	Beep	TR3	TR2	TR1	TR0
0	1	1	1	1	1	1	1	0	1	1	1	1	0
1	1	1	1	1	1	1	0	1	1				
2	1	1	1	1	1	0	1	1	1				
3	1	1	1	1	0	1	1	1	1				
4	1	1	1	0	1	1	1	1	1				
5	1	1	0	1	1	1	1	1	1				
6	1	0	1	1	1	1	1	1	1				
7	1	1	1	1	1	1	1	1	0				

비동기식 reset + 8진 segment counter + Beep 시뮬레이션화면



비동기식 reset + 8진 segment counter + Beep 회로화면



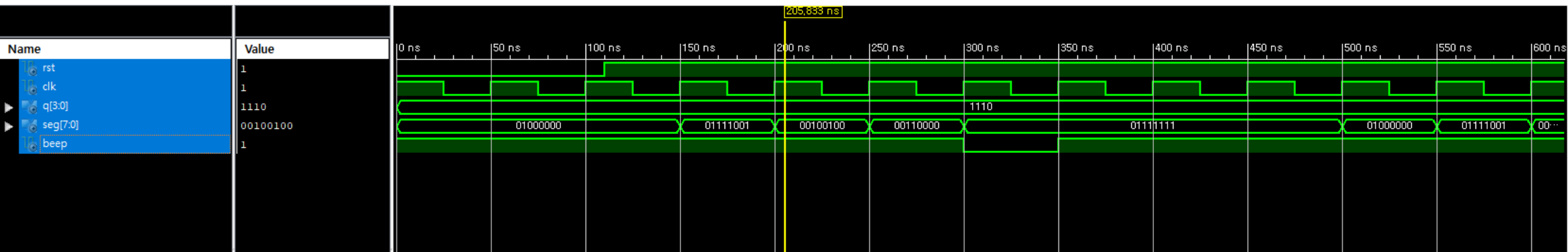
비동기식 reset + 5진 BCD + Beep 동작테이블

비동기식 Reset	Clock	beep	5진	INPUT-BCD		
EV3	CLK	Beep	내부 Count	C	B	A
J12	C9	E11		0	0	0
1	↑	1	0	0	0	0
1		1	0	0	0	1
1		1	1	0	1	0
1		1	2	0	1	1
1		1	3	1	0	0
1		1	4	1	0	1
0		0	0	0	0	0

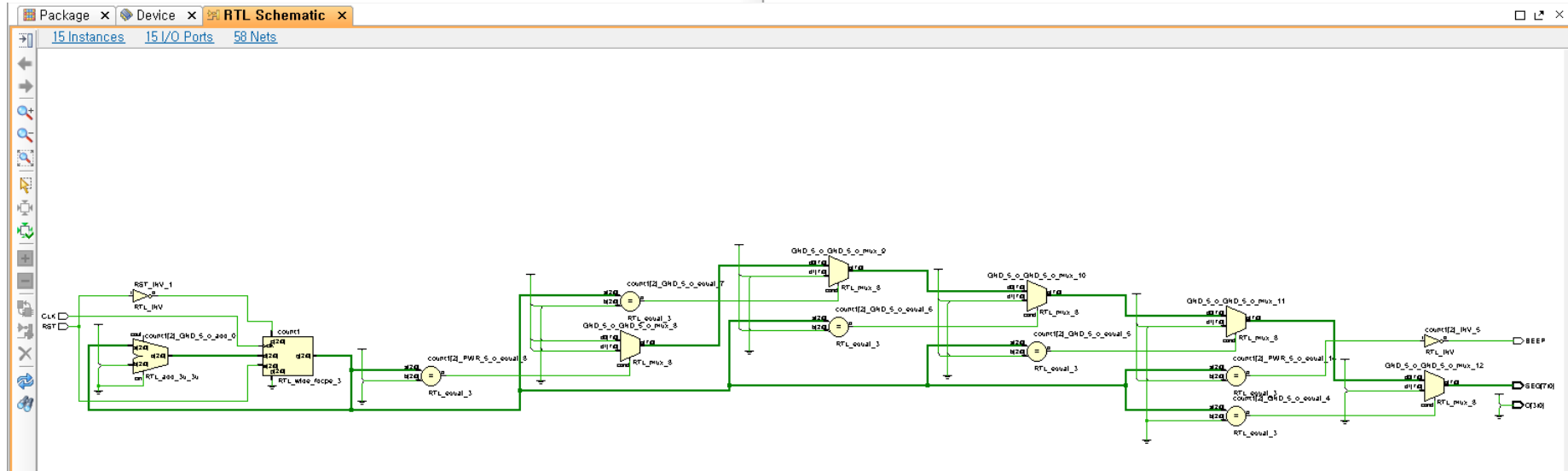
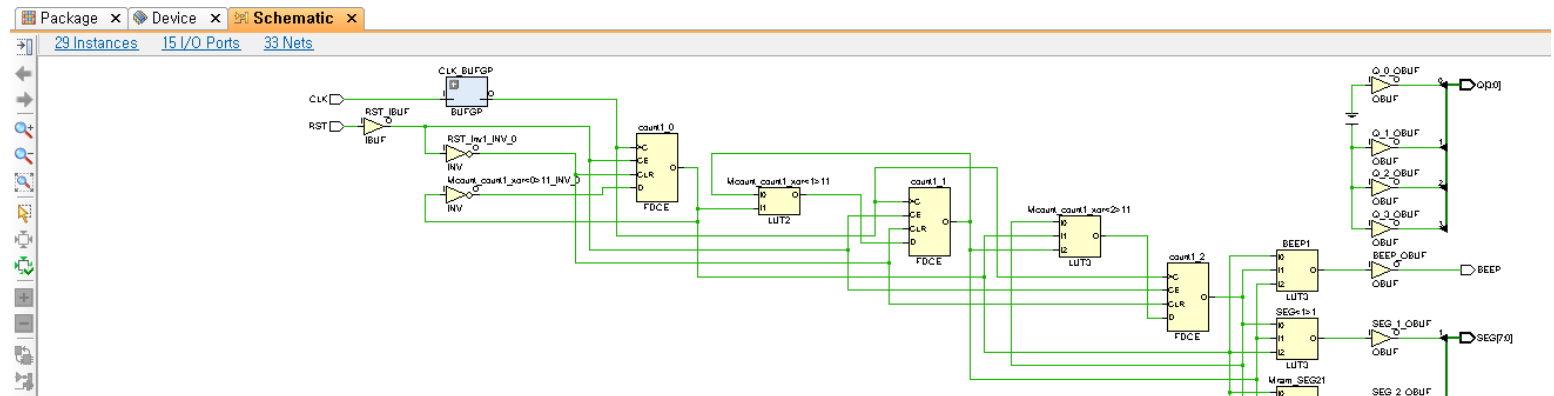
비동기식 reset + 5진 segment counter + Beep 동작테이블

16진 수	Common Anode 7-세그먼트LED 코드									Common Anode 7-세그먼트			
	dp	g	f	e	d	c	b	a	Beep	TR3	TR2	TR1	TR0
0	1	1	1	1	1	1	1	0	1	1	1	1	0
1	1	1	1	1	1	1	0	1	1				
2	1	1	1	1	1	0	1	1	1				
3	1	1	1	1	0	1	1	1	1				
4	1	1	1	0	1	1	1	1	1				

비동기식 reset + 5진 segment counter + Beep 시뮬레이션화면



비동기식 reset + 5진 segment counter + Beep 회로화면

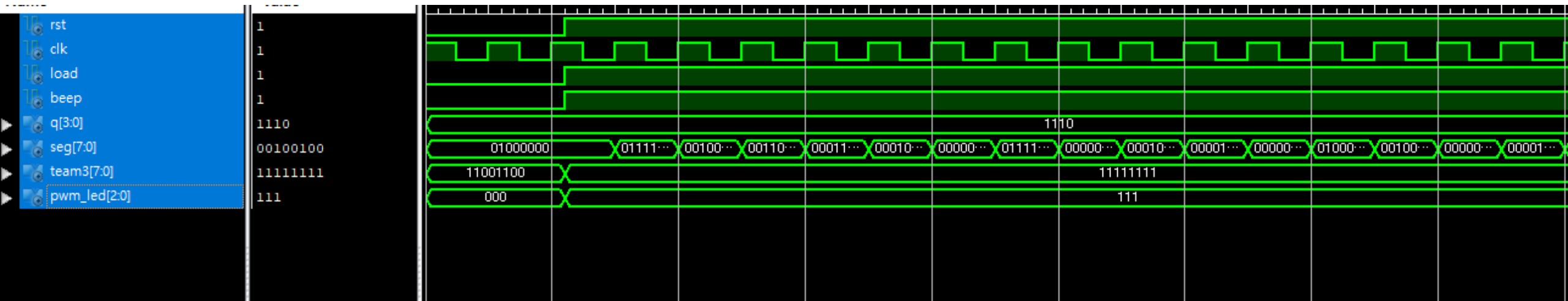


실습1 + 과제LOAD 기능추가 동작테이블

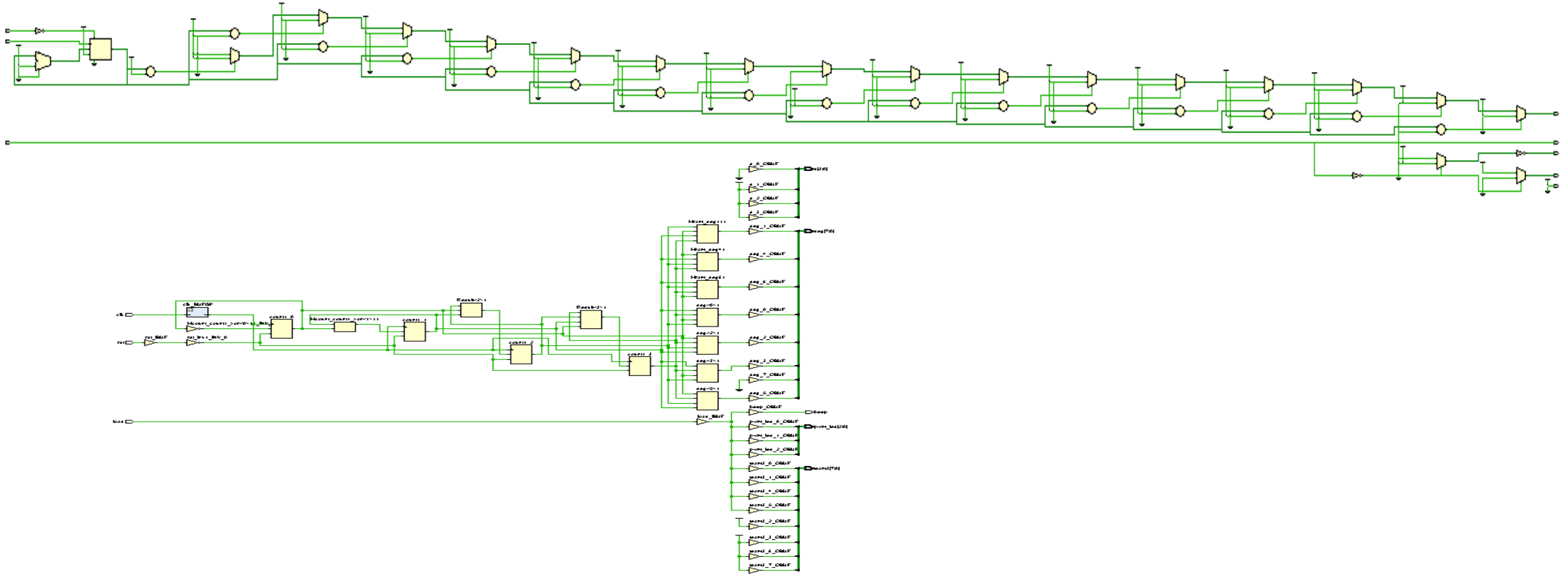
CLOCK	RESET	LOAD	COUNT	DP	G	F	E	D	C	B	A	TR3	TR2	TR1	TR0
CLK	RST	LOAD	COUNT1	A11	A9	E6	D6	C8	C6	B8	A8	D11	C13	B14	A13
C9	J12(EV3)	J14(EV2)	내부신호									1	1	1	0
On	1	1	0	0	1	0	0	0	0	0	0				
			1	0	1	1	1	1	0	0	1				
			2	0	0	1	0	0	1	0	0				
			3	0	0	1	1	0	0	0	0				
			4	0	0	0	1	1	0	0	1				
			5	0	0	0	1	0	0	1	0				
			6	0	0	0	0	0	0	1	0				
			7	0	1	1	1	1	0	0	0				
			8	0	0	0	0	0	0	0	0				
			9	0	0	0	1	0	0	1	1				
			a	0	0	0	0	1	1	1	0				

		D	G	F	E	D	C	B	A
		P							
a	1	0	0	0	0	1	0	0	0
b		0	0	0	0	0	0	1	1
c		0	0	1	0	0	1	1	1
d		0	0	1	0	0	0	0	1
e		0	0	0	0	0	1	1	0
f		0	0	0	0	1	1	1	0

실습1 + 과제LOAD 기능추가 시뮬레이션 화면



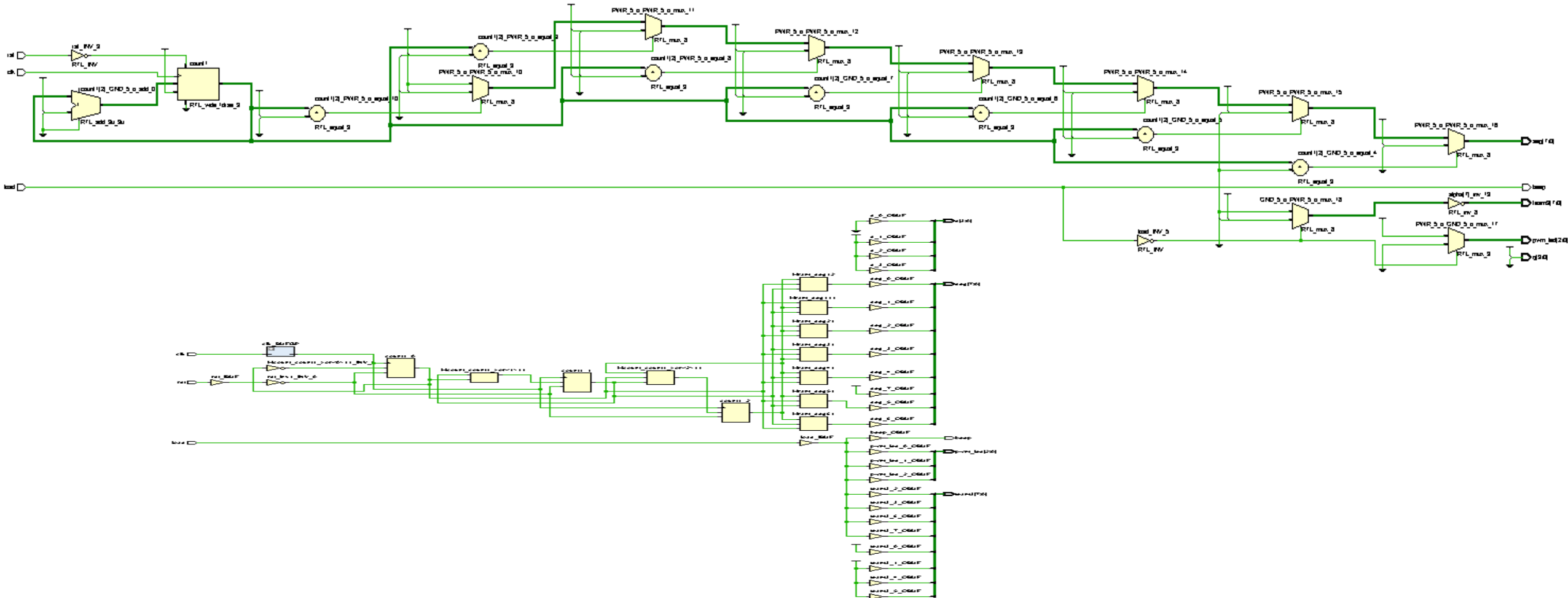
실습1 + 과제LOAD 기능추가 회로화면



실습2 + 과제LOAD 기능추가 동작테이블

CLOCK	RESET	LOAD	COUNT	DP	G	F	E	D	C	B	A	TR3	TR2	TR1	TR0	load	y5	y4	y1	y0
CLK	RST	LOAD	COUNT1	A11	A9	E6	D6	C8	C6	B8	A8	D11	C13	B14	A13	0	0	0	0	0
C9	J12(EV3)	J14(EV2)	내부신호									1	1	1	0					
On	1	1	0	1	1	1	1	1	1	1	0									
			1	1	1	1	1	1	1	0	1									
			2	1	1	1	1	1	0	1	1									
			3	1	1	1	1	0	1	1	1									
			4	1	1	1	0	1	1	1	1									
			5	1	1	0	1	1	1	1	1									
			6	1	0	1	1	1	1	1	1									
			7	0	1	1	1	1	1	1	1									

실습2 + 과제LOAD 기능추가 시뮬레이션 화면



실습2 + 과제LOAD 기능추가 회로화면

